

Mass and covering exponents for fixed-length honeycomb walks

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Abstract

For uniform self-avoiding walks of every sufficiently large integer length on the honeycomb lattice, we prove diameter exponent $3/4$ and simultaneous local-mass and covering exponent $4/3$, with arbitrary positive exponent slack and arbitrary polynomial failure probability.

1 Introduction

A self-avoiding walk is a nearest-neighbor lattice path that visits no vertex twice. Its length is fixed combinatorially, but its spatial extent is random. The exponent $3/4$ expresses a predicted relation between those two quantities in two dimensions; its reciprocal $4/3$ expresses the corresponding mass dimension. A precise statement requires both a probability law and a choice of spatial statistic. This paper concerns the regular honeycomb lattice and gives such a statement for diameter, local mass, and covering numbers.

1.1 The model and the meaning of the exponents

Let $\mathbb{H}$ be the planar dual of the equilateral triangular tiling with triangle side length one. Fix a vertex o . Let $\mathcal{W}_n$ be the set of paths

$$\gamma = (\gamma_0, \dots, \gamma_n), \quad \gamma_0 = o,$$

whose consecutive vertices are adjacent in $\mathbb{H}$ and whose vertices are all distinct. Write $c_n = |\mathcal{W}_n|$, and let $\mathbb{P}_n$ be the uniform law on $\mathcal{W}_n$. The endpoint γ_n is free. We use the Euclidean embedding and set

$$V(\gamma) = \{\gamma_0, \dots, \gamma_n\}, \quad D(\gamma) = \text{diam } V(\gamma), \quad M_\gamma(z, s) = |V(\gamma) \cap \overline{B}(z, s)|.$$

Let $\mathcal{N}_\gamma(s)$ be the least number of closed Euclidean balls of radius s that cover $V(\gamma)$; their centers may lie anywhere in the plane.

Theorem 1.1 (Fixed-length mass and covering exponents). *For every $\delta > 0$ and $k > 0$, there are constants $C_{\delta,k} < \infty$ and $n_0(\delta, k)$ such that, for every integer $n \geq n_0(\delta, k)$, an event of $\mathbb{P}_n$ -probability at least $1 - C_{\delta,k}n^{-k}$ has all the following properties:*

$$n^{3/4-\delta} \leq D(\gamma) \leq n^{3/4+\delta}, \tag{1}$$

$$n^{-\delta} \min\{n, s^{4/3}\} \leq M_\gamma(z, s) \leq n^\delta \min\{n, s^{4/3}\}, \tag{2}$$

$$n^{-\delta}(1 + ns^{-4/3}) \leq \mathcal{N}_\gamma(s) \leq n^\delta(1 + ns^{-4/3}). \tag{3}$$

The last two bounds hold simultaneously for every $z \in V(\gamma)$ and every real $s \in [1, n]$ (with z relevant only to (2)).

The power slack absorbs constants and the difference between n edges and $n + 1$ vertices. In particular, the notation $n^{a+o(1)}$ below always has an explicitly stated probability interpretation; it does not assert a limiting multiplicative constant.

Corollary 1.2 (Span in every lattice normal). *For every $\delta, k > 0$, outside an event of $\mathbb{P}_n$ -probability $O_{\delta,k}(n^{-k})$, the projection of $V(\gamma)$ onto each unit normal to a direction of triangular-lattice edges has span at least $n^{3/4-\delta}$. This holds for every sufficiently large integer n .*

This lower bound in each of the three normal directions gives more information than a lower bound on the largest spatial extent.

Corollary 1.3 (Covering dimension at polynomially separated scales). *For each fixed $\epsilon \in (0, 3/4)$,*

$$\sup_{1 \leq s \leq n^{3/4-\epsilon}} \left| \frac{\log \mathcal{N}_\gamma(s)}{\log(D(\gamma)/s)} - \frac{4}{3} \right| \rightarrow 0 \quad \text{in } \mathbb{P}_n\text{-probability.}$$

The statistic can be defined arbitrarily when $D(\gamma) \leq n^{3/4-\epsilon}$.

Corollary 1.4 (Radius of gyration and moments). *Let*

$$\bar{\gamma} = \frac{1}{n+1} \sum_{j=0}^n \gamma_j, \quad R_g(\gamma)^2 = \frac{1}{n+1} \sum_{j=0}^n |\gamma_j - \bar{\gamma}|^2.$$

Then $R_g = n^{3/4+o(1)}$ with the same arbitrary polynomial probability convention as in Theorem 1.1. For every fixed $p > 0$,

$$\mathbb{E}_n[D^p] = n^{3p/4+o(1)}, \quad \mathbb{E}_n[R_g^p] = n^{3p/4+o(1)}.$$

We prove the deductions from the geometric estimates, including the Corollaries, in Section 9. These statements concern the mass and covering behavior of the lattice trace as its length grows. The assertions do not include convergence to a continuum curve, a continuum Hausdorff dimension, a universality theorem on changing lattice, or a fixed-length lower bound on $|\gamma_n - o|$. A lower bound for the diameter does not imply that the two endpoints are far apart.

1.2 Critical weights and the geometric argument

The proof first works with the critical activity $\rho = (2 + \sqrt{2})^{-1/2}$. A path then carries weight ρ^ℓ , where ℓ counts edges for a vertex-rooted walk and visited vertices for a path whose endpoints are ports, the midpoints of honeycomb edges. A *mass* is a sum of these weights before normalization. The distinction lets pieces in adjacent open strips concatenate with exactly the product weight. A bridge crosses a strip without visiting either boundary elsewhere. It is irreducible if no intermediate lattice line parallel to the strip boundaries is crossed exactly once; such a line would split it into two bridges.

Exact lattice observables provide the mass crossing a strip, nesting around a face, the length-square mass of polygons, and a uniform bound for arcs joining two marked ports on a cylinder. We state their precise positive outputs in Section 2. The analytic proofs belong to the complete companion articles identified there. Here we prove the geometric passage from those inputs to an unrestricted fixed-length trace.

There are three obstacles. A bridge used to modify a path must fit in a corridor that avoids the remaining pieces; a strip mass or mean length alone gives no such guarantee. We sew smaller bridges and recover the sewing lines from the resulting path, controlling the multiplicity of that operation.

This produces localized first and second moments and, by Cauchy–Schwarz, a subpower chance of seeing a long bridge. Independent renewal trials amplify that chance. An exact conditioning identity then transfers the estimate to a bridge of a specified height.

Next, one bridge estimate must control all subpaths of an arbitrary walk. Ordered tests of adjacent renewal paths suppress both unusually fast travel and excessive length in a narrow region. When the two tests meet the same irreducible piece, its entire weight enters once. This joint estimate is essential when the number of turns grows with the scale. Finally, a separate estimate for many disjoint crossings controls repeated visits to one ball. A subpath bound on its own would leave these separate visits uncontrolled.

The estimates are proved for rooted walks of any length confined to diameter at most a fixed multiple of H . For every $\tau > 0$, they simultaneously bound the length of every subwalk σ by $H^\tau(1 + \text{diam } \sigma)^{4/3}$, its diameter by $H^\tau(1 + |\sigma|)^{3/4}$, and the number of visited vertices in a ball of radius s about a visited vertex by $H^\tau s^{4/3}$ for $1 \leq s \leq H$. Here $|\sigma|$ counts edges. For every $A > 0$, the total critical mass of walks violating any of these bounds is at most $C_{\tau,A}H^{-A}$. This absolute estimate is strong enough to survive summation over all the possible subpaths.

To pass to every integer length, submultiplicativity and the polynomial strip lower bound give $c_n \rho^n \geq 1$. Taking H proportional to n and dividing an exceptional mass by this fixed-length partition function preserves arbitrary polynomial decay. The diameter, local mass and covering conclusions then follow on one simultaneous event. The elementary renewal process needed here is developed locally; no change-of-ensemble theorem is assumed.

The proof follows these steps in order. Section 3 first bounds the mass of the outside pieces that a cut leaves behind. Section 4 proves the confined bridge length moments and then constructs turning corridors for the later probes. Section 5 amplifies the moment bound at a specified height. Section 6 then controls ordered adjacent paths, including a shared long irreducible. Sections 7 and 8 use those estimates to bound fast travel, excessive local length, and repeated visits. Section 9 performs the exact-length transfer and proves the stated geometric consequences. The independent asynchronous avoidance argument is retained in Appendix A.

1.3 Historical context

The predicted exponent $3/4$ belongs to the two-dimensional polymer theory developed through the dilute $O(n)$ model. Nienhuis’s solid-on-solid and Coulomb-gas analysis gives this exponent at $n = 0$ [10]. These predictions concern more than the exponential growth rate of the number of walks. Duminil-Copin and Smirnov proved the exact honeycomb connective constant by introducing a parafermionic observable and summing its local cancellation over a strip [3]. Their argument gives $c/T \leq B_T \leq 1$ for the critical strip-crossing mass; the sharper exponent $-1/4$ is recorded there as a prediction.

Subsequent work sharpened the information about this mass. Beaton, Bousquet-Mélou, de Gier, Duminil-Copin and Guttmann proved $B_T \rightarrow 0$ in their study of surface adsorption [1]. Glazman and Manolescu gave a shorter proof and a logarithmic subsequence bound, and established invariance of the boundary two-point function for a specified class of columnwise rhombic half-plane tilings with integrable weights [5]. More recently, Krachun and Panagiotis proved a polynomial upper bound on B_T and a quantitative sub-ballistic estimate: a uniform n -step honeycomb walk stays within distance $Cn/\log(n+1)$ of its origin except on an event of probability at most $\exp(-cn^{2/3})$ [7].

The analytic companions used here provide the further quantitative strip and marked-polygon estimates needed for the geometric argument.

The bridge factorization follows Kesten’s renewal method [6]; see also Madras and Slade [9] and the appendix of Lawler, Schramm and Werner [8]. Dyhr, Gilbert, Kennedy, Lawler and Passon [4,

Proposition 2.10] identify the critical free-endpoint strip law on the square lattice by conditioning the infinite half-plane walk to have a cut at a prescribed height and taking its prefix through the upper strip boundary. We derive the renewal normalization and this finite-height identity in the present port convention. For the final full-plane law at each integer length, we use absolute exceptional masses and the partition-function bound $c_n \rho^n \geq 1$.

Lawler, Schramm and Werner also proposed a continuum description through $\text{SLE}_{8/3}$, whose Hausdorff dimension they identified as $4/3$ [8, Proposition 2(ii)]. Beffara subsequently proved the general formula $\min\{2, 1 + \kappa/8\}$ for the dimension of SLE_κ [2]. These continuum results explain the predicted mass dimension; a lattice scaling limit is neither an input nor an output of the present proof.

2 Critical masses and the analytic inputs

Our geometric arguments use positive sums of walk and polygon weights. This section fixes their endpoint conventions and states the precise analytic estimates proved in the companion papers. A free endpoint always means a sum over the actual possible endpoint ports from one fixed source; it never introduces a second translation sum.

2.1 Ports, strips and local endpoint changes

A port is the midpoint of an edge of the triangular tiling. A port path follows honeycomb edges, with an initial and a final half-edge, and visits each dual vertex at most once. A nonempty port path has distinct endpoint ports; the empty height-zero bridge below is the sole exception. Its length counts visited vertices and its weight is $w(\gamma) = \rho^{|\gamma|}$, with $\rho = (2 + \sqrt{2})^{-1/2}$. The trace of a port path includes its two ports. Removing these ports changes its diameter by a bounded constant. Unrestricted vertex paths keep the edge-count convention of the introduction.

A horizontal band is the region between two successive horizontal lines of tiling edges. Its Euclidean height is $d_0 = \sqrt{3}/2$. A strict bridge of height h goes from a bottom port to a top port of the open h -band strip and has no other visit to the boundary. An arch has two distinct ports on the same boundary. Because ports carry no weight and adjacent open strips have disjoint vertices, concatenating strict bridges multiplies their weights exactly. Every horizontal seam port is a translate of the initial port by a symmetry of the triangular tiling, with its inward half-edge pointing into the next band. Thus each translated continuation has exactly the same admissible shapes and weights; there is no residual port type or parity state in this concatenation. Tangential displacements of height- h bridges lie in $h/2 + \mathbb{Z}$. Write

$$b_h(x) = \sum_{\gamma: 0 \rightarrow (x, h)} w(\gamma), \quad B_h = \sum_x b_h(x), \quad B_0 = 1.$$

We write $\mathbb{P}_h^{\text{br}}$ for the probability measure obtained by dividing these bridge weights by B_h . Let $K_h(j)$ be the arch mass between bottom ports separated by the nonzero integer j , and put $A_h = \sum_{j \neq 0} K_h(j)$ and $m_h = \sum_{j > 0} j K_h(j)$.

A weak bridge is a vertex path whose endpoints attain its minimum and maximum heights; intermediate visits to these levels are permitted. The following bounded modification will be used whenever a geometric cut produces a vertex endpoint.

Lemma 2.1 (Endpoint extension). *A weak bridge in any lattice normal direction can be extended to a strict port bridge between the first lattice boundary lines outside its height range. The height*

and length change by $O(1)$, the weight ratio is bounded above and below by positive constants, and the inverse operation has bounded multiplicity after recording its endpoint types.

Proof. At a lower endpoint whose triangle center is nearest its lower side, append the vertical half-edge to that side's port. At the other triangle type, first append its fixed downward edge and then the vertical half-edge. The reflected operation applies at the upper endpoint. Every newly added vertex lies strictly outside the old height range, so it was unvisited. The two additions are disjoint for positive macroscopic span. Bounded spans have only finitely many endpoint types and use the same construction; a zero-span piece is trivial. At most three extra vertex factors occur relative to the original edge count. Removing the recorded one- or two-edge endpoint patterns recovers the original path, proving all bounds. Rotations and reflections give the other lattice normal directions. $\square$

Throughout, a bound $F(h) \leq h^{a+o(1)}$ means that for each $\epsilon > 0$, $F(h) \leq C_\epsilon h^{a+\epsilon}$ for all sufficiently large h . The corresponding lower bound has the analogous meaning. Constants may depend on declared fixed geometry and on exponent slack. Uniform statements retain one choice of constant over their full stated parameter range.

2.2 Strip flux and nesting

Set $c = \cos(3\pi/8)$. The first input is the positive boundary consequence of the parafermionic observable, together with its quantitative strip refinement. We use the complete statements of [13, Theorem 1.1 and Lemma 2.1].

Theorem 2.2 (Boundary flux and strip mass). *For every finite simply connected union D of triangular tiles with simple polygonal boundary and each boundary source port a ,*

$$\sum_{b \in \partial D} \sum_{\gamma: a \rightarrow b} \rho^{|\gamma|} e^{3iW(\gamma)/8} = 1,$$

where W is total signed turning from the initial inward direction to the final outward direction. If D is convex, its unsigned exit mass is at most $1/c$. For each $h \geq 1$, all strip sums defined above are finite and

$$cA_h + B_h = 1, \quad B_{h+1} \leq B_h, \quad B_h \asymp (1+h)^{-1/4}, \quad m_h \asymp h^{3/4}, \quad m_{h+1} - m_h \asymp B_h.$$

The constants are independent of h , and $b_h(x) = b_h(-x)$.

The exact identity and monotonicity, rather than the bridge power alone, will control convex cuts and variation of the endpoint kernel.

For an internal honeycomb mid-edge p in a convex domain D , let $L_p(D)$ be the sum over ordered boundary chords through p , weighted by $\rho^{|\gamma|} \cos(3W(\gamma)/8)$. A polygon nest around a face center f is a finite set of pairwise vertex-disjoint simple polygons, each strictly enclosing f . Its weight is the product of $2\rho^{|\ell|}$ over its polygons, and the empty nest has weight one. Write $Z_f(D)$ for the nesting mass in D . Let $P_2(r)$ denote the whole-plane nesting mass with every polygon's diameter at most r , and let S_k denote the middle-face nesting mass in a strip of $2k$ bands.

Theorem 2.3 (Length and nesting). *For the two face centers f_1, f_2 adjacent to p ,*

$$\frac{1}{\sqrt{2}}(Z_{f_1}(D) + Z_{f_2}(D)) \leq L_p(D) \leq Z_{f_1}(D) + Z_{f_2}(D).$$

The same comparison holds with finite quantities in every infinite strip of fixed positive height. Moreover, $P_2(r) = r^{1/12+o(1)}$ and $S_k = k^{1/12+o(1)}$.

These are the length-nest and planar-normalization theorems of [11, Lemma 11.4, Theorem 11.2 and Corollary 11.3]. We will place diameter-truncated nests in the interior of a finite triangle; an untruncated planar partition function would not give that inclusion.

Theorem 2.4 (Raw strip first-length mass). *From one fixed bottom port of the h -band strip,*

$$\sum_{\gamma \text{ arch or bridge}} \rho^{|\gamma|} |\gamma| \leq h^{13/12+o(1)}.$$

The sum is unnormalized and includes paths ending on either boundary. It is Proposition 11.5 of [11]; it is stronger information for our purpose than a mean conditional on reaching a deep line. The main geometric proof will supply a matching lower first moment on bridges confined to diameter at most a fixed multiple of their height.

2.3 Marked polygons and arcs

The marked-polygon theorem of [12, Theorem 1.1] gives

Theorem 2.5 (Polygon length-square mass). *For unrooted, unoriented simple honeycomb polygons modulo triangular-lattice translations,*

$$\sum_{[\ell]: \text{diam } \ell \leq H} \rho^{|\ell|} |\ell|^2 \leq H^{2/3+o(1)}.$$

Here $|\ell|$ is its vertex count. The translation-class convention is needed when an exterior closure is translated back to a rooted path.

We also use the cylinder arc estimate of [12, Theorem 8.3]. Its cylinder is formed by a row cut in cyclic order $[p, X, p', Y]$, where p, p' are two marked ports and X, Y are sets of ordinary sites. Put $s = |X|$, $b = |Y|$, and require $s + b + 2$ even. The physical row uses rhombus parameters $\lambda = \pi/8$ or 2λ : these two parameter values label the two lattice side directions forming the positive cone. After a common shift the logarithmic parameters a_i have imaginary heights 0 or $\beta = \pi/4$ in X and at the marks, and height zero in Y . At these physical parameters all real parts of the shifted a_i are zero. An upper site is one of height β . Let $A(p, p')$ be the unnormalized critical mass of self-avoiding cylinder arcs departing west from p and arriving east at p' , making no other visit to either mark and leaving their unused half-edges vacant.

Theorem 2.6 (Uniform one-arc bound). *There are constants $\delta_1 > 0$, $L_* > 0$, and $C < \infty$ such that $A(p, p') \leq b^C$ whenever*

$$L_Y \geq L_X \geq L_*, \quad s \leq b, \quad L_X = \frac{3}{4} \log s, \quad L_Y = \frac{3}{4} \log b,$$

and at least fraction $1/2 - \delta_1$ of the ordinary X -sites are upper. There is no positive lower bound on L_X/L_Y .

The exponent C is fixed throughout this domain. In the next section we construct such a row cut from an arbitrary walk in a box. There s may stay bounded while b grows with the box, which explains why a comparable-wall estimate would not suffice.

2.4 A tail for every polygon count

The final input is different from the length-square bound. On the balanced physical two-marker cylinder with even $N = 2m$, use the parameter 2λ on its first m bands and λ on the remaining m bands. Let $w_N(a)$ be the critical mass of families of exactly a disjoint separating polygons, namely

$$w_N(a) = \sum_{\substack{\mathcal{F} \text{ disjoint, separating} \\ \#\mathcal{F}=a}} \prod_{P \in \mathcal{F}} \rho^{|P|}.$$

There is no factor two per polygon in this definition. The all-count tail bound of [11, Corollary 8.2] states that, uniformly for every integer $a \geq 0$,

$$w_N(a) \leq \exp \left(C \log N - \frac{ca^2}{\log N} \right) \quad (4)$$

for sufficiently large even N . Its proof uses the all-real-fugacity estimate of [11, Theorem 8.1], namely the bound $Z_N(2 \cosh v) \leq \exp(C \log N(1 + v^2))$ for every real $v \geq 0$. An asymptotic for bounded fugacity would not justify arbitrarily large a . We use (4) only after an explicit reflection and concatenation has turned prescribed slab crossings into such polygon families.

3 The critical mass of paths in a box

The later proofs extract a special subpath and sum freely over the two outside pieces. We therefore need a bound that sums *all* lengths and endpoints in the containing box, with one fixed polynomial exponent. The one-arc estimate in Theorem 2.6 supplies that bound after a geometric reduction. Its uniformity when one group of bands stays fixed while the other grows is used in the last step.

Proposition 3.1 (Full-box susceptibility). *Let $S(H)$ be the maximum, over starting vertices in a box of side H , of the critical mass of walks confined to that box, with arbitrary endpoint and length. There are constants C_0, C_1 such that $S(H) \leq C_0(1 + H)^{C_1}$.*

Proof. The suffix test. We first choose a small constant $a > 0$, and later choose a sufficiently large W , once and independently of H . Put $m = \lfloor aW \rfloor$. Walks shorter than m cost $\exp(O(W))$. Consider the finitely many triangular bases, including reflections, whose upward side directions span a 60° cone. In each basis require a westward step among the last m steps. If this fails in one basis, there is at most one eastward and one bisector-parallel choice at each vertex. Two consecutive parallel choices would immediately backtrack. Thus the number of length- l continuations avoiding west is at most $C\varphi^l$, where $\varphi = (1 + \sqrt{5})/2$. Since $\rho\varphi < 1$, the total mass failing at least one of these suffix tests is at most

$$\epsilon_W S(H), \quad \epsilon_W = C(\rho\varphi)^m. \quad (5)$$

The prefix ending where the suffix starts is included in $S(H)$; ignoring its avoidance constraints only enlarges this bound.

Finding a cylinder arc. We seek a middle arc from a suitable earlier step to a westward step in the suffix. The criterion below makes this arc eligible for the one-arc theorem. Choosing the first such step leaves a prefix with no earlier qualifying step; after treating the arc, we will bound the mass of these prefixes by a constant depending only on W . Fix the endpoint z , costing $O((1 + H)^2)$ choices, and scan the step midpoints before this suffix. Let ℓ_1, ℓ_2, ℓ_3 be the three lattice-side lines through

z , and put $d(p) = \min_i \text{dist}(p, \ell_i)$. If $d(p) > W$, the vector $q = z - p$ lies in one open positive cone. Write

$$q = u_1 e_1 + u_2 e_2, \quad u_1, u_2 \geq 2W/\sqrt{3}, \quad t = \frac{u_1}{u_1 + u_2},$$

where e_1, e_2 are its adjacent unit triangular side vectors. The coefficient bounds follow by taking distances to the two cone-side lines. Assignment A makes e_1 the upper-angle direction and is eligible when $t \geq r$, where $r = 1/2 - \delta_1/2$. The reflected assignment B makes e_2 upper and is eligible when $t \leq 1 - r$. Reflection fixes the cone bisector and reverses the transverse coordinate. In particular, a step with negative transverse increment is westward in A , a step with positive increment is westward in B , and a bisector-parallel step is westward in neither. Call a step successful if it is westward in some eligible assignment.

Suppose there is a success, and use the first one, with midpoint p . Choose a westward suffix step in its eligible basis, with midpoint p' . The intervening arc departs west from p and arrives east at p' . Moreover $p' - p = q + O(m)$. The two marked triangular sides have types $\tau, \tau' \in \{1, 2\}$; write $p = v + e_\tau/2$ and $p' = v' + e_{\tau'}/2$, where v, v' are triangular lattice vertices. The displacement between the two marked half-sides is

$$n_1 e_1 + n_2 e_2 = (p' - p) - \frac{e_\tau + e_{\tau'}}{2} = v' - v - e_\tau. \quad (6)$$

Thus n_1, n_2 are integers, since adjacent side vectors form an integer lattice basis. They satisfy $n_i \geq cW - Cm - 1$, and $s = n_1 + n_2 \leq C(H + 1)$. The change in the upper coefficient fraction is at most $C(m + 1)/W$. Choose a small relative to δ_1 , and then W large. Both coefficients are positive, $s \geq cW$, and the ordinary upper fraction is at least $1/2 - \delta_1$.

Figure 1 shows the cone and the marked half-sides in this displacement calculation.

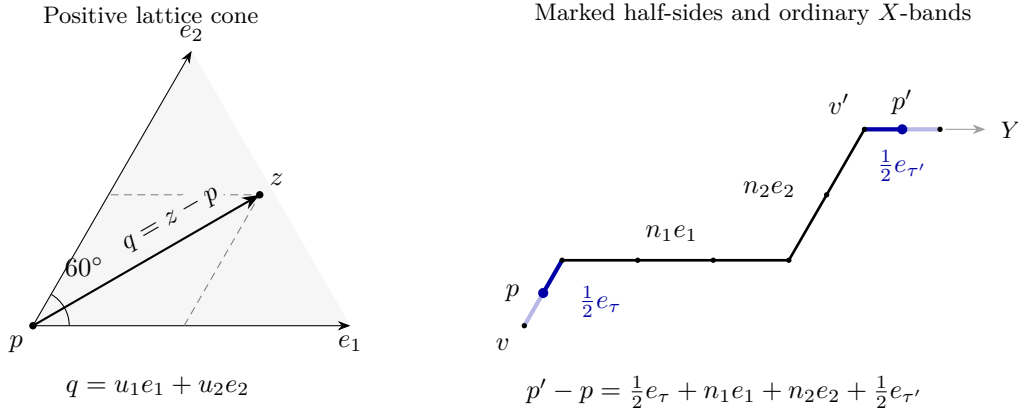


Figure 1: The row construction in a positive 60° lattice cone. On the right, an example canonical order uses $n_1 = 3$ ordinary sides in direction e_1 , followed by $n_2 = 2$ in direction e_2 ; the marked sides have types $\tau = 2$ and $\tau' = 1$. The segment from p to p' contains one half of each marked side, so subtracting these halves leaves the integer displacement $n_1 e_1 + n_2 e_2$. The faint halves complete the marked sides but do not belong to this segment. This is a row cut, not the extracted honeycomb arc; the arc may cross ordinary parts of the row. The angles drawn are Euclidean lattice angles, not the spectral parameters.

Use these integers as the ordinary X -band counts in the row $[p, X, p', Y]$, in a fixed canonical order. We do not sum over band orders. Choose all b ordinary Y -bands lower, with $b \geq K(H + s + 1)$, adjusting by at most one to make $s + b + 2$ even. We may take $b \leq C(H + s + 1)$. The period vector,

including the two full marked sides, has projection at least cb on the cone bisector. Taking K large makes its norm larger than the box diameter with the bounded endpoint margin. Projection to this cylinder is consequently injective on the box. The selected arc has no repeated marked port, and its unused marked half-edges are vacant. Ordinary crossings of the row are permitted. The physical row therefore gives exactly an arc in Theorem 2.6, with unchanged vertex weight.

The anchored port, basis, two marked-side types and two ordinary band counts determine the row and its lift once b has been chosen; there are only polynomially many such data in the box. Choose W also so that $s \geq cW$ implies $L_X \geq L_*$. The arc mass is then at most $b^C \leq C(1+H)^C$, while its suffix remnant costs $\exp(O(W))$. Cutting at the two midpoints changes the vertex-path weight by bounded factors. It remains to bound the mass of the unsuccessful prefix uniformly in H and z . If no step succeeds, the same prefix estimate applies to the whole part before the suffix.

Summing the unsuccessful prefixes. We first record a quantitative strip estimate. From one fixed start, the total free-end mass of all lengths in a straight strip of width $O(W)$, parallel to a lattice side, is at most

$$U(W) = C \exp(CW^{3/4}). \quad (7)$$

Indeed, split at the first global minimum in the normal direction, reverse the first tail, and retain the second. Decompose each tail chronologically at successive last alternating extrema of its remaining piece. The spans satisfy $a_1 \geq a_2 > a_3 > \dots$ on the one-third-band grid; each positive span occurs at most twice per tail. By Lemma 2.1 and Theorem 2.2, a weak bridge of span $j/3$ has free-end mass at most $f(j) = C'(1+j)^{-1/4}$. Zero-span pieces are trivial. Bounded endpoint and cut records can be incorporated in C' . Summing the span lists for the two tails gives

$$\prod_{1 \leq j \leq CW} (1 + f(j))^4 \leq \exp(C''W^{3/4}).$$

Anchor the minimum at one of the two honeycomb translation-class representatives. The two complete tails determine the split time, and the endpoint of the reversed first tail determines the unique translation returning to the prescribed original start. Thus there is no additional sum over lengths, split times or minimum positions. This proves (7).

Next consider a maximal run of unsuccessful midpoints with $d(p) > W$. Its cone is fixed. The cone coefficients change by a bounded amount per step and remain at least cW , so consecutive values satisfy $|t_{i+1} - t_i| \leq C/W$. Choose W so large that this is less than $\delta_1/4$. A run visiting both the sole- A region $t > 1 - r$ and the sole- B region $t < r$ would have at least two consecutive midpoints in the overlap $[r, 1 - r]$. There both assignments are eligible. An unsuccessful step then has neither positive nor negative transverse increment and must be parallel to the cone bisector. Every honeycomb vertex has exactly one incident edge in that direction, so two consecutive such steps backtrack. This is impossible. Hence the run visits at most one sole-choice region; its corresponding assignment (or either assignment if there is no sole-choice region) is eligible throughout. Every step avoids west in this one fixed assignment. Its length- l mass is therefore at most $C(\rho\varphi)^l$, by the same Fibonacci count used in (5).

Retain only those maximal runs with $\max d(p) \geq 2W$. Since d changes by a bounded amount per step, every retained interior run has length at least cW . Only the first and last retained runs can lack this lower bound. Summing the lengths of each interior run costs at most Ce^{-cW} ; either end run has bounded total mass. All other midpoints lie within $2W$ of one of the three lines. Outside a disk of radius C_0W about z , these three strips are disjoint and separated by more than a step. A remaining piece stays in one straight strip until it returns to the disk or begins a retained run. The disk contains $O(W^2)$ vertices, so self-avoidance permits only $O(W^2)$ disk steps, which we enumerate individually.

With j retained interior runs there are at most $j + C_1 W^2 + C_1$ strip pieces. Record their ordered types, strip labels and fixed assignments, each with boundedly many choices. Intermediate positions and lengths are already summed by enumerating each piece from its current start; there is no factor depending on H at a join. By (7), the unsuccessful-prefix mass is bounded by

$$P_W \leq N(W) e^{C W^2} \sum_{j \geq 0} [K U(W) e^{-c' W}]^j, \quad N(W) = C U(W)^{C_1 W^2 + C_1}. \quad (8)$$

The factor $N(W)$ also absorbs the at most two end runs. Its possible size $\exp(O(W^{11/4}))$ is harmless: W is fixed. The series converges for sufficiently large W , since $W^{3/4} = o(W)$.

After choosing a , choose W once to satisfy all the preceding coefficient, fraction, strip and one-arc requirements, and also $\epsilon_W < 1/2$. Boxes with $H \leq C W$ have a finite uniformly bounded number of self-avoiding walks and can be absorbed in a constant depending on W . For larger boxes, sum the endpoint, row and lift data, apply the prefix bound and the one-arc estimate, and then add (5). This gives, with a fixed exponent K_0 ,

$$S(H) \leq A_W (1 + H)^{K_0} + \epsilon_W S(H).$$

The box has finitely many vertices, so $S(H) < \infty$. Absorbing the last term proves the proposition. $\square$

4 Localized bridges, recoverable cuts, and length moments

We first seek a positive chance of a long bridge that remains in a box of size comparable to its height. Fix its bottom source port. For a fixed confinement constant C , write

$$m_i(h) = \sum_{\substack{\gamma \text{ strict bridge of height } h \\ \text{diam } \gamma \leq C h}} \rho^{|\gamma|} |\gamma|^i.$$

Our first goal is to choose C so that

$$m_1(h) \geq h^{13/12 - o(1)}, \quad m_2(h) \leq h^{29/12 + o(1)}.$$

Together with $m_0(h) \leq B_h \asymp h^{-1/4}$, these estimates will give subpower probability of length at least $h^{4/3 - \xi}$ for every fixed $\xi > 0$. Section 5 will amplify that probability.

The two moments come from different constructions. To bound m_2 , we close a confined path outside its strip and use the polygon length-square estimate. To bound m_1 below, we begin with the bulk length supplied by the nesting comparison and attach exterior paths to make a bridge. Both constructions require localized path mass and a way to recover the sewing lines from their output. We develop those tools first, using the port and band conventions of Section 2. All sums below are unnormalized critical masses.

After establishing the length estimate, the final subsection proves a second geometric fact needed for the ordered adjacent paths in Section 6: a path can turn around an obstacle with a free terminal port while retaining mass at least $c H^{-1/4}$. That construction needs the exact power, without exponent slack, and uses a second moment of its representations.

4.1 Localization and recoverable seams

Lemma 4.1 (Convex cuts and lateral localization). *In a convex lattice domain, removing the part beyond a lattice cut compares the mass of old boundary exits lost with the mass of exits on the new*

cut, up to absolute positive constants. In particular, paths from a side which reach a parallel cut r bands away have boundary-exit mass at most CB_r . In a strip of height h , the arch or bridge mass from a fixed bottom port lost by requiring lateral extent at most M is at most

$$ChM^{-5/4}, \quad M \geq C_1h.$$

Proof. Take real parts in the boundary winding identity before and after making the cut. Contributions of paths contained in the near subdomain cancel. For every other boundary exit the cosine is between the same positive constant and one, since both domains are convex. Thus the lost old exit mass and the new-cut exit mass are comparable. For a parallel cut, its exits are a subset of the bridges across the intervening strip. This also explains the lower comparison used later: a family of paths in the old domain which necessarily passes the cut gives a lower bound, up to a constant, for the mass of exits on that cut. No bound on the continuation of an individual prefix is required.

For localization truncate the strip to a parallelogram whose width is a small fixed multiple of M , and average the winding comparison over bottom sources in its central half. There are $\asymp M$ such sources. Reverse paths from a side exit to one of these sources. For each side port they reach a cut parallel to that side at distance at least cM , so their total mass is at most $CB_{[cM]}$. There are $O(h)$ side ports. The average loss is therefore at most $ChM^{-1}B_{[cM]} \leq ChM^{-5/4}$. If M/h is sufficiently large, the parallelogram lies within lateral distance M of every central source. For each such source, exceeding lateral distance M requires leaving this parallelogram. The former mass is the same for all these sources by translation invariance, so it is bounded by the average exit loss. Infinite strips are obtained by increasing finite truncations. $\square$

Lemma 4.2 (Variation of the bridge kernel). *With consecutive top ports identified by their tangential coordinates,*

$$\sum_x |b_h(x+1) - b_h(x)| \leq C(B_h - B_{h+1}).$$

Proof. Put $\lambda = \pi/8$. Add the two adjacent rhombi shown in Figure 2 above ports $a = x$ and $b = x+1$. Each rhombus consists of two ordinary triangles. Their four new dual vertices form the chain u_0, v_0, u_1, v_1 , with a incident to u_0 and b incident to u_1 . Denote the left and right step exits by L, R . The horizontal top exits have winding zero, while L and R have windings $+\pi/3$ and $-\pi/3$, respectively: closing to the source along the exterior boundary determines these turns.

A path to L or R crosses the two-port interface an odd number of times, hence exactly once.

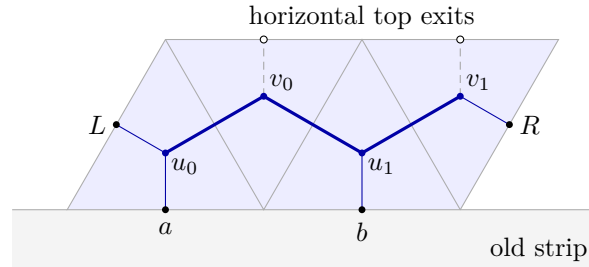


Figure 2: The exact two-rhombus addition for the kernel variation. The four blue vertices are triangle centers; a, b are the old top ports. A path to L or R enters this chain through exactly one of a, b . The dashed half-edges lead to horizontal exits, whose imaginary contributions vanish.

The four possible continuations have weights

	L	R
a	ρ	ρ^4
b	ρ^3	ρ^2

according to the number of new vertices they visit. Thus their combined contribution to the imaginary identity is

$$\sin \lambda (\rho b_h(x) + \rho^3 b_h(x+1)) - \sin(2\lambda) (\rho^4 b_h(x) + \rho^2 b_h(x+1)).$$

Using $\sin \lambda \rho = \sin(2\lambda) \rho^2$, this equals

$$\sin \lambda \rho (1 - \rho^2) (b_h(x) - b_h(x+1)).$$

There can be no other occupied segment in the raised part: another visit would require at least three interface crossings, while its horizontal top ports are vacant for a path ending at a step exit.

Subtract the old imaginary winding identity. The absolute value of the remaining bottom contribution is bounded by a constant times the mass of new arches using the bump. For distinct bump positions these arch families are disjoint: a member enters and leaves the last band through that specific adjacent pair, and uses no other port there. Their sum is bounded by $A_{h+1} - A_h$. The real strip identity $\cos(3\lambda)A_h + B_h = 1$, with $\lambda = \pi/8$, gives the assertion. To justify the infinite-strip identities, first fix a finite set of bump positions and perform each calculation in a wide finite parallelogram. Lemma 4.1 bounds every side-error mass by $C_h M^{-5/4}$, which tends to zero as the lateral cutoff M increases with h fixed. After this limit, increase the finite set of bump positions. The nonnegative, disjoint new-arch masses sum monotonically to at most $A_{h+1} - A_h$, whose finiteness is part of Theorem 2.2. $\square$

A *seam* is a lattice line with a specified crossing property that allows the pieces of a sewn path to be recovered. There is no tangency convention to choose: a visited internal port is traversed normally, and crossing a lattice line means traversing such a port. A line crossed once splits a bridge into two bridges. A line crossed twice, with both endpoints on its near side, cuts off one arch excursion. Excursions beyond ordered lines of the latter kind are nested.

Lemma 4.3 (Seam moment bounds). *Let $\mathcal{F}$ be a family of port paths, and let $N(\gamma)$ count specified seams in an interval of R parallel lattice levels. They are either once-crossed levels, or twice-crossed levels with both endpoints on their near side; in the latter case the excursions beyond the counted levels are nested. Suppose that, after cutting at any ordered list of such seams, the end pieces have bounded strip or convex boundary sums when summed successively from their cut ports. Allow a factor R_0^C for source and placement choices, where C is independent of the number of selected seams and $R \leq C_0 R_0$. Then, for every fixed positive integer j , the weighted count over $\mathcal{F}$ satisfies*

$$\sum_{\gamma} \rho^{|\gamma|} N(\gamma)^j \leq R_0^C C_j \begin{cases} R^{1+3(j-1)/4}, & \text{one crossing,} \\ R^{1+(j-1)/2}, & \text{one arch excursion.} \end{cases}$$

Consequently counts exceeding $R^{3/4} R_0^\varepsilon$, respectively $R^{1/2} R_0^\varepsilon$, have arbitrarily polynomially small weighted mass as $R_0 \rightarrow \infty$, including after multiplication by any fixed polynomial in R_0 .

Proof. Expand N^j and order the selected levels, allowing repetitions. Each internal gap g requires one bridge in the first case and two in the second. Removing mutual avoidance can only increase

the resulting upper bound. The gap factors are therefore at most $C(1+g)^{-1/4}$ and $C(1+g)^{-1/2}$, respectively. Summing the first level in R ways and each following gap gives the displayed bounds, with the ordering and repetitions absorbed in C_j . For a threshold Q , use

$$\sum \rho^{|\gamma|} N^a \mathbf{1}_{\{N>Q\}} \leq Q^{-(j-a)} \sum \rho^{|\gamma|} N^j$$

for a fixed a and then choose j sufficiently large. All paths in a box of diameter $O(R_0)$ have length $O(R_0^2)$, so length powers and raw geometric multiplicities can be included in the polynomial factor. $\square$

This is a statement about an unnormalized weighted measure. Each subsequent use specifies the end pieces; no probability estimate is inferred merely by calling a sewing level typical. For a nonnegative representation count F , we also use the elementary weighted Cauchy–Schwarz inequality

$$\sum_{F>0} \rho^{|\gamma|} \geq \frac{(\sum \rho^{|\gamma|} F)^2}{\sum \rho^{|\gamma|} F^2}.$$

We will also use the two-crossing argument for polygons of bounded diameter, modulo translation. Choose an extreme counted line and one crossing port, including these polynomially many choices in the placement factor. Cutting there makes two port arches. The outer arch has bounded strip mass, while further counted lines in the other arch produce the pairs of bridge gaps above. This reduces the polygon count to the same path calculation without introducing an unrestricted root sum.

4.2 Prescribed and free endpoints in narrow tubes

Lemma 4.4 (Fixed endpoints in thin tubes). *There is $v > 0$ such that, for every fixed $\delta > 0$, every attainable endpoint displacement $|x| \leq vh$ has bridge mass at least $h^{-5/4-o(1)}$ in a tube of transverse radius δh about its straight join. The estimate is uniform in the endpoint and large h .*

Proof. We first allow a sufficiently large fixed tube radius. Let H be even, choose $|k - H/4| \leq H/40$, and write $l = H/2 - k$. Lemma 4.2, summed over these heights, implies that a positive fraction of the choices have both kernel variations at most $CH^{-5/4}$. Their suprema have the same bound, since each summable kernel tends to zero at infinity. Reflection gives $b_k(x) = b_k(-x)$, and likewise for l .

For either good kernel f , localization leaves mass $\geq cH^{-1/4}$ on $O(H)$ endpoint positions. Thus $\sum f(x)^2 \geq cH^{-3/2}$. Moreover

$$\left| \sum_x f(x)f(x+t) - \sum_x f(x)^2 \right| \leq \|f\|_\infty \|f(\cdot+t) - f\|_1 \leq C|t|H^{-5/2}.$$

For $|t| \leq v_0 H$, with $v_0 > 0$ sufficiently small, the correlation is at least $c'H^{-3/2}$. Symmetry turns this into the same lower bound for the self-convolution. If both constituent bridges are confined to a lateral box of radius DH , the convolution loss is at most twice their lost mass times $\|f\|_\infty$, hence at most $CD^{-5/4}H^{-3/2}$. Fix D large enough.

Concatenate bridges of successive heights k, k, l, l . Convolving the two pair bounds over $\asymp H$ intermediate displacements gives mass $\geq cH^{-2}$ to each prescribed $|x| \leq v_0 H/2$, for every good k , inside a fixed large tube. Let F_* count admissible indices k for which all three levels $k, 2k, H/2 + k$ are crossed once. Then

$$\sum_{\gamma \text{ bridge of height } H} \rho^{|\gamma|} F_*(\gamma)^j \leq C_j H^{(j-1)/4}.$$

To verify this stronger version of the seam bound, order the chosen indices. Their levels lie in three disjoint macroscopic bands. A gap between successive indices creates three bridge gaps, with combined factor $(1+g)^{-3/4}$. The four outer and interband gaps all have height $\asymp H$, contributing H^{-1} . Thus the level sum is $H^{-1}HH^{(j-1)/4}$, as asserted, also when some indices coincide. The sum over good k of representations of a fixed endpoint has mass $\geq cH^{-1}$. By the high moment bound its part with $F_* > H^{1/4+\varepsilon}$ is negligible compared with H^{-1} . Dividing the remaining representation mass by this threshold gives $H^{-5/4-\varepsilon}$. An odd height is handled by appending a one-band bridge of fixed allowable displacement and positive weight; retain slack in the permitted slope.

We finish with the quantitative iteration. Suppose the endpoint estimate, with arbitrary exponent slack, holds at radius $D_j h$ and all slopes at most v_j . Set $D_{j+1} = 3D_j/4$. Choose

$$0 < \eta_j \leq \min \left\{ \frac{1}{16}, \frac{D_j}{8(1+D_j)}, \frac{v'_0 2^{-j}}{64} \right\}, \quad v_{j+1} = v_j - 8\eta_j,$$

where $v'_0 > 0$ is the slope range obtained in the preceding construction. Start with $v_0 = v'_0$ in this iteration. Since $\sum_{j \geq 0} 8\eta_j \leq v'_0/4$, all v_j remain at least $3v'_0/4$.

Take $|x| \leq v_{j+1}h$. Choose a joining level r with $|r - h/2| \leq \eta_j h$, and an attainable port $y \in r/2 + \mathbb{Z}$ with $|y - xr/h| \leq \eta_j h$. There are at least $c_j h^2$ such pairs for all large h . The two slopes differ from x/h by at most $3\eta_j$, so both lie in the previously allowed range. The difference between either piece's straight join and the whole join is at most $\eta_j h$. Its tube therefore lies within distance

$$D_j(1/2 + \eta_j)h + \eta_j h \leq (3D_j/4)h$$

of the whole join. The second displacement is compatible because $x - y \in (h - r)/2 + \mathbb{Z}$.

For any $\tau > 0$, the product of the two endpoint mass bounds is at least $c_{j,\tau} h^{-5/2-2\tau}$. After summing over the joining choices, representation mass is at least $c_{j,\tau} h^{-1/2-2\tau}$. A representation is recovered from its joining seam alone: that seam determines the joining port and both pieces. Its multiplicity is thus bounded by the count N of once-crossed joining levels. Lemma 4.3, with $a = 1$, removes the part where $N > h^{3/4+\tau}$ at a cost smaller than any prescribed negative power of h . Dividing the retained representation mass by this threshold gives

$$c'_{j,\tau} h^{-5/4-3\tau}.$$

As τ is arbitrary, this proves the endpoint estimate at radius D_{j+1} with arbitrary exponent slack. Each step uses finitely many previous constants; they may depend on j .

For fixed $\delta > 0$, a finite number of steps gives $D_j \leq \delta$. The uniform lower bound on all v_j permits a single choice, for example $v = v'_0/2$, for every δ . Choose the exponent slacks successively small enough for the desired final ε . This proves the assertion. $\square$

Lemma 4.5 (Free endpoints and widening clearance). *For each $\delta > 0$ there are $c, P > 0$ such that a height h bridge, with free terminal port, has mass at least*

$$ch^{-1/4}(s/h)^P$$

under the requirement that its lateral deviation at height y is at most $\delta(s+y)$, whenever $s \leq h$ exceeds a fixed threshold. Heights along an edge may be interpolated. In particular a fixed-proportion tube about the starting normal has mass at least $c_\delta h^{-1/4}$.

Proof. First construct bridges in a large fixed box with terminal displacement $|x| \leq v'h$, where $v' > 0$ can be any fixed small number. Use the four-piece construction above without restricting to

good heights. The localized kernels can be chosen symmetric and retain mass $\geq ch^{-1/4}$. Partition their $O(h)$ endpoint positions into intervals of width a small multiple of $v'h$. Pair an interval with its reflection. Cauchy–Schwarz on the finitely many interval masses shows that an equal-height pair with small total displacement has mass $\geq c(v')h^{-1/2}$. The two pairs therefore give ch^{-1} per k , and their representation mass summed over $\asymp h$ indices is bounded below by a positive constant. The previous estimate $\sum \rho^{|\gamma|} F_*^2 \leq Ch^{1/4}$ and weighted Cauchy–Schwarz give mass $\geq ch^{-1/4}$. Odd heights again cost only a fixed factor.

For a tube of radius δh , divide the height into a sufficiently large fixed number m of pieces. Vary the $m - 1$ joining levels in disjoint small macroscopic intervals near equal division. Apply the large-box construction to each piece, taking its permitted terminal slope sufficiently small that cumulative endpoint shifts remain below $\delta h/2$, and then m sufficiently large that each piece's box fits the remaining clearance. The representation mass is at least

$$c_m h^{-1/4} h^{3(m-1)/4}.$$

The representation count is bounded by the product of the seam counts in the $m - 1$ intervals. Its second moment is at most

$$C_m h^{-1/4} h^{3(m-1)/2}.$$

Indeed in each interval the two selected levels introduce one bridge gap, whose ordered sum is $O(h^{7/4})$, while the m other gaps are macroscopic and contribute $h^{-m/4}$. Cauchy–Schwarz proves the fixed-proportion tube assertion.

For the widening tube use seam intervals $[r_i, 5r_i/4]$, where $r_i = 4^i s$, stopping before $h/2$. The proportional width for each piece is chosen small enough that the geometric sum of all previous endpoint shifts and current fluctuations is at most $\delta(s + y)$. If there are no intervals, one fixed-proportion tube suffices. With k intervals the representation mass and second moment are bounded respectively below and above by

$$c^{k+1} h^{-1/4} \prod_i r_i^{3/4}, \quad C^{k+1} h^{-1/4} \prod_i r_i^{3/2}.$$

The moment estimate follows by selecting two seams in each interval: its gap factor sums to $O(r_i^{7/4})$, and the outer gaps contribute $h^{-1/4} \prod_i r_i^{-1/4}$. Constants are uniform in the number of intervals because the scales are separated by four. Thus Cauchy–Schwarz leaves $(c^2/C)^{k+1} h^{-1/4}$. Since $k = O(1 + \log(h/s))$, this is the asserted bound for some fixed finite P . $\square$

Figure 3 contrasts the prescribed and free terminal constraints in the two tube estimates.

4.3 Closing paths and controlling their second length moment

A bridge crossing mass is not a second length moment. To obtain the latter, we close each path by an exterior path of controlled positive mass and compare with the polygon second moment. The closing line must remain recoverable: its multiplicity is the source of the extra exponents below.

Lemma 4.6 (Localized arch kernel). *Two distinct ports on a lattice side line, at distance r , can be joined in either prescribed half-plane by arches of diameter at most Cr , of total mass at least $r^{-5/4-o(1)}$.*

Proof. There are constants $0 < a < b < \infty$ such that height h arches, localized to a fixed box of size h , have rightward displacement in $[ah, bh]$ with mass at least $ch^{-1/4}$. To see this, start from

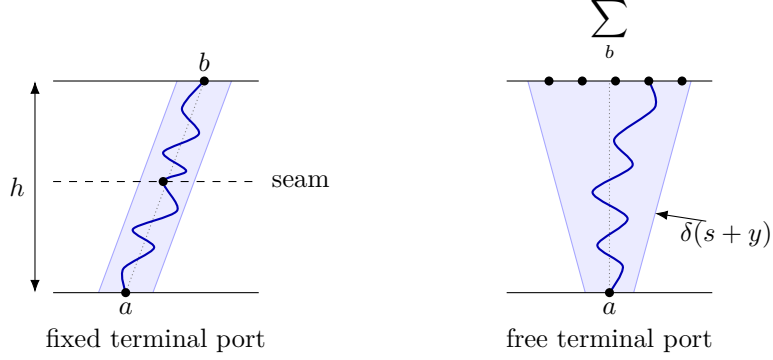


Figure 3: Schematic tube constraints. Left: a fixed endpoint contribution; the once-crossed seam recovers both pieces and their common port. Right: a widening tube with terminal ports summed over. Shading denotes spatial confinement; both lemmas bound unnormalized path mass.

$\sum_{s>0} sK_h(s) \asymp h^{3/4}$. Localization and dyadic summation bound the first moment at displacements above bh by $Ch(bh)^{-1/4}$. For $1 \leq q \leq h$, arches of displacement at least q have total mass $O(q^{-1/4})$: those leaving the height- $[q]$ strip cost $O(B_{[q]})$ by the strip winding identity, and those remaining have this bound by the first displacement moment in that smaller strip. Integrating this tail up to ah bounds its first moment by $O((ah)^{3/4})$. Choose a small and b large. Finally, removal of paths leaving a sufficiently wide box changes the first moment on $[ah, bh]$ by an arbitrarily small multiple of $h^{3/4}$, by Lemma 4.1. Division by bh proves the claim.

From the prescribed ports run two bridges into the chosen half-plane to a line at height $k \in [Dr, 2Dr]$, with D a large fixed constant. Join their terminal ports by one of the just-constructed arches beyond that line, of height r and displacement in $[ar, br]$. There are $\asymp r$ translations of this arch in a small macroscopic interval near the starting ports. For each translation and arch endpoint separation, use Lemma 4.4 for both bridges. Choose D large enough for their slopes and their tube widths small enough that they are disjoint. This is possible because the two straight segments remain in the same order and their separation is at least $\min(1, a)r$. Translation along the terminal line preserves the port parities. For each k the product mass is therefore at least $r r^{-5/2-o(1)} r^{-1/4} = r^{-7/4-o(1)}$.

Summing k gives representation mass $r^{-3/4-o(1)}$. A representation is recovered at its two-crossing seam. The end pieces are two bridges from the prescribed ports, the internal gaps require two bridges each, and the last piece is an arch; all their endpoint sums are bounded. Lemma 4.3 discards multiplicity above $r^{1/2+\varepsilon}$, proving the assertion. The construction has diameter $O(r)$. The finitely many bounded nonzero distances have positive weight by fixed ordinary arches and are absorbed in the constants. $\square$

Proposition 4.7 (Localized strip second moment). *For each fixed C_0 , and a fixed starting port,*

$$\sum_{\substack{\gamma \text{ arch or bridge in height } h \\ \text{diam } \gamma \leq C_0 h}} \rho^{|\gamma|} |\gamma|^2 \leq h^{29/12+o(1)}. \quad (9)$$

Proof. Figure 4 shows the two closures and their recovery lines. An arch needs one movable line; a bridge needs two lines that move together at fixed separation. This coupling improves the bridge's inverse multiplicity and compensates for its more costly closure.

Close an arch by an exterior arch supplied by Lemma 4.6, at mass at least $h^{-5/4-o(1)}$. The result is a simple polygon of diameter $O(h)$, and its original boundary is a two-crossing seam.

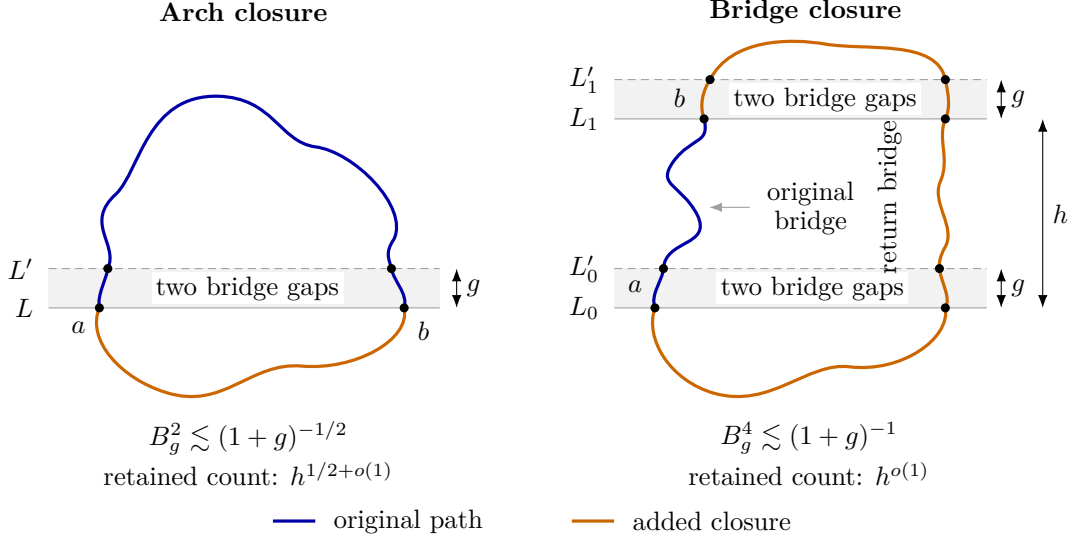


Figure 4: Closure and recovery in Proposition 4.7. Modulo translation, solid lines recover the original path up to bounded choices; dashed lines illustrate a second admissible placement. Left: moving the clean line creates two bridge gaps. Right: move the lower and upper lines together, keeping their separation h . Within a lower-line interval of width at most $h/3$, the two comparison bands are disjoint and create four bridge gaps. Their gap factors sum as a power and a logarithm, respectively, giving the displayed counts after discarding the weighted high-multiplicity exceptions. Curves are schematic; no intermediate line is required to be clean.

Modulo translation, fixing this line recovers the two arcs and the input start up to bounded choices. Its placement count has moments at most $C_j h^{C+j/2}$, with C independent of j : choose an initial line and crossing port in polynomially many ways, order the other levels, use two bridges per internal gap and bounded terminal arch sums. Thus multiplicities above $h^{1/2+\varepsilon}$ have negligible mass, even with squared length and raw multiplicity included. The polygon second moment gives $h^{5/4+1/2+2/3+o(1)} = h^{29/12+o(1)}$.

For a bridge, attach exterior arches at its bottom and top, ending far to one common side of the original path. Each new endpoint ranges over ch positions at distance $\asymp h$, with their intervals chosen so that their mutual displacement satisfies the slope condition of Lemma 4.4. Join them by a thin bridge inside the original strip, separated from the original path. The three added kernels and two endpoint sums have total mass at least $h^2 h^{-15/4-o(1)} = h^{-7/4-o(1)}$.

Now there are two clean two-crossing lines at a fixed separation h . Their joint placement count on the polygon has moments at most $C_j h^C (1 + \log h)^j$. Indeed restrict the lower line to an interval of width at most $h/3$; only boundedly many intervals are needed after anchoring a polygon of diameter $O(h)$. Each successive gap g then forces two bridges in the lower band and two in the upper, giving $(1+g)^{-1}$; its sum is $O(1 + \log h)$. The excursions are nested and all remaining pieces have bounded strip sums, with only polynomial placement costs. For fixed lines the two bridges and exterior arches are recovered up to bounded orientation choices. High moments therefore discard multiplicity above h^ε . The polygon estimate now gives $h^{7/4+2/3+o(1)} = h^{29/12+o(1)}$. In both cases the input length is at most the polygon length. Exceptional representation mass is harmless because every path stays in an $O(h)$ box and hence has length $O(h^2)$. $\square$

4.4 Transferring bulk length to bridges

Let D be a lattice equilateral triangle of side H , and let $\mathcal{C}_H$ be its boundary-to-boundary chords which visit a fixed central subtriangle at positive proportional distance from ∂D . Ordered endpoints or boundedly many port types change only constants. Lemma 4.1, applied at each of the $O(H)$ boundary sources, gives

$$\sum_{\gamma \in \mathcal{C}_H} \rho^{|\gamma|} \leq CH^{3/4}.$$

For each of $\asymp H^2$ internal ports in a smaller central triangle, the bulk identity bounds $L_p(D)$ below by a constant times $P_2(c_0H) = H^{1/12+o(1)}$, using nests about its neighboring face centers. Summing over these ports and dropping the cosine yields

$$\sum_{\gamma \in \mathcal{C}_H} \rho^{|\gamma|} |\gamma| \geq H^{25/12-o(1)}.$$

Lemma 4.8 (Separation of useful chord endpoints). *For every fixed $\eta > 0$, the preceding first-length lower bound persists after retaining only chords with the following property: endpoints on the same side have separation at least $H^{1-\eta}$; endpoints on distinct sides have maximum distance from their common corner at least $H^{1-\eta}$.*

Proof. Group the excluded chords by dyadic endpoint scale s , taking $s \asymp 1$ for bounded separations. For a common side, fix a source and close outside D with an arch of mass $s^{-5/4-o(1)}$. The two-crossing seam argument in Proposition 4.7, now using diameter $O(H)$, gives the squared-length bound, summed over sources,

$$\sum_{\substack{\gamma \text{ with same-side endpoints} \\ \text{at scale } s}} \rho^{|\gamma|} |\gamma|^2 \leq H^{1+2/3+1/2+o(1)} s^{5/4}.$$

The factor H counts sources; the factor $H^{1/2+o(1)}$ bounds seam multiplicity. All small exponent losses can be assigned to H . For distinct sides we claim

$$\sum_{\substack{\gamma \text{ with distinct-side endpoints} \\ \text{at corner scale } s}} \rho^{|\gamma|} |\gamma|^2 \leq H^{2/3+o(1)} s^{11/4}. \quad (10)$$

Here the sum includes every source and endpoint pair at that scale.

By symmetry put the sides at

$$A : y = 0, \quad B : x = y/\sqrt{3},$$

with D above A and to the right of B . Write the endpoints as $a \in A, b \in B$, choosing b farther from the shared corner. Then $y_b \asymp s$. Put $w = y/2 - d_0x$; the half-plane exterior to B is $w > 0$. We first construct a closing arc outside D with mass $s^{-3/2-o(1)}$ and diameter $O(s)$. Join a , below A , to a port $(X, 0)$ with $X \in [-8s, -7s]$. The arch kernel costs $s^{-5/4-o(1)}$. We next construct a join from $(X, 0)$ to b , within $y > 0, w > 0$, at cost $s^{-5/4-o(1)}$. Summing over the $\asymp s$ choices of X then gives the required closing mass.

For this two-face join choose small fixed $\alpha > 0$. From b first reach a horizontal line $y = t$, where $t - y_b \in [\alpha s, 2\alpha s]$. The comparison domain lies between B and a parallel line at w -depth approximately $10\alpha s$, with the additional inequalities

$$y \geq 0, \quad x \leq x_b + (\alpha/4)s - (y - y_b)/\sqrt{3}.$$

A thin bridge tube in the inward normal direction $(-d_0, 1/2)$ from B to the far parallel line fits these inequalities and passes beyond $y = t$. Its mass is $\geq cs^{-1/4}$ by Lemma 4.5. Convex cut comparison gives the same lower bound for exits on $y = t$. Every such exit z satisfies $w(z) \geq cas$, by the oblique upper bound on x . The lower bound on x , from $y \geq 0, w = O(as)$, keeps all these paths well to the right of X .

Continue from z vertically to $y = L$, where $L \in [D_2s, 2D_2s]$ and D_2 is a large fixed constant. A tube of width a sufficiently small multiple of s stays clear of B and of the future arm from X ; its mass is $\geq c_{D_2}s^{-1/4}$. For each t the product mass is therefore $\geq c_{D_2}s^{-1/2}$. Sum over $\asymp s$ choices of t . Each is recovered as a once-crossed seam. The initial segment has bounded convex-boundary mass in the two-face wedge with distant walls, the final segment has bounded strip mass, and internal gaps are bridges. Lemma 4.3 discards counts above $s^{3/4+\varepsilon}$. Thus the distinct b -arms to L have mass at least $s^{-1/4-o(1)}$.

Independently take a vertical arm from $(X, 0)$ to L in a disjoint small-width tube, with mass $\geq cs^{-1/4}$. Connect the two new ports by an arch beyond L of mass $s^{-5/4-o(1)}$. Their lateral separation is $O(s)$ independently of D_2 ; hence the arch diameter constant is independent of D_2 , and increasing D_2 makes this arch lie wholly in $w > 0$. Vary L over its interval. These are exactly two-crossing seams; the two initial arms have bounded convex boundary sums, the intermediate gaps give pairs of bridges, and the terminal arch has bounded mass. Removing multiplicity above $s^{1/2+\varepsilon}$ gives

$$s s^{-1/4-o(1)} s^{-1/4} s^{-5/4-o(1)} / s^{1/2+\varepsilon} = s^{-5/4-o(1)}$$

for the two-face join. All walls and cuts are lattice lines rounded with proportional slack; distant walls can be added to make every comparison domain finite. For bounded scales, first move b a fixed number of bands into $w > 0$ by elementary bridges with displacement $(-1/2, d_0)$, which remain in $y > 0$. Perform the construction from that translated parallel boundary. Keep the auxiliary interval on $y = 0$, shifting it tangentially relative to the new boundary's intersection with $y = 0$. This costs a positive constant and preserves the required original-side crossings.

We now count inverse images of the resulting polygon. It crosses A exactly twice, and its arc above A crosses B exactly once; other crossings of B do not matter. Given both lines, these properties recover the original chord and closing arc up to bounded choices. Each line position is within $O(s)$ levels of the corresponding directional extremum of the polygon. For A , the two-crossing moment bound gives at most $s^{1/2}H^\varepsilon$ choices, outside negligible weighted exceptions. For each A , apply the one-crossing moment bound to its positive-side arc to obtain at most $s^{3/4}H^\varepsilon$ choices of B . To justify the end factors in this second application, retain the A half-plane and the relevant extreme B half-plane for each end piece, with distant walls. Convex winding bounds its boundary sum; the opposite A -arc has bounded arch mass. Initial roots, outermost levels, and the union over A cost only a fixed polynomial in H , independent of the moment order. Thus the total inverse multiplicity is at most $s^{5/4}H^{2\varepsilon}$. Two nonparallel lines determine the placement of the fixed triangle. Squared lengths and all discarded raw multiplicities are polynomial in H , so their exceptional contribution is negligible. Dividing the polygon second moment by the added mass $s^{-3/2-o(1)}$ proves (10).

Finally apply Cauchy–Schwarz with the central-visit mass $CH^{3/4}$. The same-side contribution at scale s is at most $H^{35/24+o(1)}s^{5/8}$; the distinct-side contribution is at most $H^{17/24+o(1)}s^{11/8}$. At $s \leq CH^{1-\eta}$ these are respectively at most $H^{25/12-5\eta/8+o(1)}$ and $H^{25/12-11\eta/8+o(1)}$. Summing the $O(\log H)$ scales is power-smaller than the original first-length lower bound. $\square$

Lemma 4.9 (Exterior connectors to an outer strip). *Embed a triangle of side $H = \lfloor c_0M \rfloor$, for small fixed $c_0 > 0$, in a strip of height M . For each of the endpoint types in Lemma 4.8, choose one of the finitely many lattice strip orientations. The strip can be placed in $\asymp M$ ways with both*

boundaries at distance at least cM from D . At endpoint scale s , two disjoint exterior connectors to free ports on these boundaries have product mass at least

$$H^{-1/2-o(1)}(s/H)^P, \quad (11)$$

for a finite fixed P . They and D stay in a box of diameter CM , with C, P independent of η .

Proof. For distinct sides use the coordinates A, B, a, b, w from the preceding proof and a horizontal outer strip. Connect a to the lower boundary by a localized bridge below A . For b , take bridges from B in direction $(-d_0, 1/2)$ to a sufficiently distant parallel line, so that their straight tubes pass beyond the upper target boundary. Because $y_b \asymp s$, the widening tubes of Lemma 4.5 with initial clearance s fit inside $y > 0, w > 0$, on reducing their fixed proportional width. Include these half-plane constraints and distant bounding walls in a convex comparison domain. Cut at the upper target line. Lemma 4.1 gives mass $\geq cH^{-1/4}(s/H)^P$ for the b -arm; the a -arm has mass $\geq cH^{-1/4}$. The two arms lie on opposite sides of A , and avoid D . On the resulting bridge, A is crossed once and B is crossed once on the tail following A .

For a common horizontal side A , write $x_a < x_b$ with separation $\asymp s$. Orient the outer strip in crossing direction $p = (d_0, -1/2)$. Separate the arms by a lattice wall parallel to

$$x = (x_a + x_b)/2 + y/\sqrt{3}.$$

The separator is a p -level, since $p \cdot (x, y) = d_0x - y/2$ is constant on it. Both arms lie below A , the arm from b to the upper p -boundary on the right of the wall. Construct this arm by taking downward bridges in direction $n = (0, -1)$ in a longer convex domain and cutting at the target p -line. Since $n \cdot p = 1/2$, they reach that target. Their widening tubes, with initial scale s , remain right of the separator and inside the opposite outer strip bound. They give mass $\geq cH^{-1/4}(s/H)^P$.

The arm from a must turn toward the lower outer boundary. Put $q = (-d_0, -1/2)$. First reach a q -normal line at depth in $[\alpha s, 2\alpha s]$ from a , with small fixed $\alpha > 0$. Writing $t = -y$, use the convex domain with

$$0 \leq t \leq 10\alpha s, \quad x \geq x_a - (\alpha/4)s - t/\sqrt{3},$$

also restricted to the left of the separator and to the outer strip where needed. A thin downward bridge tube fits and passes beyond the chosen q -line, so cutting gives mass $\geq cs^{-1/4}$. Its exits z satisfy $t(z) \geq c\alpha s$ and $|x(z) - x_a| = O(\alpha s)$, using

$$q \cdot (z - a) = -d_0(x(z) - x_a) + t(z)/2.$$

From z , continue in positive q direction in the half-plane beyond this switching line. A widening tube of initial scale s remains below A , left of the separator, and clear of the opposite outer boundary: its depth below A and clearance from the separator both grow linearly along q . Since $q \cdot (-p) = 1/2$, long enough bridges in this direction cross the lower target line. Convex truncation there gives mass $\geq cH^{-1/4}(s/H)^P$.

Sum the switching line over $\asymp s$ choices. It is recovered as a once-crossed seam; the extreme end pieces have bounded convex-boundary mass after retaining the original and target boundaries and the extreme seam half-planes. The seam moment bound, allowing a polynomial factor in H , removes counts above $s^{3/4}H^\varepsilon$. The line count s cancels the short-arm factor $s^{-1/4}$ and this seam factor. The resulting left arm has mass $H^{-1/4-o(1)}(s/H)^P$, proving (11) after combining with the right arm and, if necessary, increasing P . The entire output crosses A exactly twice, with its endpoints on its exterior side.

The widening-tube construction above requires s to exceed a fixed threshold. To include bounded s , choose a fixed integer K above that threshold and first add the following elementary stems. On a

common side, take K downward one-band bridges from a , each with displacement $(-1/2, -d_0)$, and from b , each with displacement $(1/2, -d_0)$. The stems are disjoint, remain respectively in $x \leq x_a$ and $x \geq x_b$, and end on $A' = \{y = -Kd_0\}$ with separation $(x_b - x_a) + K \asymp s + K$. Apply the preceding common-side construction below A' . The new arms avoid the stems; the completed output still crosses the original A exactly twice. For distinct sides, extend b by K elementary B -normal bridges of displacement $(-1/2, d_0)$. Their endpoint b' lies on $B' = \{w = Kd_0\}$ with $y_{b'} \geq Kd_0$. Run the b -arm construction in $y > 0, w > Kd_0$. The new arm avoids the stem, the triangle, and the downward a -arm; the completed b -arm retains the required single crossing of the original B . The enlarged scales are fixed positive constants for bounded s , so their powers of s/H are comparable up to constants. The stems have fixed positive weights, and change all outer-strip clearances by $O(K)$ only.

All comparison domains here are intersections of the specified lattice half-planes, a starting and a far terminal line normal to the comparison bridge, and distant lattice walls. The fitted tube stays inside with proportional slack away from its endpoints. This verifies that every cut comparison is between convex domains and avoids assuming uniform continuation weights for individual prefixes. The allowed normal strip placements form an interval of $\asymp M$ choices; all distances and tube diameters are fixed multiples of M . $\square$

Proposition 4.10 (A localized bridge length lower bound). *For a sufficiently large fixed C , the height M bridges from a fixed source and of diameter at most CM have first-length mass $M^{13/12-o(1)}$ in the lower-exponent sense. Consequently, for every fixed $\xi > 0$, the normalized critical bridge law satisfies*

$$\mathbb{P}_M^{\text{br}}\{|\gamma| \geq M^{4/3-\xi}\} \geq M^{-o(1)}. \quad (12)$$

Proof. Fix $0 < \eta < 1$. Retain the chords from Lemma 4.8 and attach the two connectors for each outer strip placement. Their endpoint scale obeys $s \geq H^{1-\eta}$, so the added mass is at least $H^{-1/2-P\eta-o(1)}$. Translate every output bridge to a fixed bottom source. We must count how many inputs yield the same translated output, including the original placement of D .

For distinct sides, once-crossed A -levels have at most $H^{3/4+\varepsilon}$ choices outside negligible mass. For each such A , apply the same bound to candidate B -levels on the specified tail above A , obtaining another $H^{3/4+\varepsilon}$. The initial piece up to A has a bounded bridge sum. The end pieces for the B -counts have bounded convex-boundary mass with the outer strip, A half-plane, and relevant extreme B half-plane retained. The union over the first level or over A costs only a polynomial in H before taking high moments. These two nonparallel lines determine D and the junctions.

For a common side, the permissible A -levels are crossed twice, with both output endpoints on the exterior side. Their nested excursions give at most $H^{1/2+\varepsilon}$ choices. The end arms lie in the intersection of the outer strip and the exterior half-plane of the first level, and the last piece is an arch, so their sequential boundary sums are bounded. Once the seam and its two crossing ports are fixed, at most $O(H)$ tangential placements of D remain. Thus both endpoint types have total inverse multiplicity $H^{3/2+o(1)}$. This also recovers the outer strip placement by translating back relative to D . The auxiliary construction parameters within a connector were already removed in Lemma 4.9; they are not counted again. Exceptions remain negligible with length and raw placement factors, since all output paths lie in an $O(M)$ box.

The input first-length mass, the $\asymp M$ outer placements, the connector mass, and the inverse multiplicity therefore give

$$H^{25/12-o(1)} cM H^{-1/2-P\eta-o(1)} / H^{3/2+o(1)} = M^{13/12-P\eta-o(1)}.$$

There are only boundedly many orientations and endpoint orders; apply a lattice symmetry to the

selected class. The width constant is independent of η . Since $\eta > 0$ is arbitrary, this proves the asserted lower exponent.

For the localized moments defined at the start of the section, we now have $m_1 \geq M^{13/12-o(1)}$, $m_2 \leq M^{29/12+o(1)}$, and $m_0 \leq B_M \asymp M^{-1/4}$. For $a = M^{4/3-\xi}$, the first-length mass below a is at most $aB_M = O(M^{13/12-\xi})$, hence negligible. Weighted Cauchy–Schwarz gives mass at least $m_1^2/(4m_2) \geq M^{-1/4-o(1)}$ above the threshold, for large M . Division by B_M proves (12). This step alone does not give a high-probability lower bound. $\square$

4.5 Turning corridors with a free terminal port

The bridge-length input is now established. The ordered adjacent-path construction in Section 6 also needs paths that pass an obstacle or turn back to their starting boundary. We prove that these paths retain free-end mass at least $cH^{-1/4}$, with no exponent loss. To do so, we estimate a second moment of representations instead of discarding large multiplicities at a power threshold.

In this subsection the normal vectors and the coordinates x, y are Euclidean. The starting and terminal boundaries in the next statement are horizontal lattice lines, and its endpoints are boundary ports. Distances in band units differ by the fixed factor d_0 ; this only changes the constants below.

Lemma 4.11 (Sewing a turning corridor). *Fix $D < \infty$ and $\delta > 0$. There is $H_0 = H_0(D, \delta)$ such that the following holds for $H \geq H_0$. Let $a = (x, u)$ be a horizontal boundary port and let $0 \leq v - u \leq H$. Suppose a set of forbidden vertices and edges in the slab $u \leq y \leq v$ is contained in $\{x' \leq x + DH\}$ and its part in $u \leq y \leq \min(v, u + \delta H)$ is contained in $\{x' \leq x - \delta H\}$. The mass of strict bridges from a to a free port on $y = v$, avoiding this set, is at least*

$$c_{D,\delta} H^{-1/4}.$$

The bridges can be confined to a box of diameter $C_{D,\delta}H$. The empty bridge is used when $v = u$. The reflected assertion also holds.

Proof. We first construct paths along a finite chain of normal tubes and turning regions, keeping their terminal port free. Consecutive directions will have positive scalar product, and nonadjacent stages will have fixed positive clearance. After proving the mass bound for this construction, we fit its chain around the forbidden set.

We begin with a turn and its two recovery lines. Write n_1, n_2 for the incoming and outgoing unit normals. It suffices to use turns by 60° , so $n_1 \cdot n_2 = 1/2$; a longer turn is a sequence of these. Let C be the corner and put $F = C - \ell n_1$, where ℓ is a small fixed multiple of H . The line through F is normal to n_1 , and the line through C is normal to n_2 . Each line will vary in its own band of width $\varepsilon\ell$, with $\varepsilon > 0$ fixed and small. The incoming long tube ends at the F -line. A short piece turns from that line to the C -line, and the next long tube begins there.

Here is an explicit region for the short piece. Before rounding its sides to lattice lines, take the intersection of the half-planes between the F -line and the parallel far line $n_1 \cdot (p - C) = \ell/100$, and impose

$$-(.5 + .02)\ell < (n_1 - n_2) \cdot (p - C) < .02\ell. \quad (13)$$

For a source within $\varepsilon\ell$ of F , this convex region contains a thin normal tube extending past the chosen C -line. The straight center starts with the side-wall coordinate near $-\ell/2$ and finishes near $\ell/200$, leaving positive margins in (13). The free-tube lemma therefore supplies $c\ell^{-1/4}$ mass reaching the far wall. Cutting at the chosen C -line and using the convex cut comparison supplies that much

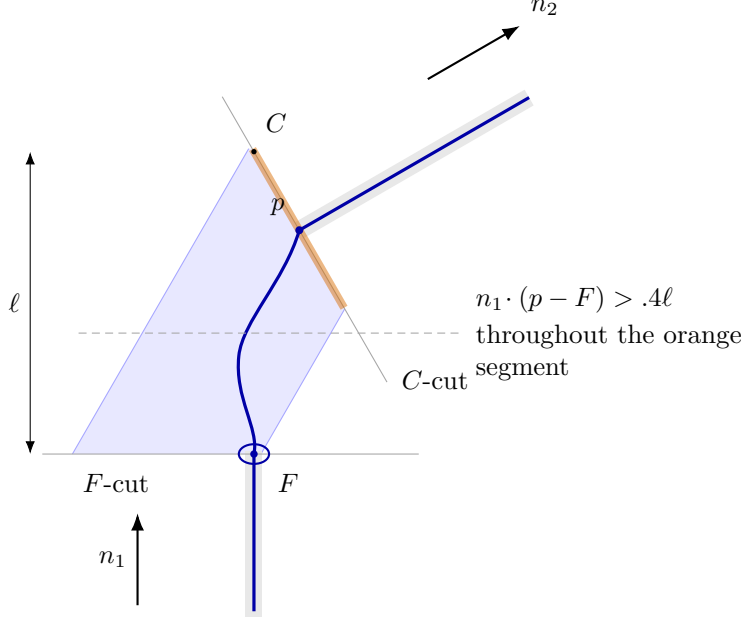


Figure 5: Two recovery cuts at a 60° turn, with $F = C - \ell n_1$ and $n_1 \cdot n_2 = 1/2$. The shaded short region is bounded by the side walls and the C -cut. Its outgoing ports (orange) lie more than $.4\ell$ above F in the n_1 direction; the next tube remains above $.3\ell$. Thus the two local port regions stay separated as the cut lines vary. Tubes and path are schematic.

mass ending on the cut. It is a lower bound on the whole exit sum, rather than a lower bound on the continuation of any individual prefix.

At an exit p on the unshifted C -line, $n_2 \cdot (p - C) = 0$, so (13) gives $n_1 \cdot (p - F) > .48\ell$. Small source errors, cut shifts and rounding leave the uniform bound

$$n_1 \cdot (p - F) > .4\ell. \quad (14)$$

Along the following normal direction n_2 the left-hand side increases. Choosing the next tube narrow enough makes it greater than $.3\ell$ throughout that tube. The preceding tube, in contrast, ends near F . Thus the two cuts have disjoint port regions and the outgoing long tube cannot revisit the earlier one. Figure 5 shows the two separated port regions.

Choose the finitely many corner lengths successively, with each much larger than the accumulated endpoint errors from the preceding stages. Choose the first length so small that the last is still a small fraction of every original segment. Then choose all cut ranges and tube widths smaller still. These are choices of fixed constants: every allowed cut band contains at least cH lattice levels when H is large. The endpoint of one stage may move over its actual lattice ports; the next stage is started at that actual endpoint. Equation (14) and the positive scalar products preserve all clearances uniformly over these choices.

Each long stage has mass at least $cH^{-1/4}$ by Lemma 4.5; the preceding convex construction gives that bound for each short corner stage. Adjacent stages occupy opposite strict half-planes of their joining line. Nonadjacent stage regions are disjoint, after making the widths sufficiently small. Consequently their concatenation is self-avoiding and its weights multiply. If there are m stages and $F(\gamma)$ counts their representations of an output path, summing the $m - 1$ joining levels and all

the intermediate ports gives

$$\sum_{\gamma} \rho^{|\gamma|} F(\gamma) \geq cH^{-m/4} H^{m-1}. \quad (15)$$

No separate endpoint-position factor occurs: it is already included in each stage's free-terminal mass.

We next give the second-moment argument, including the upper bounds for the pieces that remain after two representations are cut. Each joining port region is visited only by its two adjacent stages. At the first cut of a corner this region is close to F ; at the second it is within $O(\ell)$ of C and has the positive separation (14) from F . Hence, in any two representations, both splits at one joint precede both splits at the next joint. Otherwise the intervening portion would belong to nonadjacent stage regions, which are disjoint. At a single joint, the portion between its two split times stays strictly between the two parallel joining lines, by the opposite half-plane restrictions in the two representations. It is therefore a bridge. Equal levels give the same split and port; the corresponding bridge has weight $B_0 = 1$.

Delete these $m - 1$ short bridge gaps. Exactly m portions remain: one before the first pair of splits, one between every consecutive pair, and one after the last. Each starts on its incoming supporting line and stays strictly on its forward side. It ends on its outgoing supporting line at normal distance at least κH into that forward half-plane, where $\kappa > 0$ depends only on the fixed geometry. For a long stage this is the segment length less the two narrow cut bands. For a corner it is (14), less the cut shifts. These conclusions hold for a mixed pair of representations: retaining the later incoming split and the earlier outgoing split only moves the relevant lines within their already allowed bands.

For completeness, this depth gives an upper mass $CB_{[\kappa H]}$ for each remaining portion. Enclose it in the convex intersection of its endpoint half-planes and distant lattice walls. Its source is on the incoming wall. Cut this domain by a line parallel to that wall at depth $\kappa H/2$. Every allowed terminal port is lost in the cut. Lemma 4.1 bounds their total mass by the new-cut mass, which is at most $CB_{[\kappa H/2]} \leq CH^{-1/4}$. This estimate is uniform in the actual starting port. We can thus sum the portions in their temporal order, dropping mutual avoidance for an upper bound, without paying for another translation or for a new root.

For each joint the sum over its two levels and their bridge gap is at most

$$CH \sum_{g=0}^{CH} B_g \leq CH^{7/4}.$$

The m macroscopic portions therefore give

$$\sum_{\gamma} \rho^{|\gamma|} F(\gamma)^2 \leq CH^{-m/4} H^{7(m-1)/4}. \quad (16)$$

The finitely many choices of the order of two levels are included in C . Weighted Cauchy–Schwarz, applied to (15) and (16), yields

$$\sum_{F(\gamma) > 0} \rho^{|\gamma|} \geq cH^{-m/4+2(m-1)-7(m-1)/4} = cH^{-1/4}.$$

This proves the asserted finite-chain estimate with no exponent loss.

We now fit such a chain to the obstacle. We may decrease δ below one and increase D above one. If $v - u \leq \delta H/2$, a sufficiently narrow vertical free tube stays to the right of $x - \delta H$ and has mass at least $c(1 + v - u)^{-1/4} \geq c'H^{-1/4}$; zero and bounded gaps cost only fixed elementary paths.

Otherwise rise a small fixed fraction of δH , then take one right-up segment and a fixed number of right-down/right-up pairs, with normals $(\sqrt{3}/2, -1/2)$ and $(\sqrt{3}/2, 1/2)$. Choose equal segment lengths in each pair, much smaller than δH , and use sufficiently many pairs that the horizontal coordinate exceeds $x + (D + 1)H$. The last oblique direction is right-up, so it has scalar product $1/2$ with the final upward direction. Keep this part in the first $\delta H/2$ bands, and finish by rising vertically to $y = v$. There is fixed proportional clearance from the obstacle and from both horizontal walls except at the required endpoints. Adjacent normals have scalar product $1/2$; the stages are simple and separated at all nonadjacent segments. The two-cut construction just proved fits inside this clearance. All lengths and the enclosing box are fixed multiples of H , depending only on D, δ . This proves the lemma. $\square$

Corollary 4.12 (A long arch in a shallow strip). *For every fixed $\sigma > 0$ and all sufficiently large fixed D' , there are $c, C > 0$ and h_0 with the following property for $h \geq h_0$. From any horizontal boundary port $a = (x_A, u)$, there is arch mass between $ch^{-1/4}$ and $Ch^{-1/4}$ which stays in*

$$u < y < u + \sigma h, \quad x > x_A - \sigma h,$$

except at its two boundary endpoints, and has free terminal port on $y = u$ with abscissa exceeding $x_A + D'h$. The arches can be confined to a box of diameter Ch .

Proof. Use the same normal-direction chain as in Lemma 4.11: rise first, make equal right-up and right-down pairs inside a small fraction of the σh -high slab, and end with a downward segment beyond $x_A + (D' + 1)h$. Choose the last oblique direction right-down, so its scalar product with the final downward direction is $1/2$. All intermediate regions lie strictly above the starting line, and the final downward tube ends on that line. They remain right of $x_A - \sigma h$ with fixed proportional clearance. The finite-chain estimate gives the lower bound. The terminal displacement forces lateral extent at least $D'h$. For $D' \geq C_1\sigma$, Lemma 4.1 bounds the total such arch mass in the height- $[\sigma h]$ strip by $C\sigma h(D'h)^{-5/4} \leq C'h^{-1/4}$. Lattice rounding changes only fixed constants. $\square$

5 From localized moments to a typical bridge

Proposition 4.10 turns the confined first and second moments into the following seed estimate: for every fixed $\xi > 0$,

$$\mathbb{P}_h^{\text{br}}\{|\gamma| \geq h^{4/3-\xi}\} \geq h^{-o(1)}.$$

This probability may tend to zero. We amplify it to a high-probability length bound by running independent irreducible-bridge trials, then pay explicitly for conditioning on a specified terminal height.

The irreducible-bridge construction goes back to Kesten's renewal identity [6]; see Madras–Slade [9] and the appendix of Lawler–Schramm–Werner [8]. We derive the normalization and the conditional renewal estimates needed for the present port convention.

A bridge is *irreducible* if it has positive height and no once-crossed intermediate level. Cutting at all such levels gives a unique sequence of irreducible bridges. Conversely any sequence concatenates freely: the pieces occupy successive disjoint open strips and their port weights multiply. Let I_h be the total irreducible mass at height $h \geq 1$, and set

$$I(z) = \sum_{h \geq 1} I_h z^h, \quad B(z) = \sum_{h \geq 0} B_h z^h.$$

Uniqueness gives $B(z) = (1 - I(z))^{-1}$ for $0 \leq z < 1$. Since $B_h \asymp (1 + h)^{-1/4}$,

$$B(e^{-1/l}) \asymp l^{3/4}, \quad \sum_{h \geq 1} I_h = 1, \quad \sum_{h \geq l} I_h \leq Cl^{-3/4}.$$

For the tail, use $(1 - e^{-1}) \sum_{h \geq l} I_h \leq 1 - I(e^{-1/l})$; normalization follows by letting $z \uparrow 1$.

We may therefore sample independent irreducible bridges, giving each individual bridge probability equal to its critical weight. Their concatenation is an infinite self-avoiding walk in the half-plane. Write $U_0 = 0 < U_1 < U_2 < \dots$ for its renewal heights, and let

$$T(l) = \min\{k : U_k > l\}, \quad G_l = \text{the prefix ending at } U_{T(l)}.$$

The renewal heights tend to infinity because their increments are positive integers. The event $\mathcal{H}_h = \{h = U_k \text{ for some } k\}$ has probability B_h . Conditional on $\mathcal{H}_h$, the prefix ending at height h has exactly the critical height- h bridge law, normalized by B_h . This follows path by path from the unique irreducible decomposition. The analogous strip-conditioning identity in the square-lattice vertex convention appears in [4, Proposition 2.10]; its final bond lies above the conditioning height.

Lemma 5.1 (Renewal overshoots and a successful trial). *For $x \geq 2l \geq 2$,*

$$\mathbb{P}\{U_{T(l)} > x\} \leq C(l/x)^{3/4}.$$

For every fixed $\xi > 0$, a trial beginning at a renewal and ending at the first renewal more than h bands above it has probability at least $h^{-o(1)}$ of containing $h^{4/3-\xi}$ vertices.

Proof. The last renewal before crossing l is some $i \leq l$, and its next increment must exceed $x - i$. Hence

$$\mathbb{P}\{U_{T(l)} > x\} = \sum_{i=0}^l B_i \sum_{j > x-i} I_j \leq Cx^{-3/4} \sum_{i=0}^l B_i \leq C(l/x)^{3/4}.$$

For the second assertion let Z_h count renewals $m \in [h/2, h]$ whose prefixes have length at least $h^{4/3-\xi}$. Applying (12) with slack $\xi/2$, which absorbs the constant factor between m and h , gives $\mathbb{E}Z_h \geq h^{3/4-o(1)}$. The unrestricted number N_h of renewals up to h satisfies

$$\mathbb{E}N_h^2 \leq \sum_{m \leq h} B_m + 2 \sum_{m < n \leq h} B_m B_{n-m} \leq Ch^{3/2}.$$

The product in the second sum follows from independence after a renewal. Thus $\mathbb{P}\{Z_h > 0\} \geq (\mathbb{E}Z_h)^2 / \mathbb{E}N_h^2 \geq h^{-o(1)}$. Every prefix counted by Z_h lies in the trial. $\square$

Theorem 5.2 (Typical finite-bridge length). *For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that a critical height- H bridge satisfies*

$$\mathbb{P}_H^{\text{br}}\{H^{4/3-\varepsilon} \leq |\gamma| \leq H^{4/3+\varepsilon}\} \geq 1 - O_\varepsilon(H^{-c_\varepsilon}).$$

Its diameter is at least $d_0 H$ and at most $CH^{1+\varepsilon}$ with polynomially small exceptional probability.

Proof. Fix small $\theta, \xi > 0$, and put

$$h = \lfloor H^{1-\theta} \rfloor, \quad R = \lfloor H^{1-\theta/2} \rfloor, \quad K = \lfloor H^{\theta/16} \rfloor.$$

Run K successive trials of Lemma 5.1, restarting at each completion. Independence of irreducible pieces after these stopping renewal indices makes the trials independent. If Δ_j is their height advance, then

$$\mathbb{P}\left\{\sum_{j=1}^K \Delta_j > R\right\} \leq \sum_{j=1}^K \mathbb{P}\{\Delta_j > R/K\} \leq CK(hK/R)^{3/4} \leq CH^{-17\theta/64}.$$

For large H every trial succeeds with probability at least $H^{-\theta/32}$, by its $h^{-o(1)}$ lower bound. Thus all trials fail with probability at most $\exp(-cH^{\theta/32})$. Outside these two events, a successful trial is contained in G_R . Choose θ, ξ so that $(1 - \theta)(4/3 - \xi) > 4/3 - \varepsilon$. Then

$$\mathbb{P}\{|G_R| < H^{4/3-\varepsilon}\} \leq CH^{-17\theta/64} + \exp(-cH^{\theta/32}).$$

We transfer this estimate to conditioning on $\mathcal{H}_H$. Put $\tau = T(R)$ and extend B_j by zero for $j < 0$. For any event E determined by the prefix through τ , renewal independence gives the exact identity

$$\mathbb{P}(E \mid \mathcal{H}_H) = \frac{\mathbb{E}[\mathbf{1}_E B_{H-U_\tau}]}{B_H}.$$

Here $R < H$, so a path which overshoots H at τ could not have hit H earlier. On $U_\tau \leq H/2$ the density ratio is bounded by a constant. For all sufficiently large H , we have $R \leq H/4$. For the remaining conditional probability, let $i \leq R$ be the last renewal before crossing R , and let l be the height of the crossing irreducible. If $i + l > H/2$, then $l \geq H/2 - R$. Summing over these possibilities gives the upper bound

$$\frac{1}{B_H} \sum_{l \geq H/2-R} I_l \sum_{i=0}^{\min(R, H-l)} B_i B_{H-l-i} \leq C(R/H)^{1/2}.$$

For completeness, writing $C_R(n) = \sum_{i=0}^{\min(R, n)} B_i B_{n-i}$, the full convolution bound gives $C_R(n) \leq C(n+1)^{1/2} \leq CR^{1/2}$ when $n \leq 2R$. When $n > 2R$, use $C_R(n) \leq CR^{3/4}n^{-1/4} \leq CR^{1/2}$. Multiply this uniform bound by $\sum_{l \geq H/2-R} I_l \leq CH^{-3/4}$ and $B_H^{-1} \leq CH^{1/4}$. This proves the displayed estimate. The prefix G_R is part of the conditioned height- H bridge whenever its endpoint is at most $H/2$, so the lower-length failure is $O(H^{-\theta/4})$. For example, for $0 < \varepsilon < 1$, $\theta = \varepsilon/8$ and $\xi = \varepsilon/4$ give the explicit bound $O_\varepsilon(H^{-\varepsilon/32})$.

The full strip first-length estimate in Theorem 2.4, divided by B_H , gives $\mathbb{E}_H^{\text{br}}|\gamma| \leq H^{4/3+o(1)}$. Markov's inequality proves the upper bound with polynomially small failure. Finally, dividing the lateral loss $CHM^{-5/4}$ of Lemma 4.1 by B_H gives $\mathbb{P}_H^{\text{br}}\{\text{lateral extent} > M\} \leq C(H/M)^{5/4}$. Take $M = H^{1+\varepsilon}$. The deterministic height span is d_0H , which also proves the lower diameter bound. $\square$

6 Ordered renewal probes

Two adjacent pieces of a self-avoiding path cannot pass through each other. We quantify this elementary obstruction under the independent renewal law before imposing the eventual height or length of either piece. The estimate must also allow a single long irreducible bridge to be inspected from both ends. Those two inspections share the same critical weight, so their costs cannot be multiplied as independent probabilities.

Throughout this section, heights count lattice bands, port paths carry weight $\rho^{|\gamma|}$, and $|\gamma|$ counts visited vertices. Put $\alpha = 3/4$. We use the bridge masses $B_j \asymp (1+j)^{\alpha-1}$, with $B_0 = 1$, the

localization and kernel variation estimates of Lemmas 4.1 and 4.2, the free-end tubes of Lemma 4.5, and the corridor construction of Corollary 4.12. A free endpoint is summed over its lattice ports; once its shape is specified, attaching it to another path makes its translation unique.

The argument has three parts. First we fix the probability and unnormalized measures used in every probe. Next a recoverable hairpin construction bounds masses of pairs with suitably separated renewal endpoints. Finally we show that, except for a controlled exceptional part, an ordered probe supplies many such endpoints. This last step uses its conditional failure bound before conditioning on continued avoidance.

6.1 Stopped paths and renewal occurrences

Write $\nu(I) = \rho^{|I|}$ for the probability mass of a rooted irreducible strict port bridge, with the endpoint convention already used in bridge factorization. Thus $\sum_I \nu(I) = 1$. Let $\mathbf{P}$ be the product law of an infinite sequence of such bridges, with renewal heights $U_0 = 0, U_1, \dots$. For every real threshold $t \geq 0$, let $\mathcal{Q}_t$ be the law of its prefix through the first renewal at or above t . For $t = 0$ this is the empty prefix. In contrast, define the renewal-occurrence measure $\mathcal{R}$ on finite strings by

$$\int G(P) d\mathcal{R}(P) = \mathbf{E} \sum_{k \geq 0} G(I_1 \cdots I_k).$$

At fixed height j this is exactly the unnormalized bridge measure; its endpoint marginal is $b_j(x)$, summed with counting measure over lattice ports x , and its total mass is B_j . In particular $\mathcal{Q}_t$ is a probability measure, whereas $\mathcal{R}\{P : \text{ht}(P) \leq t\} \asymp (1+t)^\alpha$.

Whenever a renewal prefix of height $v \geq t$ is marked, it can be cut at its first renewal $U \geq t$. Its exact measure is $d\mathcal{Q}_t(P) d\mathcal{R}(R)$, with the residual-height condition $U + \text{ht}(R) = v$. For a sum over $v \leq Ct$, the residual mass is uniformly at most $C(1+t)^\alpha$. The empty residual string is included. This identity, rather than conditioning a bridge to hit a specified terminal height, is the change of measure used below.

Lemma 6.1 (Jump charges and renewal counts). *For an irreducible bridge let J be its height and W its maximum absolute horizontal displacement from its start. For dyadic $a \geq 1$ and real $d \geq 0$,*

$$\mathbf{P}\{a \leq J < 2a, W \geq da\} \leq Ca^{-\alpha} \Phi(d), \quad \Phi(d) = (1+d)^{-5/4}.$$

Moreover, for $b \geq 0$ and $x \geq 1$,

$$\mathbf{E} \sum_{k: U_{k-1} < b} \min\{1, (J_k + W_k)/x\} \leq C(1+b)^\alpha x^{-\alpha}.$$

For every positive integer p , the number of renewals in any height interval of length $v \geq 0$, after a stopping renewal, has p -th moment at most $C_p(1+v)^{p\alpha}$, uniformly in the stopping history and interval location.

Proof. Prepend and append bridges with heights in $[a, 2a]$ to a marked irreducible. Their product mass is at least $ca^{2\alpha}$. Output heights are $O(a)$, and a marked irreducible of height at least a has bounded multiplicity in such an output. Its lateral excursion forces comparable output range. The strip lateral estimate bounds output mass by $Ca^\alpha \Phi(d)$, proving the first assertion. Summing dyadic a and width bins gives $\mathbf{E} \min\{1, (J+W)/x\} \leq Cx^{-\alpha}$. The event $U_{k-1} < b$ is decided before jump k ; multiplying by the fresh-jump expectation and summing gives the charge estimate, because $\sum_{j < b} B_j \leq C(1+b)^\alpha$. This includes the last, possibly overshooting jump. For the moment estimate, order the marked renewals, restart at each, and sum their nonnegative height increments using $\sum_{j \leq v} B_j \leq C(1+v)^\alpha$. Diagonal repetitions cost only a constant depending on p . A delayed interval is handled by first stopping on entrance; overshoot can only shorten its remaining length. $\square$

Launch two independent renewal strings at neighboring ports on one line. The left and right designations specify their launch order. The test $E(h)$, measurable under $\mathcal{Q}_h \otimes \mathcal{Q}_h$, requires disjointness at depths $0 \leq y < h$ and, at every such depth,

$$\begin{aligned} \max x(\text{left arc at } y) &\leq \max x(\text{right arc at } y), \\ \min x(\text{left arc at } y) &\leq \min x(\text{right arc at } y). \end{aligned}$$

It is enough to use depths intersecting the embedded edges. Strict renewals prevent later visits below their level, so $E(h)$ implies all smaller-depth tests. This is an ordered test; disjointness alone of the two stopped strings is not its definition.

Fix constants $L_2 \geq L > 1$, to be selected below. A dyadic height is 2^k with $k \in \mathbb{Z}_{\geq 0}$. For dyadic $r \geq h$, let $F(h, r, d)$ be the probability of $E(h)$ together with a jump I of a designated arm satisfying $r \leq J(I) < 2r$, $W(I) \geq dr$, and born before height h/L . Such a jump is unique: after it the renewal height is at least h , so a second jump cannot be born in the specified window. Equivalently,

$$F(h, r, d) = \int_{\text{ht}(P) < h/L} d\mathcal{R}(P) \sum_{\substack{r \leq J(I) < 2r \\ W(I) \geq dr}} \nu(I) \int \mathbf{1}_{E(h)}(PI, A) d\mathcal{Q}_h(A). \quad (17)$$

The obstacle PI already contains its stop at h .

For the paired quantity use the same irreducible I from its two ends, reversing and reflecting it as needed. Precede it, in these separate inward orientations, by prefixes P_i of height $f_i < h_i/L_2$, and launch independent workers A_i at adjacent ports. There is no avoidance condition between the two sides. Its precise definition is

$$\begin{aligned} F_{12}(h_1, h_2, r) = & \sum_{r \leq J(I) < 2r} \nu(I) \int_{f_1 < h_1/L_2} d\mathcal{R}(P_1) \int_{f_2 < h_2/L_2} d\mathcal{R}(P_2) \\ & \times \prod_{i=1}^2 \left[\int \mathbf{1}_{E(h_i)}(P_i I^{(i)}, A_i) d\mathcal{Q}_{h_i}(A_i) \right]. \end{aligned} \quad (18)$$

Here $I^{(1)}, I^{(2)}$ are the two normalized orientations of one shape, not independent jumps. Each side is normalized by its own near launch; its other ports are determined by concatenation. In particular no horizontal position is summed in addition to the endpoints already present in $d\mathcal{R}$ and ν .

Figure 6 displays the shared and separate integrations in this definition.

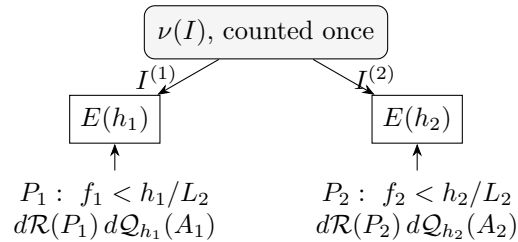


Figure 6: The integrations defining F_{12} . Both tests use one irreducible bridge. Conditional on that shape, their prefix and worker integrations separate. Prefix endpoint positions are already summed by $\mathcal{R}$; there is no additional translation factor.

Theorem 6.2 (Ordered probes, with one or two giant ends). *The constants L, L_2, K_1, K_2, K_3 can be chosen so that, for all integer $1 \leq h, h_1, h_2 \leq r$, with r dyadic and $d \geq 0$,*

$$\mathbf{P}(E(h)) \leq K_1 h^{-\alpha}, \quad F(h, r, d) \leq K_2 r^{-\alpha} \Phi(d), \quad F_{12}(h_1, h_2, r) \leq K_3 r^{-\alpha}. \quad (19)$$

Either adjacent-port choice and each reflected orientation obey the same bounds.

6.2 Recovering one or two hairpins

Fix $e > 0$, $D < \infty$, and $0 < \sigma < e/10$. A worker bridge A , launched to the right of an obstacle, is *qualified at scale h* if its endpoint height u lies in $[eh, 2h/5]$, the two shapes are disjoint, $|x| \leq Dh$ on A , and the obstacle satisfies

$$x \leq Dh \quad (0 \leq y \leq u + \sigma h), \quad x \leq x_A - \sigma h \quad (u \leq y \leq u + \sigma h).$$

Coordinates are relative to the near launch, and x_A is the worker endpoint abscissa. The obstacle may extend arbitrarily far left. Reflected qualifications are allowed throughout.

Write $\chi_h(O, A)$ for the indicator that the worker bridge A is qualified against the obstacle O , with the prescribed neighboring launches. The three unnormalized masses used below are

$$\begin{aligned} Y(h) &= \iint \chi_h(O, A) \mathbf{1}_{\{\text{ht}(A) + 2\sigma h \leq \text{ht}(O) \leq h\}} d\mathcal{R}(O) d\mathcal{R}(A), \\ X_d(h, r) &= \sum_{\substack{r \leq J(I) < 2r \\ W(I) \geq dr}} \nu(I) \int_{\text{ht}(P) \leq h/8} d\mathcal{R}(P) \int \chi_h(PI, A) d\mathcal{R}(A), \\ X_{12}(h_1, h_2, r) &= \sum_{r \leq J(I) < 2r} \nu(I) \prod_{i=1}^2 \left[\int_{\text{ht}(P_i) \leq h_i/8} d\mathcal{R}(P_i) \int \chi_{h_i}(P_i I^{(i)}, A_i) d\mathcal{R}(A_i) \right]. \end{aligned}$$

The last line uses the two inward orientations of one irreducible, exactly as in (18). Its factor $\nu(I)$ occurs once. Worker endpoints are summed by $\mathcal{R}$; these quantities are masses of endpoint occurrences, not probabilities of stopped paths. There is no extra horizontal-translation sum.

Lemma 6.3 (Hairpin mass bounds). *For fixed qualification parameters and all sufficiently large scales, with r dyadic, $1 \leq h, h_1, h_2 \leq r$, and $d \geq 0$,*

$$Y(h) \leq Ch^\alpha, \quad X_d(h, r) \leq Ch^\alpha r^{-\alpha} \Phi(d), \quad X_{12}(h_1, h_2, r) \leq Ch_1^\alpha h_2^\alpha r^{-\alpha}.$$

Bounded scales can be included by changing constants.

Proof. Constructing the hairpin. Apply Corollary 4.12 at the worker endpoint. It supplies an arch contained in the next σh bands, to the right of $x_A - \sigma h$, whose far endpoint lies beyond $D'h$, for any sufficiently large fixed D' . To pass from the relative displacement in that lemma to this absolute bound, increase its fixed displacement constant by D , using $|x_A| \leq Dh$. Its total critical mass is between $ch^{-1/4}$ and $Ch^{-1/4}$. The lower bound has no exponent loss; the fixed-endpoint tube bound would not suffice here. The corridor lemma also establishes the endpoint half-plane bounds needed when this arch is cut in the second-moment calculation below.

Choose $z \in [eh/5, eh/3]$. From the far end of the long arch descend by a narrow bridge to level z , with mass at least $ch^{-1/4}$. Continue downward by a bridge of free height in $[2r, 4r]$; use $r = h$ for Y . Require its portion through depth $z + 1$ from its starting port to stay within horizontal distance $D''h$ of that port. This restriction retains mass at least cr^α . Indeed its failure probability in the renewal law is small for large D'' , by Lemma 6.1. If N counts renewals in the desired terminal interval, then $\mathbf{E}N \geq cr^\alpha$ and $\mathbf{E}N^2 \leq Cr^{2\alpha}$; consequently $\mathbf{E}[N \mathbf{1}_{\text{failure}}] \leq Cr^\alpha \mathbf{P}(\text{failure})^{1/2}$. Take D'' large and then D' much larger.

Join the two near launches by a fixed short arch in the one band on the opposite side. The resulting path rises from its new lower end, goes round the long arch, descends along the worker to this short turn, then rises along the obstacle. Append an upper continuation with height in $[2r, 4r]$. For X_{12} also make the reflected hairpin at the other end. Qualification gives all lateral clearances.

The two near-end structures protrude by at most $(2/5 + \sigma)h_i$; since $h_i \leq r$ and the giant has height at least r , they are separated in height. These operations produce strict bridges at a fixed translated root. Local connections change weights by fixed factors. Figure 7 shows the one-giant construction; there τ is the absolute height of the near launch.

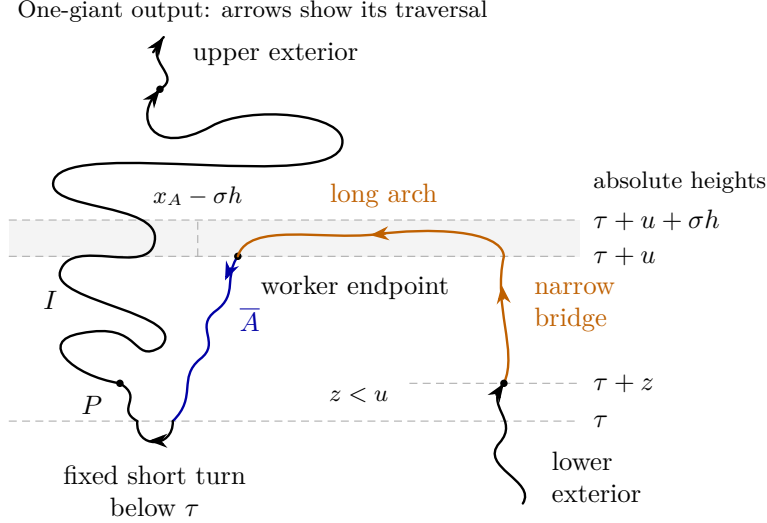


Figure 7: The one-giant hairpin, schematically. From its new lower end the output follows the right exterior to height $\tau + z$, the narrow bridge and long arch to the worker endpoint at $\tau + u$, the reversed worker, the fixed short turn, the prefix P , the giant I , and the upper exterior. Here $z \in [eh/5, eh/3]$ and $u \in [eh, 2h/5]$ are relative heights; τ is the absolute near-launch level. In the shaded slab the obstacle is to the left of $x_A - \sigma h$ and the long arch is to its right. The curves are not to scale and indicate joining order and clearance, not lattice edges or monotonicity. In particular I has no width bound, and the drawn prefix endpoint has no required height order relative to u .

Representation mass and nearby turn lines. For the one-giant construction, let $m(\Gamma)$ count the possible input tuples and joining choices that produce the rooted output bridge Γ . Define

$$Z_1 = \sum_{\Gamma} \rho^{|\Gamma|} m(\Gamma), \quad V_1 = \sum_{m(\Gamma) > 0} \rho^{|\Gamma|}, \quad M_2 = \sum_{\Gamma} \rho^{|\Gamma|} m(\Gamma)^2.$$

The weight of an input tuple is comparable to the output weight, because its pieces partition Γ apart from the bounded local port changes. Thus Z_1 is comparable to the summed construction weight. Weighted Cauchy–Schwarz gives $Z_1^2 \leq V_1 M_2$. For $X_d = X_d(h, r)$, the preceding construction and the strip lateral estimate therefore give

$$Z_1 \geq cX_d r^{2\alpha} h^{1/2}, \quad V_1 \leq Cr^\alpha \Phi(d).$$

The factor $h^{1/2}$ is the product of the $\asymp h$ choices of z and the two $h^{-1/4}$ connector masses. The marked giant’s width forces comparable output range, giving $\Phi(d)$.

We estimate the second moment by pairs of representations of the same output. First bin their turn lines τ , in absolute output coordinates, into intervals of width $r/50$. There are only boundedly many such bins, because output heights are $O(r)$. Cauchy–Schwarz permits us to sum same-bin second moments at bounded cost. For turn-line difference at most Nh , the contribution is

$$C_N r^{2\alpha} X_d(h, r) h^{1/2} h^{5/4}. \tag{20}$$

Here is the full cost allocation. Subdivide into intervals of width ceh , with a sufficiently small fixed c . For nearby cells, $ab \leq (a^2 + b^2)/2$ reduces pair counting to same-cell pairs with a factor depending on N, e , but no factor for the absolute cell index. A line strictly between all $\tau + z$ and all $\tau + u$ in the cell has the following property: after its first visit, the lowest point of the path is the short turn. Before the descending worker reaches that turn, the initial rising branch stays above $\tau + z$; afterwards everything other than the fixed short arch stays above the turn line. Thus the short turn, its line, and its two bounding visits coincide in the two representations. The upward branch determines the giant mark up to bounded choices, since its prefix is at most $h/8$. Retain the marked input associated to the smaller worker height u .

Between the two worker heights there are two bridges, one on either side of the long arch. Their jointly summed cost is $\sum_{j \leq Ch} B_j^2 \leq Ch^{1/2}$. The higher long arch costs $Ch^{-1/4}$. The far rising bridge from the higher z to the smaller u has height comparable to h , hence costs $Ch^{-1/4}$. The pair of z -levels and their gap cost $Ch \sum_{j \leq Ch} B_j \leq Ch^{1+\alpha}$. Multiplying these four costs gives $Ch^{1/2+5/4}$, as in (20). The two exterior continuations supply $r^{2\alpha}$. Ties use $B_0 = 1$. Every cut is determined by its level: each selected u -level is crossed twice on the pre-turn part, while the z -cuts are first visits before these crossings. A rooted tuple of these relative shapes has one translation to the prescribed joining port, also when attached by its far endpoint. Thus no horizontal multiplicity is omitted.

Distant turn lines. For distant turn lines in the same coarse bin, write $\Delta = \tau_2 - \tau_1 > Nh$, and let $p_i = \tau_i + f_i$ be their giant junctions. Everything before p_1 has height at most $\tau_1 + (2/5 + \sigma)h < \tau_2 + z_2$, since $\Delta > Nh$. Thus the first visit to $\tau_2 + z_2$ occurs after p_1 . Moreover, $\Delta < r/50$ and $z_2 = O(h)$, with $h < \Delta/N$, put this cut below the terminal level of the first giant. The portion from p_1 to this cut is therefore contained in that giant. It is strict above its lower level by the giant's lower wall, and strict below its upper level by the first-visit property. After the cut the second representation never falls below $\tau_2 - O(1)$. If this subbridge had a single-crossing level below $\tau_2 - O(1)$, it would also be a seam of the first giant. Its first irreducible I' consequently has height

$$J(I') \in [\Delta - f_1 - O(1), \Delta - f_1 + O(h)],$$

and the rest is a bridge of free height $O(h)$. Replacing the first giant by I' preserves the earlier qualification, which concerns only its near range. Retain this reduced first construction and then the second construction with its original giant. The very first and last exterior continuations are retained as well. The removed endpoint of the first giant is recoverable as its first ensuing renewal in the upward continuation; it has no unbounded marking multiplicity.

The second construction costs $h^{1/2}X_d(h, r)$; the reduced first one at dyad a costs $h^{1/2}X_0(h, a)$; the intervening bridge costs h^α . Hence the distant contribution is at most

$$Cr^{2\alpha}h^{1/2}X_d(h, r)h^{\alpha+1/2} \sum_{Nh/C \leq a < r} X_0(h, a).$$

Relative offsets of the two turn lines are already determined by the reduced shapes. For $d = 0$, induct on the dyadic r with the bound $X_0(h, a) \leq Kh^\alpha a^{-\alpha}$. After division by $r^{2\alpha}h^{1/2}X_0(h, r)h^{5/4}$, the distant contribution is at most $CKN^{-\alpha}$. Cauchy–Schwarz, $Z_1^2 \leq V_1$ times the second moment, therefore gives

$$X_0(h, r) \leq h^\alpha r^{-\alpha}(C_N + CKN^{-\alpha}).$$

Choose N first to make the coefficient of K smaller than one half, then K to absorb C_N and the bounded initial scales. For $d > 0$ use this proved X_0 bound in the distant sum and retain the factor $\Phi(d)$ in V_1 . This proves the one-giant bound.

For Y , only nearby bins are needed, with $r = h$. The obstacle's terminal cut v is now movable on the common upward branch. Retain the smaller worker height and smaller obstacle height, even

if they come from different representations. Shortening preserves qualification and $v - u \geq 2\sigma h$. The larger v costs one additional summed bridge gap, at most Ch^α . The same Cauchy–Schwarz calculation gives $Y \leq Ch^\alpha$.

Two ends of a shared giant. It remains to bound both ends of one giant simultaneously. Use the analogous representation count for the paired construction. Its first moment is at least $cX_{12}r^{2\alpha} \prod_i h_i^{1/2}$, and output mass is at most Cr^α . Bin both turn lines into intervals of width $r/50$. On side i , color the fine cells of width ceh_i periodically with a sufficiently large fixed number of colors. Matching colors mean either the same cell or separation exceeding $100h_i$. Splitting into these finitely many classes and applying Cauchy–Schwarz lets us use only matching-color pairs.

Call a same-cell side local. The preceding recovery of the short turn also recovers its giant junction: all seams within the first $h_i/8$ bands are unaffected by the other-end hairpin, which lies far beyond that range. Distinct junctions would put an internal seam in one of the two giants. On a distant side the outer junction precedes the inner one in the inward orientation, and the preceding I' -replacement applies.

We describe the retained subpaths before counting them. Write a_j, b_j for the lower and upper giant junctions of representation j , and write $\prec$ for order along the oriented output path. At the lower end denote the two junctions by $a_{\text{out}} \preceq a_{\text{in}}$, and at the upper end by $b_{\text{in}} \preceq b_{\text{out}}$; equality means that side is local. The subpath between a_{in} and b_{in} will be the central giant. When both sides are distant, the two possible orders, after interchanging representation names or reflecting, are

$$a_1 \prec a_2 \prec b_1 \prec b_2 \quad \text{and} \quad a_1 \prec a_2 \prec b_2 \prec b_1.$$

Thus the central subpath is $[a_2, b_1]$ in the crossed case and $[a_2, b_2]$ in the nested case. Figure 8 shows these path orders; it does not depict the spatial positions of the junctions.

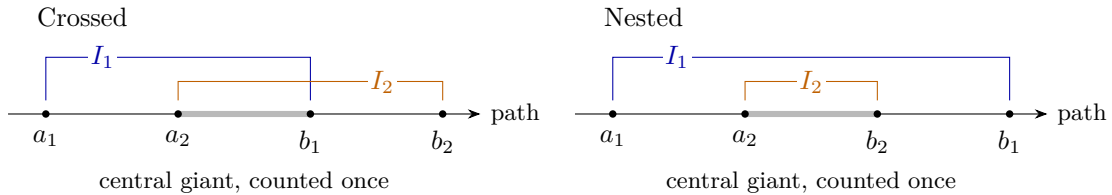


Figure 8: Two representations of the same output when both sides are distant, shown in path order, not in spatial coordinates. In the crossed case $a_1 \prec a_2 \prec b_1 \prec b_2$, the central giant is $[a_2, b_1]$; in the nested case $a_1 \prec a_2 \prec b_2 \prec b_1$, it is $[a_2, b_2]$. The brackets identify the original giants $I_j = [a_j, b_j]$ as recovered subpaths. The two outside junctions are retained by the reduced inputs. Neither the bracket crossings nor their vertical positions represent intersections or heights of the walk.

The central subpath is indeed an irreducible bridge of scale r or $r/2$. When both its junctions come from one representation, it is that representation's giant. In the crossed case it is contained in both giants: the giant starting at a_2 keeps it strictly above a_2 , and the giant ending at b_1 keeps it strictly below b_1 . Divide its height range into lower and upper halves. A seam in the lower half would extend to a seam of the giant $[a_2, b_2]$: between b_1 and b_2 , the upper hairpin can descend below b_1 by at most $(2/5 + \sigma)h_{\text{up}} + O(1)$, where h_{up} is the upper-side test scale. A seam in its upper half is excluded by the reflected argument using $[a_1, b_1]$. Each turn-line difference is less than $r/50$, and a distant difference exceeds $100h_i$, so the near-end excursions cannot invalidate either extension. The central height can fall below r only in this two-distant case, where necessarily $h_i \leq r/2$. On a distant side retain the inner representation's worker and prefix; on a local side retain the smaller

worker height. They qualify against the central giant by containment, and so give a central paired input.

Recovering the paired representations. In the crossed case, the lower reduced construction retains the first representation's worker and prefix, followed by its first irreducible I'_- ; thus a_1 is still the prefix-irreducible junction of that counted input. On the upper side, read inward from the upper end, the reduced construction retains the second representation's worker and prefix, followed by its first irreducible I'_+ ; its prefix-irreducible junction is b_2 . The remaining endpoints a_2, b_1 are the two endpoints of the central giant. These are existing input junctions, rather than additional marks placed on an unmarked bridge.

Here is the order in which the lower pieces are joined. For $j = 1, 2$, write A_j, P_j, S_j for the lower worker, obstacle prefix, and fixed short turn, respectively. Let K_j be the connector from the cut at $\tau_j + z_j$, through the narrow bridge and long arch, to the worker endpoint, retaining their counted joining ports and local connection data; here τ_j denotes the turn-line height. Let R_- be the bridge remainder after I'_- , and let E_- be the first exterior continuation. With a bar denoting path reversal, the lower part of the common output is assembled in the order

$$E_- K_1 \overline{A_1} S_1 P_1 I'_- R_- K_2 \overline{A_2} S_2 P_2,$$

and then the central giant is attached. The upper part is assembled by the reflected reverse procedure, with representation 2 outer and representation 1 inner. In the forward cutting operation, I'_- ends at the first positive renewal of the strict subbridge from a_1 to the second connector's first-visit cut; R_- is the rest of that subbridge. The upper cut uses the same rule in the reversed orientation. Consequently the reduced-irreducible cuts require no further renewal index. The connector cuts themselves are the first-visit and ordered worker cuts already specified above.

Each item in this list is a rooted relative shape: part of the central or a reduced outer input, a connector, a bridge remainder, or an exterior continuation. Once one joining port is placed, the next shape has a unique translation to it, including when the shape is attached by its terminal port. Its height and horizontal displacement are then fixed. Starting at the prescribed output root therefore places every piece, and in particular all four retained junctions, without a separate sum over turn-line differences or horizontal offsets. The second lower exterior continuation is the assembled prefix $E_- K_1 \overline{A_1} S_1 P_1 I'_- R_-$ ending at the exterior port of K_2 ; the remaining upper exterior is the corresponding recovered suffix. Thus each original exterior is recovered by cutting at its retained connector port. Finally, extract the ordered subpaths $[a_1, b_1]$ and $[a_2, b_2]$, and assign the retained workers, prefixes and connectors to their recorded representation. This recovers both original constructions. Interchanging the representation names, choosing the reflected orientation, and the bounded local connection choices require only finitely many records.

For the nested two-distant case, the order is $a_1 \prec a_2 \prec b_2 \prec b_1$. The central paired input is bounded by a_2, b_2 , both from representation 2, and the two reduced inputs retain a_1, b_1 , both from representation 1. A finite matching record assigns the four retained boundaries $a_{\text{out}}, a_{\text{in}}, b_{\text{in}}, b_{\text{out}}$ to the two representations; the same ordered assembly and exterior-port cuts recover the nested case. On a local side the two junctions coincide as proved above. The tuple pieces partition the common output, apart from bounded port modifications. Their product critical weight is therefore comparable to the single critical weight of that output in the second moment. In particular the central giant is charged once; the two original giants are recovered subpaths of the assembled output and do not contribute additional weight factors.

The reconstruction has bounded ambiguity, uniformly in all scales. We can therefore integrate these pieces. Each exterior continuation costs Cr^α . A local side has the four gap and arch costs already computed, namely $Ch_i^{1/2+5/4}$. On a distant side the inner connector costs $Ch_i^{1/2}$, the reduced

outer construction costs $Ch_i^{1/2}X_0(h_i, a)$, and the bridge remainder costs Ch_i^α . The proved one-sided estimate gives

$$h_i^{1/2}h_i^{1/2}h_i^\alpha \sum_{a \geq ch_i} X_0(h_i, a) \leq Ch_i^{1+\alpha} = Ch_i^{1/2+5/4}.$$

The central paired input has scale r or $r/2$, and its giant weight is charged once. Combining these costs with the first moment and output mass yields

$$\begin{aligned} & \left(cX_{12}(h_1, h_2, r)r^{2\alpha} \prod_i h_i^{1/2} \right)^2 \\ & \leq Cr^{3\alpha} \prod_i h_i^{1/2+5/4} \{X_{12}(h_1, h_2, r) + X_{12}(h_1, h_2, r/2)\}, \end{aligned}$$

omitting an out-of-domain last term. Writing $V_r = X_{12}(h_1, h_2, r)/(h_1^\alpha h_2^\alpha r^{-\alpha})$, this becomes $V_r^2 \leq C(V_r + 2^\alpha V_{r/2})$. A sufficiently large constant majorant closes the dyadic induction, proving the paired bound. $\square$

6.3 Finding separated renewal endpoints

Lemma 6.4 (Usable renewals after a regular history). *Stop a worker at its first renewal at or above s , where s/h is a sufficiently small fixed constant. Call this history regular when the sum of $J + W$, including its last jump, is less than $h/16$. For every $\delta > 0$ one can choose fixed qualification parameters so that, for all sufficiently large h depending on these choices, conditionally on any such history and any fixed opposite path, with failure probability at most δ , continued $E(h)$ implies at least ch^α qualified worker renewals in $[eh, 2h/5]$. For two workers with regular histories, outside total conditional failure probability at most 2δ , continued $E(h)$ supplies at least $ch^{2\alpha}$ separated renewal pairs. In each pair the lower renewal is a qualified worker against the opposite prefix ending at the higher renewal, whose height exceeds it by at least $2\sigma h$. The failure probabilities are taken before conditioning on continued $E(h)$.*

Proof. Write (t, x_0) for the worker stop. A regular history has $t < h/16$ and horizontal range less than $h/16$. For $m = \lceil 3c'h^\alpha \rceil$, the probability that the next m jumps advance more than $h/4$ is $O(c')$: bound jumps larger than $h/4$ by a union bound and the remaining sum by its truncated mean. Choose c' small. There are then at least $2c'h^\alpha$ future renewals before $t + h/4 < 2h/5$, except with arbitrarily small probability. Renewals within eh of the stop have expected count $O((1+eh)^\alpha)$, so remove them by taking e small.

No short bin contains a substantial fraction of these renewals with high probability. Indeed partition the candidate interval into $O(\theta^{-1})$ bins of length θh . By Lemma 6.1, for any fixed $p > 1/\alpha$, the probability that some bin contains at least εh^α renewals is at most $C_{p,\varepsilon} \theta^{p\alpha-1}$, once θh is large. Use neighboring bins as well to control every interval of length $2\theta h$. The charge bound also makes the entire worker's lateral range through its test stop at h at most Dh , with failure probability $O(D^{-\alpha})$.

Freeze the opposite path. For a right worker let $R(u)$ be the rightmost opposite abscissa in $[u, u + \sigma h]$. From (t, x_0) the renewal-occurrence density at (u, x) is exactly $b_{u-t}(x - x_0)$. It is not divided by B_{u-t} , since we count renewal occurrences. Thus the expected number of renewals, at distance at least eh above the stop, with $|x - R(u)| \leq 2e'h$, is at most

$$\begin{aligned} C(e'h + 1) \sum_{j \geq eh} \|b_j\|_\infty & \leq C(e'h + 1) \sum_{j \geq eh} (B_j - B_{j+1}) \\ & \leq C_e(e' + h^{-1})h^\alpha. \end{aligned}$$

The middle inequality follows from Lemma 4.2: a summable nonnegative kernel tends to zero at infinity, so its supremum is at most its total variation. The sum of the resulting differences telescopes. No height-by-height derivative bound on B_j is needed. At each renewal u , restart and discard it if the path through its first renewal strictly above $u + \sigma h$ moves more than $e'h$ laterally. Its conditional probability is at most $C((1 + \sigma h)/(e'h))^\alpha$; multiplying by the expected candidate count bounds the expected number discarded by this quantity times Ch^α .

On continued order, every opposite point in this slab lies to the left of the worker's rightmost point at that level. For a retained candidate that point is at most $x + e'h$, because strict renewals exclude earlier and later pieces from the slab. Hence $R(u) \leq x + e'h$. Since $|x - R(u)| > 2e'h$, necessarily $R(u) < x - 2e'h$. Choose $\sigma < e'$. This is the required clearance; the obstacle's global one-sided bound below $u + \sigma h$ follows from order and the worker's bounded range. Reflection gives the other orientation.

Choose the constants successively: the desired failure tolerance, c' , then e, θ , then D large and e' small, and finally σ small relative to e', e, θ . The expectation bounds and Markov's inequality leave ch^α candidates. For two workers, apply this uniform estimate first with the opposite future frozen and then with roles reversed, integrating by Tonelli. A union bound suffices; independence of the two good events is unnecessary. Bin nonconcentration ensures that only a small fraction of all cross-pairs have heights within $2\sigma h$. The remaining pairs have the claimed lower/higher qualification. $\square$

The preceding lemma converts a regular stopped history into many qualified endpoints. To use it without biasing the fresh jumps, we will retain a smaller test that is measurable at the stopping time, then bound failures under the original conditional law. Before doing so, we record the dyadic estimates that control irregular histories.

6.4 The ordered-probe induction

We record the elementary dyadic sums that close the proof. Let a range over positive dyadic heights and let l range over $0, 2, 4, 8, \dots$; the bin $l = 0$ means $W/a < 2$. Put

$$T_h(a, l) = \min\{1, (1 + l)a/h\}.$$

Since $\sum_l (1 + l)\Phi(l) < \infty$ on these dyads,

$$\sum_{a, l} a^{-\alpha} \Phi(l) T_h(a, l) \leq Ch^{-\alpha}, \quad (21)$$

$$\sum_{a < \kappa h, l} a^{-\alpha} \Phi(l) T_h(a, l) [1 + \log^+(\kappa h/a)] \leq C\kappa^{1-\alpha} h^{-\alpha}. \quad (22)$$

For example, first sum in l , obtaining $C \min(1, a/h)$, then sum the geometric series below and above h . The logarithm in the second line is summable against $a^{1-\alpha}$. Also, if $s = h/L$,

$$(1 + s)^\alpha \sum_{a, l} a^{-\alpha} \Phi(l) T_h(a, l) [\log(2L) + \log^+((1 + s)/a)] \leq CL^{-\alpha} \log(2L) \quad (23)$$

for h sufficiently large relative to the fixed L . The extra logarithm for $a < s$ contributes $O(s/h)$, which is bounded by the right-hand side.

Proof of Theorem 6.2. We first prove the E and F bounds simultaneously by strong induction on their test height, with r, d unrestricted. Floors below one mean the trivial test, with constant cost. All uses of nontrivial smaller tests have strictly smaller test height.

For $F(h, r, d)$, stop the worker at height $s = h/L$. If its history is irregular, then

$$\mathbf{1}_{\text{irregular}} \leq 16 \sum_{k: U_{k-1} < s} \min\{1, (J_k + W_k)/h\}.$$

Integrate this inequality in (17), retaining the large test only where it is needed to imply a smaller test. Let f, g be the dyads of one plus the birth depths of the main giant and a charged worker jump; let a, l be that worker jump's height and width bins. Its charge is at most $CT_h(a, l)$. Both birth dyads are at most $C(1 + s)$. Constants in what follows do not grow with L .

When $f < g/(16L)$, retain the whole smaller event $F(\lfloor g/2 \rfloor, r, d)$. Do not also sum separately over f . On contributing paths the worker's stop for this smaller test is no later than the charged birth, which itself is a renewal. Restarting there, the expected total charges born in the g -bin are at most $Cg^\alpha h^{-\alpha}$, by Lemma 6.1. After dropping the large test, this is a uniform bound on the conditional future integral. Hence the contribution, divided by $r^{-\alpha}\Phi(d)$, is at most

$$CK_2 h^{-\alpha} \sum_{g \leq C(1+s)} g^\alpha \leq CK_2 L^{-\alpha}.$$

When $g < \min(f, a)/(16L)$, retain instead $F(\lfloor \min(f, a)/2 \rfloor, a, l)$, with the workers' roles exchanged. The charged jump is now the unique marked jump of that smaller test; there is no separate g -birth factor. The original giant is born later than its arm's smaller test stop. Its subsequent birth count in the f -bin is at most Cf^α , uniformly in the stop, including a possible birth at the stop itself. The ensuing fresh jump costs $Cr^{-\alpha}\Phi(d)$. Thus the relative contribution is bounded by

$$CK_2 \sum_{f \leq C(1+s)} f^\alpha \sum_{a, l} a^{-\alpha} \Phi(l) T_h(a, l) \leq CK_2 L^{-\alpha}.$$

This formula is a product of a smaller tested integral, a renewal count, and a fresh-jump integral; it is not a conditional probability estimate under the larger F event.

In the remaining case retain $E(\lfloor \min(f, g)/2 \rfloor)$. Both subsequent births are renewals at or beyond their respective smaller stops. Once their earlier histories are fixed, their count integrals are bounded by Cf^α and Cg^α , and the new jump integrals are independent. The smaller test costs $CK_1 \min(f, g)^{-\alpha}$, leaving $CK_1 \max(f, g)^\alpha$. The restrictions defining this case are $f \geq g/(16L)$ and $g \geq \min(f, a)/(16L)$. Their dyadic sum obeys

$$\sum_{f, g} \max(f, g)^\alpha \leq C(1 + s)^\alpha [\log(2L) + \log^+((1 + s)/a)].$$

Indeed for $f \leq a$ the two birth dyads are within factor $16L$, so at each larger dyad there are $O(\log(2L))$ partners. For $f > a$, the permitted smaller partner is at least $a/(16L)$, adding at most $O(\log^+((1 + s)/a))$ choices. Geometric summation of the larger dyad proves the displayed bound. Equation (23) makes the relative contribution at most $CK_1 L^{-\alpha} \log(2L)$.

Consider regular histories next. To apply Lemma 6.4 without paying the unrestricted prefix mass, retain a smaller-test envelope. If the main birth is before $s/(4L)$, use $F(\lfloor s/2 \rfloor, r, d)$. Otherwise use $E(\lfloor s/(8L) \rfloor)$, followed by the main birth count $O(s^\alpha)$ and its jump weight. In the later-birth branch the obstacle's smaller stop occurs no later than the counted birth. In both branches the worker's smaller stop occurs no later than its stop at s . The total envelope mass is therefore at most

$$C_L(K_1 + K_2)r^{-\alpha}\Phi(d).$$

Condition on the worker history to s and on the full opposite obstacle before invoking the lemma. Its failure probability is uniformly at most δ , without conditioning on continued $E(h)$. The envelope

is measurable with respect to this exposed information; integrating it bounds the failure contribution by δ times the last display.

On success there are at least ch^α qualified renewal choices. Count them with $\mathcal{R}$, using its occurrence identity. Each choice leaves a qualified worker bridge and the original marked obstacle; all unused worker futures have probability at most one. Tonelli therefore bounds the total choice mass by $X_d(h, r)$. Lemma 6.3, divided by the guaranteed number of choices, bounds this success contribution by $Cr^{-\alpha}\Phi(d)$. Altogether,

$$\frac{F(h, r, d)}{r^{-\alpha}\Phi(d)} \leq C + CL^{-\alpha}[K_2 + K_1 \log(2L)] + \delta C_L(K_1 + K_2). \quad (24)$$

For $E(h)$, stop both workers at $s' = h/L'$, where L' will be large compared to L/κ . If either early history has a jump of height dyad $a \geq \kappa h$, retain the smaller event $F(\lfloor \kappa h \rfloor, a, 0)$; its birth is early enough since $s' < \kappa h/L$. Summing over $a \geq \kappa h$ costs at most $CK_2\kappa^{-\alpha}h^{-\alpha}$. For all other charged jumps, $a < \kappa h$. If their birth is before a/L , use $F(a, a, l)$, with no further birth-count factor. Otherwise retain a smaller E -test below the birth dyad and then count subsequent births and the new jump as above. The number of possible late birth dyads contributes $1 + \log^+(s'L/a)$. Consequently the irregular-history contribution, relative to $h^{-\alpha}$, is at most

$$\begin{aligned} Ch^\alpha \sum_{a < \kappa h, l} [K_2 + K_1\{1 + \log^+(s'L/a)\}] a^{-\alpha} \Phi(l) T_h(a, l) \\ \leq C(K_1 + K_2)\kappa^{1-\alpha}, \end{aligned}$$

by (22) and $s'L \leq \kappa h$.

For regular histories retain the envelope $E(\lfloor s' \rfloor)$, whose mass is at most $CK_1(s')^{-\alpha}$. Apply the usable-renewal lemma to each worker while freezing the other path; integrate each failure estimate and take a union bound. Their total cost is $C\delta K_1(s')^{-\alpha}$. On success count the $ch^{2\alpha}$ separated cross-pairs. Each pair, according to which renewal is lower, contributes to $Y(h)$ in one of two reflected orientations. The occurrence identity integrates unused futures to at most one, so the total success probability is at most $CY(h)/h^{2\alpha} \leq Ch^{-\alpha}$. Thus

$$\frac{\mathbf{P}(E(h))}{h^{-\alpha}} \leq C + CK_2\kappa^{-\alpha} + C(K_1 + K_2)\kappa^{1-\alpha} + C\delta K_1(L')^\alpha. \quad (25)$$

Choose κ so small that the coefficient of $(K_1 + K_2)$ in the third term of (25) is small. Choose the ratio K_1/K_2 sufficiently large compared to $\kappa^{-\alpha}$. Next choose L so that the charge terms of (24) are a small fraction of K_2 , and then $L' \gg L/\kappa$. Choose the two failure tolerances small enough for these fixed constants. Finally increase K_1, K_2 together in their fixed ratio to absorb the success constants and bounded test heights. At bounded heights, the prefix sum is bounded and the giant tail supplies $r^{-\alpha}\Phi(d)$. Both recursions close.

It remains to prove the paired bound. Regard the two established constants as fixed and induct on $h_1 + h_2$, using $s_i = h_i/L_2$. If either test height is bounded, integrate its prefix and worker at bounded cost and use the single F bound on the other side; since $L_2 \geq L$, its prefix restriction is stronger than that of F . This gives the initial cases.

For an irregular worker on side 1, use the same three birth-order regimes, now with L_2 . In the very early main-birth regime, retain the smaller paired test on side 1 and the original side-2 test. Restart only the side-1 worker after its smaller stop. This gives relative charge $CK_3L_2^{-\alpha}$. In the very early worker-birth regime, the smaller single F marks the charged worker jump and only uses side 1's obstacle before its later shared-giant birth. Once this tested prefix and the subsequent side-1

birth count have been integrated, the remaining shared-giant and side-2 integral is

$$\sum_{r \leq J(I) < 2r} \nu(I) \int_{f_2 < h_2/L_2} d\mathcal{R}(P_2) \int \mathbf{1}_{E(h_2)}(P_2 I^{(2)}, A_2) d\mathcal{Q}_{h_2}(A_2) \leq K_2 r^{-\alpha}.$$

It is bounded by the single F mass because $L_2 \geq L$. The pre-giant side-1 factors are independent of this remaining integral: translating I to the endpoint of the side-1 prefix adds no weight and no choice. This yields relative charge $CK_2^2 L_2^{-\alpha}$. In the comparable-birth regime the smaller side-1 test is E , followed by its two birth counts and fresh charged jump, then exactly the same remaining integral. Its charge is $CK_1 K_2 L_2^{-\alpha} \log(2L_2)$. The giant has been charged once in all three regimes. The full side configurations are not independent; their only dependence left after peeling the pre-giant factors is precisely the one shared giant in the last integral. Repeating the argument on side 2 gives total relative bad-history cost

$$CL_2^{-\alpha} [K_3 + (K_2^2 + K_1 K_2) \log(2L_2)].$$

For regular histories keep measurable smaller-test envelopes before integrating failure probabilities. If both main births are very early, retain a paired test with heights $\lfloor s_i/2 \rfloor$. On each later-birth side retain instead $E(\lfloor s_i/(8L_2) \rfloor)$ and its subsequent prefix count. If one early side remains, keep its smaller single F test. If neither remains, charge the giant by its one-jump tail. Thus the total exposed envelope mass is at most

$$CL_2(K_3 + K_1 K_2 + K_1^2) r^{-\alpha}.$$

All retained worker test stops occur no later than the corresponding worker stops at s_i ; retained obstacle data belong to the exposed paths. Hence these envelopes are measurable before either tested worker future is integrated. Apply the conditional failure bound separately to the two independent worker futures and integrate by Tonelli; a union bound makes its cost at most δ times a fixed multiple of this display. This does not assert independence of the usability events or condition either future on continued survival.

On success each side has at least ch_i^α usable renewals. Their simultaneous choices number at least $ch_1^\alpha h_2^\alpha$, since there is no condition between the two side tests. In (18), sum both renewal occurrences before integrating. The shared giant remains a single factor $\nu(I)$; each worker prefix becomes its critical bridge measure and its unused future has mass at most one. The resulting integral is bounded by $X_{12}(h_1, h_2, r)$, so the paired hairpin bound makes the success contribution at most $Cr^{-\alpha}$. We have proved

$$\frac{F_{12}(h_1, h_2, r)}{r^{-\alpha}} \leq C + CL_2^{-\alpha} [K_3 + (K_2^2 + K_1 K_2) \log(2L_2)] + \delta CL_2 (K_3 + K_1 K_2 + K_1^2).$$

Choose $L_2 \geq L$ large, then δ small, and finally K_3 large. This closes the induction and proves (19). $\square$

6.5 Avoidance up to the first hit

The next application forgets the order condition in the definition of $E(h)$ and stops each path when it first visits a level, rather than at a renewal above that level. Define $S_2(h)$ as the probability that two independent infinite renewal walks from neighboring horizontal ports are disjoint up to their respective first visits to height h . The renewal stops may lie beyond this height. We handle those overshoots explicitly.

Corollary 6.5 (First-hit avoidance with logarithmic loss). *For every integer $h \geq 1$, the first-hit probability just defined satisfies*

$$S_2(h) \leq Ch^{-\alpha}(1 + \log h)^2. \quad (26)$$

Proof. Stop both renewal strings at their first renewals at or above $h/4$. If both stops are below h , disjoint first-hit arcs to h imply $E(\lfloor h/4 \rfloor)$. Indeed those renewal stops are single crossings, so later portions cannot revisit the tested lower region; the two first-hit crosscuts have their launch order throughout it. Equation (19) bounds this case.

Otherwise mark each jump born below $h/4$ that reaches or exceeds h . There are one or two such marked jumps, each of height at least $3h/4$. Bin their birth depths plus one into dyads f_i . Retain the ordered test at half the smallest birth dyad, or the trivial test at bounded depth. Its stops precede the corresponding births, and in the unmarked arm precede its stop below h , so they precede both first hits at h . The same crosscut argument makes this test valid. It costs $Cf_{\min}^{-\alpha}$. Each subsequent marked birth count costs Cf_i^α by restarting, and each ensuing jump costs $Ch^{-\alpha}$ by the irreducible tail. Thus one marked bin costs at most $Ch^{-\alpha}$, and a pair of marked bins costs at most

$$Cf_{\min}^{-\alpha}(f_1/h)^\alpha(f_2/h)^\alpha = Cf_{\max}^\alpha h^{-2\alpha} \leq Ch^{-\alpha}.$$

There are $O(1 + \log h)$ bins per marked jump. Summing proves (26). $\square$

Corollary 6.6 (Adjacent-arm avoidance). *For every $\epsilon > 0$, the probability $S_2(H)$ of disjoint first-hit paths from neighboring ports satisfies $S_2(H) \leq C_\epsilon H^{-3/4+\epsilon}$.*

Proof. Corollary 6.5 gives $S_2(H) \leq CH^{-3/4}(1 + \log H)^2$. For fixed $\epsilon > 0$, the logarithmic factor is bounded by $C_\epsilon H^\epsilon$. $\square$

The weaker power estimate is enough for the fixed number of turns in the next section. Appendix A gives its independent proof by extending asynchronous renewal pairs to a common height.

7 Amplification against fast travel

The fixed-height renewal theorem gives one polynomial failure power for a short bridge. We need every polynomial power in order to sum over the positions and outside pieces of arbitrary subpaths. The following argument uses a large but *fixed* number of turns, chosen after the desired failure exponent. Adjacent-arm avoidance pays for most turns; a long irreducible shared by the two ends is integrated once.

7.1 A length deficit and recoverable turns

For an event E of height- h bridges, write $B_h[E]$ for its unnormalized critical mass. A weak bridge is a path whose endpoints attain its extreme heights; intermediate contacts with their levels are allowed.

Lemma 7.1 (An integrated short-bridge estimate). *For every sufficiently small $\varepsilon > 0$, there are $\kappa, a_0 > 0$ such that, for each fixed $C < \infty$,*

$$\sum_{0 \leq j \leq Cr} B_j[|\gamma| \leq 2H^{4/3-\varepsilon}] \leq C'r^{3/4}H^{-a_0}, \quad H^{1-\kappa} \leq r \leq CH. \quad (27)$$

Proof. The polynomial lower-length failure bound in Theorem 5.2 gives, for some $a > 0$, $B_j[|\gamma| \leq j^{4/3-\varepsilon/2}] \leq CB_j j^{-a}$. Choose $\kappa > 0$ so small that $(1 - 2\kappa)(4/3 - \varepsilon/2) > 4/3 - \varepsilon$. For large H , if $j \geq H^{1-2\kappa}$, the length cap in Equation (27) is below that threshold. Such heights contribute at most $Cr^{3/4}H^{-a(1-2\kappa)}$. Smaller heights, including zero, have total mass $O(H^{3(1-2\kappa)/4})$, at most $Cr^{3/4}H^{-3\kappa/4}$ in the stated range. Taking $a_0 \leq \min\{a(1 - 2\kappa), 3\kappa/4\}$ proves the lemma. $\square$

Lemma 7.2 (Turning-extremum decomposition). *Fix $d \geq 1$ and put $k = 2^d$. A weak up bridge of height in $[H, 2H]$ admits an averaged decomposition into k weak up pieces, each of span in $[c_d H, 2H]$, separated by $k - 1$ retreats. If the retreat spans have dyadic scales s_ℓ , with scale 1 including empty and bounded retreats, the total level-choice weight of a fixed decomposition is at most $C_d \prod_\ell (s_\ell/H)$.*

Every piece can be recoded as a strict port bridge, with bounded weight change and finitely many local inverse records per cut. It gains only $O(1)$ length. At a turn whose retreat span grows as a positive power of H , the two departing arms start at neighboring ports and remain disjoint through first visits to any common distance $o(s_\ell)$ and $o(H)$ into their shared side.

Proof. Average a cut level m over lattice lines in the middle third of the bridge. If the line is crossed once, cut there and insert an empty retreat. Otherwise, after the first hit of m , let i be the last point attaining the minimum height of the remaining path, and let j be a point attaining the maximum height before i , with a fixed tie convention. The multiple crossing implies $i < m < j$, and j occurs after the first hit. The three pieces are the up bridge to j , the down retreat to i , and the remaining up bridge. They are weak bridges: the defining extrema bound all their intermediate heights. Repeat on both up pieces through depth d . Each up span loses at most a fixed factor per generation, so its final span is at least $c_d H$.

For a fixed final decomposition, the cut at an internal tree node must lie in its retreat's height interval, or at the single-crossing cut if the retreat is empty. Its number of choices is $O(s_\ell)$. The averaging denominator at that node is $\asymp H$, with constants depending on d . The tree order is fixed, so multiplying these bounds proves the asserted placement factor.

We describe the local recoding because neighboring starts are needed for Corollary 6.6. A nonempty peak is a down-pointing triangle, two thirds up its band, visited via its two lower neighbors. Assign that triangle to the retreat, entering through its top port. The reversed up piece can start at the neighboring top port on its branch side and pass through the adjacent down-pointing triangle to the same lower neighbor of the peak. If it already used that triangle, the visit must be immediately next to this lower neighbor: it cannot cross above its extremum, and the piece continues macroscopically below. In that case start directly there and remove the short peak portion. The adjacent triangle cannot be in the relevant initial retreat arc, because any such visit would use that same lower neighbor, contradicting the original disjointness.

Reflect this construction at minima. If an original endpoint is a vertex, extend it to the nearest exterior band line by the bounded construction in the proof of Lemma 2.1. The added vertices lie outside the old height range; the extension changes length and height by $O(1)$, has bounded multiplicity, and changes weight by a bounded factor. Original port endpoints and single-crossing port cuts already lie on boundary lines. These local operations produce strict port bridges within the outer band lines of their extrema. At a macroscopic turn, modifications at remote ends are outside the first-hit portions of distance $o(s_\ell)$ and $o(H)$; hence those neighboring arms remain disjoint. A minimum attained again at the starting height is still a local minimum between upper neighbors, so the same construction applies there. Each cut has only finitely many possible local records and weight ratios. Relative shapes and these records recover the original pieces, their translations being fixed by concatenation up to bounded lattice choices. In upper bounds all ending heights may therefore be summed freely over their indicated scale ranges. $\square$

7.2 Integrating stopped and terminally marked ends

The local path modification has made the two arms at a turn start from neighboring ports. Their endpoint laws still differ: one arm may stop at first passage, while the other is summed over a terminal renewal. The next lemma isolates that change of measure before any avoidance factors are multiplied.

Lemma 7.3 (Avoidance with marked terminal renewals). *Consider two independent renewal end strings from neighboring ports. For either string use one of the following measures: its probability law stopped at the first renewal above q , or its bridge measure summed over a marked terminal renewal with $f \leq 1 + U < 2f$, where $f \geq 1$. Call the first kind good and the second kind marked. A good end has mass scale 1 and probe scale q ; a marked end has mass scale $f^{3/4}$ and probe scale f . Requiring the two strings to avoid through their first visits to a small fixed multiple of the smaller probe scale costs at most $S_2(t)$, times a constant and the two mass scales. Here t is that small multiple, rounded to an integer; bounded scales use the bound 1. The assertion remains valid at all turns of a chain simultaneously when different turns use distinct end strings and all other incidence constraints are dropped.*

Proof. For unbounded scales choose t less than, say, one tenth of both probe scales, reducing the constant to handle fixed scale conventions. Expose each infinite end walk through its first renewal at or above t . The first-visit avoidance event is measurable with respect to these two stopped strings. A good string has conditional remaining mass at most 1. For a marked string, condition on its exposed endpoint height a . If $a \geq 2f$ there is no admissible marker, except for harmless bounded boundary cases. Otherwise its conditional expected number of possible marked renewals is at most

$$1 + \sum_{j \leq 2f} B_j \leq C f^{3/4}.$$

This includes the stopping renewal itself and is uniform in overshoot. The infinite renewal law up to a marked renewal has exactly the unnormalized bridge weights, so the argument bounds the required finite-string measure. After multiplying the two conditional bounds, the remaining event has probability $S_2(t)$. This proves all good/marked combinations without conditioning a terminal height in advance.

For simultaneous turns, first express every end in relative coordinates. The local records at a turn fix its neighboring starting ports up to a bounded choice. Once middle bridges and large jumps have been summed, discard consistency of their translations and heights with these relative strings. This enlarges the set being counted. Each end then appears in exactly one turn, so the pairs of independent string variables used by different avoidance tests are disjoint. The preceding conditional integration can be performed at every turn, and its factors multiply. In particular, no avoidance estimate is being applied to a string already biased by a different turn. $\square$

7.3 The gain from many turns

We now combine the short-bridge deficit with adjacent-arm avoidance. The partition into disjoint atomic end strings is part of the proof: it is what permits multiplication of turn costs without conditioning one test on the success of another.

Theorem 7.4 (Fast bridges). *For every $\varepsilon > 0$ and every $A < \infty$,*

$$\sum_{H \leq h \leq 2H} B_h[|\gamma| \leq H^{4/3-\varepsilon}] \leq C_{\varepsilon, A} H^{-A}. \quad (28)$$

Proof. It suffices to take small $\varepsilon > 0$. Use Lemma 7.2 with $k = 2^d$, where d is fixed now and chosen large at the end. Local recodings increase the total length by $O(k)$, so every resulting portion has length at most $2H^{4/3-\varepsilon}$ for large H . Let κ, a_0 come from Lemma 7.1, and choose

$$0 < \Delta < \min\{1/40, \kappa/10, a_0/10\}.$$

An up piece has scale $r = H$, while a retreat has scale $r = s_\ell$. The unrestricted summed-height mass of either is $O(r^{3/4})$. Retreats with $s_\ell \leq H^{1/20}$ will use only this bound.

For any other piece inspect from both ends its first renewal beyond $q = \lfloor rH^{-\Delta} \rfloor$, a threshold less than half its span. The irreducible decomposition is unchanged by reversal. If the two stopped strings do not overlap, allowing a shared cut level, call the piece type M. Sum its two end strings separately and give its middle bridge the full mass bound $Cr^{3/4}$. If $r \geq H^{1-\kappa}$, the middle also has the length cap, so Equation (27) supplies the extra factor H^{-a_0} , regardless of its actual remaining span.

An M end is good, denoted G, if its stopping jump lands within distance cr of that end, for a small fixed $c > 0$. Ignore this extra constraint and use its stopped renewal probability law. Otherwise the end is bad. Group 1+the height before its stopping jump in a dyadic bin of scale $f \leq Cq$. Its preceding marked-renewal prefix has mass $O(f^{3/4})$, and its next independent jump has height at least $cr/2$, with probability $O(r^{-3/4})$. Thus the bad end costs $C(f/r)^{3/4}$ in addition to the middle's baseline.

If the end strings overlap, call the piece type Z. They contain the same irreducible jump between the two near-end ranges. There cannot be two such jumps, since the ranges are disjoint and the irreducibles are linearly ordered in height. Mark both ends bad. Summing their preceding prefixes and this single shared jump gives

$$Cf_1^{3/4}f_2^{3/4}r^{-3/4} = Cr^{3/4}(f_1/r)^{3/4}(f_2/r)^{3/4}. \quad (29)$$

The shared jump has height comparable to r . There is no length gain for this piece, and its tail is charged only once.

For each turn of a retreat with $s > H^{1/20}$, impose adjacent-arm avoidance at a small multiple of the minimum probe scale: q at a good end, f at a bad end. These distances are $o(s)$ and $o(H)$, as required by the local recoding. When the minimum is bounded use the constant bound instead. By Lemma 7.3, the marked terminal renewal sum after exposing this avoidance still costs at most $Cf^{3/4}$. Large-jump tail factors follow these prefix portions and do not consume them. In the Z case the one shared jump has already been integrated as in Equation (29). The remaining end variables are distinct at distinct turns, so their avoidance gains multiply. Corollary 6.6 therefore supplies the minimum probe scale to power $-3/4 + o(1)$ at every such turn.

Each turn adjoins an up piece of scale H to a retreat of scale s . Combine its avoidance factor with its attached bad-end costs. Apart from $H^{o(1)}$, it supplies $s^{-3/4}$ and the following additional factors:

end types	factor after extracting $s^{-3/4}$
G, G	$H^{3\Delta/4}$
one bad	1
both bad	$H^{-3\Delta/4}$.

For two good ends, the smaller threshold is comparable to $sH^{-\Delta}$. With one bad end of scale r , if its probe f is smaller, its $f^{3/4}$ cost cancels the avoidance factor, leaving $r^{-3/4} \leq Cs^{-3/4}$. If the good probe $r'H^{-\Delta}$ is smaller, use $f/r \leq CH^{-\Delta}$ and $s \leq Cr'$. With two bad ends, writing their probes as f_1, f_2 , the factor is exactly

$$(f_1/H)^{3/4}(f_2/s)^{3/4} \min(f_1, f_2)^{-3/4} = s^{-3/4}(\max(f_1, f_2)/H)^{3/4},$$

which is at most $Cs^{-3/4}H^{-3\Delta/4}$.

Nominally extract $s_\ell^{-3/2}$ from the two turns of every retreat. Multiplying all baseline masses and placement factors gives

$$H^{3k/4} \prod_{\ell=1}^{k-1} \left[s_\ell^{3/4} \frac{s_\ell}{H} s_\ell^{-3/2} \right] = H^{3/4} \prod_{\ell=1}^{k-1} (s_\ell/H)^{1/4}. \quad (30)$$

For an untreated retreat $s \leq H^{1/20}$, restore the nominal factor $s^{3/2}$. Its net factor is at most $H^{-1/4}s^{7/4} \leq H^{-13/80}$. A treated retreat with $s < H^{1-\kappa}$ pays at least $H^{-\kappa/4}$ through its scale factor in Equation (30).

Every both-good loss can be charged to its incident M up piece, since a Z piece has both ends bad. Each up piece has at most two turns, so its own H^{-a_0} gain leaves at least $H^{-a_0/2}$, by the choice of Δ . Assign each Z up piece to one incident retreat; an endpoint up piece still has one such retreat because $k \geq 2$. A retreat receives at most two assignments. If its scale is below $H^{1-\kappa}$, its scale saving pays those assignments. If it is larger and type M, its own H^{-a_0} length gain pays them. If it is larger and type Z, the assigned turn is both bad and pays $H^{-3\Delta/4}$. Thus there is a constant $b = b(\varepsilon) > 0$, independent of fixed k , such that all up pieces together save H^{-bk} . For example one can choose any sufficiently small positive number below $\min\{a_0/2, \kappa/8, 13/160, 3\Delta/8\}$.

For fixed k , there are finitely many type choices and only logarithmically many dyadic scales per piece. All avoidance losses are $H^{o(1)}$ for this fixed number of tests. The total mass is consequently at most $C_k H^{3/4-bk+o(1)}$. Taking a fixed power of two k sufficiently large in terms of A , and then enlarging the constant for bounded H , proves Equation (28). $\square$

8 Length in small regions and repeated crossings

The fast-path estimate controls the distance a subwalk can travel in a short time. We need the converse control: a subwalk cannot accumulate much more than the $4/3$ power of its spatial span. This alone does not bound all visits to a ball, since a walk may leave and return. After proving the subwalk bound, we control the number of such returns by joining disjoint crossings into cylinder polygons.

Throughout this section, $\alpha = 3/4$, heights are measured in band units, and B_j denotes the total critical mass of strict port bridges of height j , with $B_0 = 1$. We use

$$B_j \leq C(1+j)^{\alpha-1}, \quad \sum_{j \leq R} B_j \asymp (1+R)^\alpha,$$

the irreducible-renewal probability law, the strip first-length moment, and the full-box susceptibility bound established above. A bridge is *weak* if its endpoints attain its extreme heights; bounded extensions and shortenings turn it into a strict port bridge. All such operations below have bounded inverse multiplicity and alter critical weights by bounded factors.

For a fixed lattice vertex o and fixed $C_0 < \infty$, write

$$\mu_H(\mathcal{A}) = \sum_{\substack{\gamma \text{ starts at } o \\ \text{diam}(\gamma) \leq C_0 H}} \rho^{|\gamma|} \mathbf{1}_{\mathcal{A}}(\gamma), \quad H \geq 2.$$

Here $|\gamma|$ counts edges, and vertex-subwalk lengths also count edges. Port recodings change these counts and weights by bounded factors. This is a finite, unnormalized measure. Its total mass is polynomial in H by the full-box bound. Simplicity also gives $|\gamma| \leq CH^2$ throughout its support. Constants may depend on C_0 . A bound described below as superpolynomial means that for every fixed $A > 0$ its mass is at most $C_A H^{-A}$, uniformly in $H \geq 2$.

Theorem 8.1 (Uniform local length bound). *For every $\tau > 0$ and $A > 0$ there is $C_{\tau,A} < \infty$ such that, for every $H \geq 2$, outside a set of μ_H -mass at most $C_{\tau,A}H^{-A}$, every vertex subwalk σ satisfies*

$$|\sigma| \leq H^\tau (1 + s(\sigma))^{4/3}, \quad s(\sigma) = \text{its vertical span in band units.} \quad (31)$$

The same conclusion holds for each of the finitely many lattice normal directions simultaneously.

The proof uses repeated cuts at turning extrema. A long path in a small height range forces either a long chain of comparable pieces or many separate pieces whose lengths are already large. Ordered renewal probes make either configuration rare. We first handle two estimates needed when a probe returns unusually close to its own start or a piece has very small height.

8.1 Returns inside one irreducible bridge

Lemma 8.2 (Small returns). *For every $c > 0$ and $P > 0$ there are $D < \infty$ and $C < \infty$ such that if I is sampled from the irreducible-renewal law and J is its height, then for every $r \geq 2$,*

$$\mathbb{P} \left\{ \begin{array}{l} J \in [r, 2r), \text{ } I \text{ returns to depth at most } r(1 + \log r)^{-D} \\ \text{after first reaching depth } cr \end{array} \right\} \leq Cr^{-\alpha}(1 + \log r)^{-P}. \quad (32)$$

The statement holds in either orientation of I .

Proof. Decreasing c only enlarges the event, so assume that c is a small fixed constant. The middle-third extremal decomposition used for (28), now at the line of height $\lfloor cr \rfloor$, gives an initial rising bridge A_1 , a retreat T , and a final rising bridge A_2 . Place the initial endpoint at height zero. Let y denote the height of the last minimum after the first hit of this line, and let z denote the height of a chosen maximum before that minimum, with a fixed tie rule for its location along the path. Before recoding, their heights are z , $z - y$, and $J - y$. For large r , the return assumption gives $y = o(r)$ while $z \geq cr + O(1)$ and $r \leq J < 2r$, so all three spans are comparable to r . After recoding at the two turns, write their heights as m_1, m_T, m_2 . Choose dyads ar and br for $1 + y$ and $1 + J - z$, respectively. Thus $1/r \leq a, b \leq C$, $a \leq C(1 + \log r)^{-D}$, and

$$|m_1 - m_T| \leq Car, \quad |m_2 - m_T| \leq Cbr.$$

The first irreducible of A_1 has height at least a constant times ar , and the last of A_2 has height at least a constant times br . A single-crossing seam below y , or above z , would otherwise be a seam of I itself: the bounded changes at the remote turn lie on the far side of that seam. This would contradict irreducibility.

At each turn inspect the incident end strings to their first renewals at depth at least q , where $q \asymp r(1 + \log r)^{-Q}$, with Q fixed. Separate the required outer irreducible at each unprobed end of A_1, A_2 . For an outer bridge, type M means that this outer jump and the string inspected from the turn do not overlap; type Z means that they share a jump, which must be the required outer jump. For T , inspect from both ends: type M means the two stopped strings are disjoint, allowing a common cut, and type Z means they share one irreducible. The three baseline masses are

$$r^\alpha(ar)^{-\alpha}, \quad r^\alpha, \quad r^\alpha(br)^{-\alpha}.$$

Each M piece leaves an independent residual bridge with freely summed height at most Cr , supplying the factor r^α in its baseline.

An inspected end is good, denoted G, if its stopping renewal lands below $c'r$, where c' is sufficiently small depending on c . Otherwise it is bad. A type Z retreat has two bad ends: the two

birth depths are below $q = o(r)$, so the jump they share spans a fixed proportion of the retreat. If f is the dyadic scale of one plus the birth depth of its stopping jump, the prefix mass is at most Cf^α and the jump has mass at most $Cr^{-\alpha}$. Consequently a bad end costs $(f/r)^\alpha$ beyond baseline. There is only one jump variable when the two inspections overlap. For a type Z retreat its actual mass factor is

$$f_1^\alpha f_2^\alpha r^{-\alpha} = r^\alpha (f_1/r)^\alpha (f_2/r)^\alpha.$$

For the first outer piece of type Z, the shared jump is macroscopic and its mass factor is instead

$$f^\alpha r^{-\alpha} = [r^\alpha (ar)^{-\alpha}] (f/r)^\alpha a^\alpha.$$

Thus the first outer piece has an extra factor a^α beyond its baseline and bad-end factor; the second has the corresponding b^α . These equalities allocate the weight of one shared jump, without replacing it by independent copies. The same additional a^α is available when A_1 is M and its required first jump has height at least $c'r$.

Use the logarithmic first-hit estimate (26) at each turn, at a sufficiently small fraction of the smaller available depth: q on a good end and f on a bad end. At bounded depth use the trivial bound instead. The test precedes every counted bad jump; after its stopping renewal, the remaining prefix occurrences still cost at most Cf^α by the renewal property. The two end prefixes at a turn are distinct, and the recoding preserves their first-hit disjointness. The two turn tests therefore give $r^{-\alpha}$ each, with the following additional factors: $(r/q)^\alpha$ for two good ends, a constant for one good and one bad end, and $(q/r)^\alpha$ for two bad ends. All other logarithmic losses, including birth dyads and the choices of a, b , are bounded by $(1 + \log r)^{C_1}$, where C_1 does not depend on D, Q .

It remains to use the height constraints without conditioning the probe laws. Put

$$K_r(j) = r^{-\alpha} B_j \mathbf{1}_{\{0 \leq j \leq Cr\}}.$$

Uniformly under arbitrary translations, one kernel restricted to an interval of width $w \geq 1$ has mass at most $C(w/r)^\alpha$. For $m = 2, 3$, Hölder's inequality gives

$$\sum_j \prod_{\nu=1}^m K_r(j + u_\nu) \leq \prod_{\nu=1}^m \left(\sum_j K_r(j + u_\nu)^m \right)^{1/m} \leq Cr^{1-m}.$$

Here $m(\alpha - 1) > -1$, so the last estimate follows by summing the power kernel. Summing the allowed height differences now shows that two residual kernels matching within Car cost Ca , and three kernels with the two displayed height constraints cost Cab . These bounds hold after every outer jump and every probe has been fixed. If a residual height is itself at least $c''r$, its normalized kernel is at most C/r , so restriction to width Car costs Ca .

We apply these estimates in the following order. Fix all end strings and distinct jump shapes, and first sum the residual bridge heights with their translated restrictions. The kernel estimates just proved are uniform in the fixed data. Then integrate every distinct separated or shared jump once. Finally apply the first-hit estimates to the remaining pre-jump end strings. Different turns use distinct such strings. Their untested continuations contribute the uniform renewal occurrence bounds already stated. This order proves simultaneous gains without conditioning a probe law on a completed bridge height.

After the turn factors the baseline is $r^{-\alpha}(ab)^{-\alpha}$. It remains to gain either a positive power of the small return scale a or a positive power of q/r . In the all-M case,

$$(ab)^{-\alpha} ab = a^{1-\alpha} b^{1-\alpha} \leq Ca^{1/4}.$$

The other cases distribute the same kernel restrictions differently. If T and both outer pieces are M, the triple-kernel bound leaves $a^{1-\alpha}b^{1-\alpha}$. If T, A_1 are M and A_2 is Z, use the two-kernel bound a and its extra b^α . If T is M and A_1 is Z, use its extra a^α , restrict T to width Car , and obtain b^α from either a single restriction on A_2 or its Z improvement. In these three cases the remaining factor is bounded by Ca^κ , where $\kappa = \min(\alpha, 1 - \alpha)$, before possible good-good losses.

If T is Z, neither turn has a good-good loss. Obtain b^α from A_2 as above. When A_1 is Z, its extra a^α comes from the macroscopic shared-jump tail. When A_1 is M with a bad turn probe, fix the Z retreat's height and use $|m_1 - m_T| \leq Car$ to restrict the free residual kernel of A_1 , obtaining a^α . In both cases the tail and restriction factors cancel $(ab)^{-\alpha}$; the remaining first-turn factor is explicitly $(q/r)^\alpha$. When A_1 has a good turn probe and a macroscopic first jump, the tail improvement and the single residual restriction give $a^{2\alpha}$. In the remaining case its first jump and good turn string both have height below $c'r$, whereas A_1 has macroscopic height. Its residual height is therefore macroscopic; the sharper single-kernel restriction gives a . Thus, with $\kappa = \min(\alpha, 1 - \alpha) = 1/4$, the total mass is at most

$$Cr^{-\alpha}(1 + \log r)^{C_1} \left[(1 + \log r)^{-D\kappa + 2Q\alpha} + (1 + \log r)^{-Q\alpha} \right].$$

Choose Q first so that $Q\alpha > P + C_1$, and then D so that $D\kappa > P + C_1 + 2Q\alpha$. Bounded scales are absorbed by increasing C . $\square$

The arbitrary logarithmic gain in this lemma will pay for early-birth dyads at the many turns of a long chain. Without it, a power of $\log H$ at every turn would overwhelm the later height-matching gain.

8.2 Thin excursions

Lemma 8.3 (Thin excursions). *For every $\lambda > 0$ and $A > 0$, there is $C_{\lambda,A} < \infty$ such that*

$$\mu_H \left\{ \begin{array}{l} \text{some subarc has vertical span at most } H^{\lambda/4} \\ \text{and length greater than } H^\lambda \end{array} \right\} \leq C_{\lambda,A} H^{-A}, \quad H \geq 2.$$

Subarcs may have vertex or horizontal-port endpoints, with bounded endpoint changes in their lengths and spans.

Proof. If such an arc exists, simplicity and the lattice area bound force horizontal range at least $cH^{3\lambda/4}$. One of the oblique lattice normal projections consequently has range $R \geq cH^{3\lambda/4}$. Between suitable extreme vertices of that projection take a weak bridge, and extend its ends at bounded cost. Its height is comparable to R and its vertical range is still $O(H^{\lambda/4})$. This is confinement in the direction tangent to an oblique cut, rather than confinement in the oblique height.

Use the turning-extremum amplification in the proof of Theorem 7.4, replacing its length cap by this confinement. That argument turns a fixed power saving for every free M residual bridge into arbitrary polynomial decay by increasing a fixed iteration depth. We verify the only changed input, namely suppression of a free M residual bridge. Let $V = O(H^{\lambda/4}) = O(R^{1/3})$, and suppose $R^{1-\kappa} \leq s \leq CR$, for a small fixed $\kappa > 0$. Residual heights $j \leq R^{1-2\kappa}$ contribute at most $CR^{\alpha(1-2\kappa)}$. For larger heights, the permitted vertical endpoint difference gives at most $C(1+V)$ lateral endpoint offsets at each height, since the tangent to the oblique cut has a nonzero vertical component. If b_j is the endpoint kernel, the variation bound therefore gives

$$\sum_{R^{1-2\kappa} < j \leq Cs} \sum_{\text{permitted offsets}} b_j \leq C(1+V) \sum_{j \geq R^{1-2\kappa}} \|b_j\|_\infty \leq C(1+V) B_{\lfloor R^{1-2\kappa} \rfloor} \leq CR^{1/12+\kappa/2}.$$

Both contributions are $O(s^\alpha R^{-a_0})$ for sufficiently small $\kappa, a_0 > 0$. Every other factor in the turning-extrema amplification is unchanged, so its arbitrarily large fixed iteration depth gives mass $O_A(R^{-A})$. Sum over dyadic R , placements, and the two exterior walk portions. Their cost is a fixed polynomial in H by the full-box bound; since R is a positive power of H , it is absorbed by choosing the preceding exponent large enough. Trimming port endpoints costs only a bounded factor. $\square$

8.3 Turning chains and simultaneous integration

We give the chain estimate with its probability bookkeeping, since the number of turns will grow with H . Pieces in the following trees may have vertex or horizontal-port endpoints. Their *additive length* counts a full edge as one and a half-edge as one half, so lengths add exactly at every cut. This agrees with the vertex-subwalk convention and with the visited-vertex count for a port-to-port path. Fix a small $\epsilon > 0$, set $\beta = \epsilon/4$, and call a weak bridge of span s *long* when its additive length exceeds $H^\epsilon(1+s)^{4/3}$. Fix a large integer p and a large constant U , to be chosen in that order. For every eligible weak-bridge subarc of span $s \geq H^\beta$, form a random trial tree as follows. At an expanded node choose a lattice line uniformly from the middle third of its height interval. A single crossing gives two forward children and an empty retreat, retained as a zero-length child. Otherwise choose the last minimum after the first hit of the line and a maximum before that minimum, with fixed tie rules, obtaining the three chronological weak bridges. Reverse or reflect coordinates when necessary. Expand each child by the same rule, stopping when that child has span less than the fixed threshold s/U , or at depth $L = \lceil \log H \rceil$. Here s remains the original root span throughout the tree. Choices at distinct expanded nodes are fresh.

There are two kinds of witnesses. A *spine witness* retains L expanded nodes along one descent. A *marked witness* is the minimal subtree down to p incomparable long nodes, which are retained as marked leaves. In either case retain every unused child as a leaf, so all expanded vertices have exactly three chronological children. If k is the number of expansions, then $k = L$ for a spine and $k \leq pL$ for a marked witness. The ordered leaves form an alternating chain.

At a node with exactly one retained expanded child, call that child *continuing*; its two siblings are leaves. If the parent span is h and the three chronological child spans are a, d, c , then $h = a - d + c$ and each child span is at most h . A small decrease from h to the continuing span forces its two sibling spans to be close, as Figure 9 shows. The witness estimate will use this constraint to obtain a small factor at many expansions.

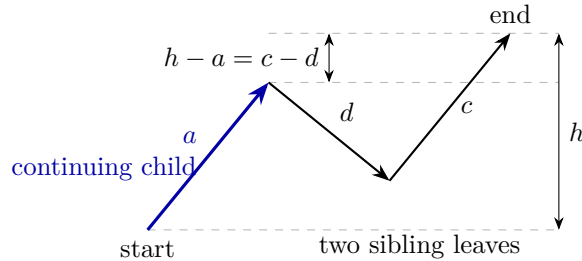


Figure 9: A unary expansion with the first child continuing. The parent and child heights obey $h = a - d + c$, so a small continuing-height drop forces the two sibling leaves to have nearly equal heights. If the retreat continues instead, $a, c \in [d, h]$ and $|a - c| \leq h - d$. Only heights are represented: the horizontal spacing and straight segments are schematic.

For a fixed path γ , let K_γ be the probability law of all these independent trial choices, over every eligible root subarc and every potential descendant node. This is a finite product: the path

has at most CH^2 steps and all trees have depth at most L . For a trial outcome ω , let $N_{r,k}^t(\gamma, \omega)$ count the witnesses of kind $t \in \{\text{spine}, \text{mark}\}$ with k expansions and root span in $[r, 2r)$. Different root subarcs and different retained subtrees are counted separately. The weighted witness mass is therefore

$$\int N_{r,k}^t(\gamma, \omega) K_\gamma(d\omega) \mu_H(d\gamma).$$

In particular, a bound on this integral bounds the mass of path-and-trial pairs having at least one such witness.

Why these witnesses suffice. Suppose that the thin-excursion exception of Lemma 8.3, with $\lambda = \epsilon$, is absent, and that a trial outcome contains neither kind of witness, for any eligible root. If

$$pU^{-4/3} < 1/4,$$

then, for all sufficiently large H , every weak-bridge subarc σ of span s satisfies

$$|\sigma| \leq H^{2\epsilon}(1+s)^{4/3}. \quad (33)$$

This implication is deterministic; it holds for every outcome of the trial choices.

Prove it by induction on three times the band span, an integer for both vertex and horizontal-port endpoints. Below H^β the thin bound gives length at most H^ϵ . For a larger root span s , take its trial tree and prune at each non-long piece. At any depth fewer than p long pieces remain, since they are incomparable and would otherwise give a marked witness. Thus at most $1 + 3pL$ non-long pieces stop the pruning. Every long terminal piece has span below s/U : a terminal at depth L would give a spine. These long terminals are also incomparable, so there are fewer than p of them. Their spans are strictly smaller than s , and the induction hypothesis and exact length additivity give

$$|\sigma| \leq (1 + 3pL)H^\epsilon(1+s)^{4/3} + pH^{2\epsilon}(1+s/U)^{4/3}.$$

After division by the right-hand side of (33), the coefficient is

$$(1 + 3pL)H^{-\epsilon} + p \left(\frac{1 + s/U}{1 + s} \right)^{4/3} < 1$$

for large H , uniformly over $s \geq H^\beta$. This proves the induction. It remains to show that the weighted mass of the two witness classes is small.

Lemma 8.4 (Turning-chain witness bound). *There are absolute $b > 0$ and $C < \infty$ such that, for fixed ϵ, p, U and a root-span dyad $s \in [r, 2r)$ with $H^\beta/2 \leq r \leq C_0H$, the weighted witness mass just defined is at most*

$$H^{C+o_{p,U,\epsilon}(1)} H^{-\epsilon m_*} [C'(1 + \log \log H)^{C'}]^k (k+1)^{-bk}. \quad (34)$$

There $m_* = 0$ for a spine and $m_* = p$ for a marked witness. The exponent C is independent of p, U, ϵ ; the constant C' may depend on them. The $o(1)$ is uniform in the indicated r, k , and denotes a quantity tending to zero for those fixed parameters. The same bound, with $m_* = 0$ and k equal to the number of pieces, holds for an alternating chain of $1 \leq k \leq pL$ pieces of nonincreasing spans in $[r, 2r)$, where $L = \lceil \log H \rceil$ and p remains fixed. For this last assertion the mass is the analogous sum over all rooted chain records, without trial probabilities.

Proof. Cuts and bridge weights. We first record the cuts and relative bridge shapes. The calculation then separates end tests from the remaining height variables. The height restrictions are summed by bounds uniform in the end data, so the tests can be integrated with their original renewal laws.

Every outer child of an expansion has span at least one third of its parent's span. Choose $0 < c_0 \leq 1/(12U)$. Every expanded parent has span at least $s/U \geq r/U$, so all leaves have spans between $c_0 r$ and $2r$, except possibly retreat leaves. There are only $O_{p,U}(1)$ small retreat leaves, and they are isolated between macroscopic leaves. To see the bound, the minimal witness has $O(p)$ branching or terminal expansions and otherwise consists of $O(p)$ unary chains. If a parent has height h and a small retreat has height $d < c_0 r$, the middle-third cut gives both forward heights at most $2h/3 + d \leq 3h/4$. The retreat itself has height below s/U and cannot continue. Along a continuing chain heights never increase and stay at least s/U until the last expansion; there can consequently be only $O_U(1)$ such reductions on each chain. The first and last leaves of an expanded subtree descend through outer children and are macroscopic, so small retreats are isolated in the chronological leaf order.

Bin each small span plus one dyadically with scale r_i , and put $r_i = r$ on macroscopic leaves. Fix the complete leaf pieces and all cut records. At the parent of a small retreat, the chosen middle-third line must belong to that retreat's height interval (one specified line when it is empty); there are $O(r_i)$ possible choices out of at least cr . Each small leaf therefore contributes Cr_i/r to the probability of these witness data. All other line-choice probabilities sum to at most one. This uses only choices recorded by the witness, without specifying any descendants outside it.

At a turn whose incident spans are both larger than $Cr^{1/20}$, recode its incident leaves to strict bridges from neighboring ports, preserving disjoint first-hit portions to distance $c_A \min(r_i, r_{i+1})$. Here c_A is small depending on c_0 . For completeness, at a peak both lower neighbors of its triangle occur on the path. One arm uses the central top port. Launch the other from the adjacent top port through the neighboring downward triangle to its original lower neighbor. If that neighboring triangle already occurs on this arm, it must occur immediately next to that lower neighbor, so shorten the initial segment instead. An internal visit would have to use the same lower neighbor. The other arm cannot use this triangle in the retained first-hit portion for the same reason. Reflect this operation at minima. Remote-end changes lie beyond the retained first-hit portions. Where no turn test is required, extend to bounding port lines; if an extreme triangle does not border that line, one extra triangle nearer the line is unused by extremality. Empty leaves have bounded bookkeeping. This gives a finite record per cut and changes lengths and spans by $O(1)$.

After fixing those records, normalized bridge shapes have their translations fixed successively by incidence. There is no additional sum over horizontal positions at each turn. A turn test involves only two relative shapes launched at fixed neighboring ports; changing absolute translations does not change the test. The root location, the two exterior portions, and endpoints cost $O(H^C)$ altogether by the full-box bound, with C independent of the number of leaves.

Baseline leaf i by r_i^α and initially seek $\min(r_i, r_{i+1})^{-\alpha}$ at every turn. Because small leaves are isolated, multiplying the baselines, these turn factors, and the placement probabilities gives at most

$$C^k r^\alpha \prod_{i \text{ small}} (r_i/r)^{1-\alpha}.$$

If $r_i \leq Cr^{1/20}$, omit both adjacent turn tests. Restoring their nominal factors multiplies this expression by at most $Cr_i^{2\alpha}$. The resulting contribution from that leaf is bounded, since

$$(r_i/r)^{1-\alpha} r_i^{2\alpha} \leq Cr^{-(1-\alpha)+(1+\alpha)/20}.$$

No probes are needed at its own ends. Call turns touching small or marked leaves *rough*; there are $O_{p,U}(1)$ of these. All other turns are *ordinary*.

End probes at ordinary turns. The factors at turns will cancel all but one of the bridge mass scales. Their dependence on k must remain explicit, because k can be of order $\log H$.

For a non-tiny leaf inspect its two strings of irreducibles through the first renewals reaching

$$q_i = \lfloor c_q r_i (k+1)^{-\delta} \rfloor,$$

with a sufficiently small fixed $\delta > 0$. The coefficient c_q is small, and common on macroscopic leaves. For large H these are positive. The strings are of type M when disjoint, allowing a common terminal cut, and then leave a free middle bridge. Otherwise they share exactly one irreducible and have type Z. Call an end good when its stop lands below $c' r_i$, and bad otherwise, with $c_q \ll c' \ll c_A c_0$. In type Z both ends are bad. A bad string is a prefix with endpoint depth below q_i , followed by a jump of height comparable to r_i . For a birth dyad f , its prefix cost is $C f^\alpha$ and its jump cost is $C r_i^{-\alpha}$. On an M leaf the residual middle supplies baseline r_i^α ; on a Z leaf its single shared jump supplies $r_i^\alpha (r_i^{-\alpha})^2 = r_i^{-\alpha}$. This equality is only bookkeeping: every distinct shared jump is integrated once. For example, among $n_0 = m_0 + z_0$ macroscopic pieces with b_0 bad ends there are $d_{\text{bad}} = b_0 - z_0$ distinct bad jumps, and

$$r^{\alpha m_0} r^{-\alpha b_0} = r^{\alpha m_0} r^{-\alpha d_{\text{bad}}}.$$

The right-hand side records the genuine independent middle and jump variables; the left-hand side assigns their costs to leaves and turns.

At an ordinary turn the desired estimates, beyond these leaf baselines, are

incident ends	turn factor
GG	$C r^{-\alpha} (k+1)^{\delta\alpha}$
one good, one bad	$C r^{-\alpha}$
two bad	$C r^{-\alpha} (k+1)^{-\delta\alpha}$.

(35)

The total additional loss over all ordinary turns will be at most $C^k (1 + \log \log H)^{Ck}$.

First fix the birth dyads. Apply the ordered test E from (19) at a small fraction of the minimum of the available depths, using q_i at a good end and f at a bad end. At bounded minimum use the trivial test. Its stopping renewals occur before the first-hit portions terminate: on a bad end they precede the specified birth, and a good stop is below $c' r$. Renewal cuts prevent later visits below the tested level. Moreover two disjoint crosscuts from the neighboring launches to the common first-hit line have the required left-right extremal order at each lower level; a point past the opposite extremum would lie on the wrong exterior side of that crosscut. Hence E is a necessary condition. After these stops the remaining birth counts are at most $C f^\alpha$, including the possibility that the birth occurs at the stop itself. The three costs for fixed dyads are respectively

$$C q^{-\alpha}, \quad C r^{-\alpha} f^\alpha \min(f, q)^{-\alpha}, \quad C r^{-2\alpha} f^\alpha g^\alpha \min(f, g)^{-\alpha}.$$

Since $f, g \leq C q$ these give the table; the last expression is more precisely $C r^{-\alpha} (\max(f, g)/r)^\alpha$.

Summing birth depths. We must sum the birth dyads without a power of $\log H$ at every turn. Set

$$t = \lfloor r(1 + \log H)^{-D_*} \rfloor, \quad T_0 = r(1 + \log H)^{-10},$$

where $D_* > 12$ is a sufficiently large fixed constant. Tag an unmarked macroscopic bad jump if, from either of its relevant ends, it returns to depth at most $2t$ after reaching $c'' r$, where $c'' > 0$ is small. By Lemma 8.2, choosing D_* larger if necessary, the mass of tagged jumps is

$$C r^{-\alpha} (1 + \log H)^{-5}.$$

The constants are uniform for $H^\beta/2 \leq r \leq C_0 H$, since $\log r \asymp_\beta \log H$. The union over two ends costs only a constant, and no designated-jump test below will be assigned to a tagged jump.

If both bad birth depths are at most T_0 , sum their dyads using the improved maximum factor. Summing over the larger dyad first gives

$$\sum_{f,g \leq CT_0} \left(\frac{\max(f,g)}{r} \right)^\alpha \leq C(1 + \log H)(T_0/r)^\alpha \leq C(k+1)^{-\delta\alpha},$$

for small δ , since $k \leq pL$. Birth dyads between t/L_2 and Cq have only $O(1 + \log \log H)$ possible values. Here L_2 is the fixed paired-test constant in (19), unrelated to the trial depth L . The remaining situation has a birth below t/L_2 opposite a good end or a bad birth larger than T_0 . If its jump is tagged, sum this early birth dyad directly. A distinct jump has at most two ends; its factor $(1 + \log H)^{-5}$ pays both possible lost dyad sums, each of size $O(1 + \log H)$.

For an untagged jump in that remaining situation, use $F(t, r', 0)$ from (19), designating the early-birth end; r' is the jump's height dyad, with only a bounded number of choices comparable to r . The opposite end supplies an independent worker string. Its test stop precedes any specified bad birth there, since that birth is above $T_0 \gg t$. On the designated arc, every visit below t up through the jump occurs before the common first-hit line. Otherwise, as the designated birth is below t/L_2 , the jump itself would reach $c''r$ and subsequently return below $2t$, forcing its tag. The crosscut argument thus gives the required $E(t)$ event. If the same jump is designated from both ends, use the single paired bound $F_{12}(t, t, r')$ instead. It integrates that one jump, both early prefixes, and both opposite workers' tests. In either case the cost is $Cr^{-\alpha}$ for the distinct jump and all its designated prefixes and tests. A worker with a later bad birth still contributes Cg^α after its test, so a two-bad turn retains its factor $(g/r)^\alpha \leq C(k+1)^{-\delta\alpha}$. The single-test birth condition holds because L_2 is at least the single-test ratio in (19).

Simultaneous integration. We will combine the turn gains with restrictions on residual middle heights. The needed restriction can be stated now. For an unmarked macroscopic M leaf, let z be the combined height of its two end strings and j its middle height. Its total height is $z + j$. The normalized middle kernel

$$K_r(j) = r^{-\alpha} B_j \mathbf{1}_{\{0 \leq j \leq Cr\}}$$

obeys, for every interval I of width $w \geq 1$ and every shift z ,

$$\sum_{j: z+j \in I} K_r(j) \leq C(w/r)^\alpha. \quad (36)$$

Indeed $B_j \leq C(1+j)^{\alpha-1}$, so the sum over any $w+1$ consecutive heights is at most Cw^α . Since $k \leq pL$ and $r \geq H^\beta/2$, requiring two leaf heights to differ by $O(1 + r/\sqrt{k})$ costs $O((k+1)^{-\alpha/2})$ if one leaf is of type M: fix the partner's height, sum that middle in its translated window, and then sum the partner's middle freely if it has one. The bound is uniform in the partner's end strings and jumps, leaving those variables available for their later probe estimates. For disjoint leaf pairs these middle integrations can be performed together. The construction of sufficiently many such pairs is given below.

We now integrate the turn tests while retaining these uniform middle-height gains. Work with nonnegative critical-weight sums and fix only discrete records: the chain shape, scales, piece types, good/bad labels, tag labels, and assignments of turn tests. Do not condition any bridge law on its total height or on the completed chain. For an M leaf its atomic variables are its two oriented end prefixes, any distinct bad stopping jumps, and its middle bridge. For a Z leaf they are the two pre-jump prefixes and one shared jump. Concatenation identifies a tuple of such variables with at most a constant number of original records. Dropping unused self-avoidance and incidence conditions therefore bounds the original sum by a product of their critical measures. The restrictions

retained in this product are local turn tests, individual jump restrictions, and the residual-height constraints just described.

Each end prefix occurs at exactly one turn, or at a free chain end. Assign every distinct bad jump to exactly one of three blocks: its individual tail bound (with its tag restriction if present), a single F block, or a paired F_{12} block. A designated block contains the designated pre-jump prefix or prefixes and the opposite worker prefixes through their test stops. An ordinary E block contains its two prefixes only through their test stops. These sets of variables are disjoint. In particular a worker at a designated turn has no early designated birth there: it is either good or its birth is above T_0 . The tested worker portion therefore excludes its own separately charged bad jump. If a shared jump has just one assigned F test, the test at its other end uses only the prefix before that jump; it belongs to a different block and never uses the shared jump a second time.

First sum the residual middles with the prescribed height restrictions, using (36) for one middle in each disjoint pair and the unrestricted bound for the others. These bounds are uniform in all the end variables. Leave every designated pre-jump prefix intact inside its assigned F or F_{12} block. For an ordinary E prefix or a worker prefix, next sum only its untested continuation. Given its test stopping history, the mass of subsequent renewal occurrences in a birth dyad g is at most Cg^α , uniformly in the stopping location and overshoot: an overshoot beyond the permitted endpoint contributes zero, and otherwise the remaining range has length at most Cg . A good completion is a probability at most one. Length insertions are handled by the same conditional argument described below. Replacing these continuation sums by their uniform bounds removes them without conditioning the preceding tests. Now integrate each disjoint E , F , or F_{12} block using (19); integrate every remaining jump using its individual tail bound. Tonelli's theorem justifies this order. Relative launches in each block are fixed neighboring ports, so absolute translations depending on variables outside the block introduce no parameter into its estimate. In a paired block this is exactly the two-end convention in the definition of F_{12} . Finally sum the discrete labels and dyads as above. Thus no estimate has been applied to a renewal law conditioned on an eventual height, a tag, another turn's survival, or full-chain avoidance. This proves (35) with the stated total loss.

Long marked leaves. The ordinary-turn estimates control the unmarked chain. A marked leaf must additionally pay $H^{-\epsilon+o(1)}$ for exceeding its length threshold. Only a fixed number of turns touch such leaves, so the logarithmic first-hit estimate suffices there.

We next include rough turns and the marked length factors. Uniformly for $1 \leq u \leq CH$, the first-length moment gives

$$\sum_{j \leq Cu} \sum_{B: \text{ht}(B)=j} \rho^{|B|} (1 + |B|) \leq u^{\alpha+4/3} H^{o(1)}.$$

The analogous summed moment for an irreducible of height at most Cu is at most $u^{4/3-\alpha} H^{o(1)}$. Indeed, in a height dyad b , prepend and append bridges of heights in $[b, 2b]$, whose summed masses each are at least cb^α . The resulting bridge has height $O(b)$ and has only boundedly many irreducibles of height at least b . Its length moment bounds the marked jump's length moment, giving $b^{4/3-\alpha} H^{o(1)}$ after division by $b^{2\alpha}$. Sum the dyads, using $4/3 > \alpha$. These moment statements mean that for every fixed $\eta > 0$ the displayed factors $H^{o(1)}$ are bounded by $C_\eta H^\eta$, uniformly over all indicated scales.

Expand the product of length-plus-one factors on marked leaves by assigning each insertion to its middle, a bad jump, a good end string, or a pre-bad-jump prefix. There are only boundedly many categories per marked leaf; for an end string sum over its individual jumps. A middle or tiny-piece insertion costs at most $r_i^{4/3} H^{o(1)}$ beyond its unweighted baseline. A bad-jump insertion has the same relative cost by the irreducible moment. No tag saving or designated F test uses a marked leaf's own jump.

For an insertion in an end string, let d be the dyadic scale of one plus the inserted jump's birth depth. Up to that birth the occurrence mass is at most Cd^α . The jump's length moment, after extracting $r_i^{4/3}H^{o(1)}$, costs $Cr_i^{-\alpha}$. Thus this insertion gives an additional factor $(d/r_i)^\alpha$. Any subsequent required bad birth of dyad f costs at most Cf^α by restarting after the inserted jump, with its bad-jump price already separately budgeted; completion to a good stop has probability at most one. This includes the terminal stopping jump of a good string. Additive constants in the length weight use the unweighted estimate.

At a non-omitted rough turn apply the first-hit bound (26) at a small fraction of the smaller available depth. Use d on an end with an insertion in its prefix, f on another bad end, and q_i on another good end. The test precedes the counted births and lies within the preserved disjointness range. Subsequent occurrence counts have the same conditional bounds as before. A side of available depth v has, after extraction of its baseline and any length price, cost at most $H^{o(1)}(v/r_i)^\alpha$. For a good side the factor is supplied at polylogarithmic cost, since q_i/r_i is a negative power of $k+1$. For two incident sides the test and these factors give

$$H^{o(1)} \frac{v_1^\alpha v_2^\alpha}{r_i^\alpha r_{i+1}^\alpha} \min(v_1, v_2)^{-\alpha} \leq H^{o(1)} \min(r_i, r_{i+1})^{-\alpha},$$

because $v_j \leq Cr_j$. Free ends and omitted turns cost at most $H^{o(1)}$. There are only $O_{p,U}(1)$ rough turns and small leaves. Their scale choices cost $(1 + \log H)^{O_{p,U}(1)}$, so all these losses remain $H^{o_{p,U}(1)}$.

This rough calculation is compatible with the preceding block integration. At a rough turn an unmarked neighboring leaf supplies only its good prefix or its prefix before a bad jump; no marked insertion or rough test consumes that unmarked bad jump. Stop the rough tests before the inserted jumps, then use the conditional moment and renewal bounds for the continuations. Distinct marked leaves have distinct inserted jumps. Their length prices hence multiply. On a long marked leaf,

$$\mathbf{1}_{\{|B| > H^\epsilon(1 + \text{ht}(B))^{4/3}\}} \leq \frac{1 + |B|}{H^\epsilon(1 + \text{ht}(B))^{4/3}}.$$

Its scale is comparable to $1 + \text{ht}(B)$, so division by this threshold gives $H^{-\epsilon+o(1)}$ per mark.

The gain from many leaves. We have obtained the turn baselines and the cost of each marked length. It remains to obtain the factor $(k+1)^{-bk}$ from the many unmarked macroscopic leaves. We now construct the close-height pairs to which (36) applies.

There are disjoint sibling pairs covering all but $O_{p,U}(\sqrt{k}+1)$ leaves for which the two leaf heights differ by $O(1 + r/\sqrt{k})$. Indeed all but $O(p)$ expansions lie on unary chains. The total drop of continuing heights along each such chain is at most $2r$, so at most $O(\sqrt{k})$ expansions on that chain have a drop larger than $r/\sqrt{k}$. At every other unary expansion the two non-continuing children are leaves. Write h for the parent height, a, c for the forward child heights, and d for the retreat height. The signed-height identity is $h = a - d + c$, and all three child heights are at most h . If the first child continues, the drop $h - a = c - d$ is exactly the difference of the other two heights; the last-child case is symmetric. If the retreat continues, then $a + c = h + d$, with $a, c \leq h$, so $a, c \in [d, h]$ and $|a - c| \leq h - d$. Consequently a continuing-height drop at most $r/\sqrt{k}$ always supplies a sibling pair whose heights differ by at most that amount, up to the bounded port recodings. Remove the boundedly many pairs touching marked or small leaves. Specifying which of these pairs meet the constraint and their types costs at most exponentially many choices in k .

In each retained pair with at least one M leaf, integrate one such middle kernel with the height constraint, holding the partner's height fixed. If the partner also has a middle, sum that middle without the constraint. The pairs are disjoint, so these integrations are simultaneous and uniform in all probe variables. These are the residual integrations used in the block argument above. There

are $N = 2k + 1$ leaves. If at least $N/10$ are M, then, after removing the $O_{p,U}(\sqrt{k} + 1)$ exceptions, at least $N/40$ retained pairs contain an M leaf for large k . Their total gain is at most $(k + 1)^{-\alpha N/80}$. The at most N good-good turns cost at most $(k + 1)^{\delta \alpha N}$, so a fixed choice $0 < \delta < 1/160$ leaves a negative multiple of $k \log(k + 1)$. If fewer than $N/10$ leaves are M, at least $4N/5 - 1 - O_{p,U}(1)$ adjacencies join two Z leaves, allowing for the bounded number of small leaves and rough exceptions. At most $N/10$ adjacencies can be good-good. For large k the excess of bad-bad over good-good ordinary turns is therefore at least $N/2$, giving the factor $(k + 1)^{-\delta \alpha N/2}$. Thus in either case the total factor is at most $(k + 1)^{-bk}$ for an absolute $b > 0$, apart from constants to the power k and the stated logarithmic losses. Bounded k are absorbed by changing the constants.

Ordered ternary tree shapes, choices of the continuing children, piece labels, and the finite recoding records have at most exponential multiplicity in k . Choices of the fixed number of marks contribute $H^{o_p(1)}$. Small-scale dyads do so as well. The factor r^α , root placements and exterior paths are all covered by a fixed polynomial H^C independent of p, U . This proves the bound in (34).

For the final assertion of the lemma, start instead with the given alternating chain of $k \leq pL$ comparable, nonincreasing spans. This range preserves the probe and dyad estimates above. There are no small leaves or placement weights. Pair successive pieces, leaving at most one unpaired. Since the total height drop is at most $2r$, all but $O(\sqrt{k})$ pairs have difference at most $Cr/\sqrt{k}$. The same recoding, tests, variable partition, and middle-kernel integrations apply, and give the stated estimate. $\square$

8.4 Slow-path amplification

Proof of Theorem 8.1. Choose $\epsilon > 0$ much smaller than τ . In the notation of Lemma 8.4, when $k = L$ the logarithm of the last two factors of (34) is

$$-bL \log L + O(L \log(2 + \log \log H)),$$

which is smaller than $-A \log H$ for every fixed A and all sufficiently large H . Thus spine witnesses have superpolynomially small mass. For arbitrary $k \leq pL$, let $V_H = C'(1 + \log \log H)^{C'}$. The maximum of $k \log V_H - bk \log(k + 1)$ is $O(V_H^{1/b}) = o(\log H)$. Consequently the sum over k costs $H^{o(1)}$. Choose p sufficiently large that the factor $H^{-p\epsilon}$ absorbs the fixed polynomial and the desired failure power. Next choose U large enough that $pU^{-4/3} < 1/4$. Constants depending on these now fixed parameters do not affect either conclusion. Summing the $O(\log H)$ root dyads makes all unwanted witnesses negligible at any prescribed power.

Sum the witness bounds over their sizes and use the kernel K_γ defined above. Since each witness count dominates its existence indicator, the resulting bound controls the $\mu_H(d\gamma)K_\gamma(d\omega)$ -mass of pairs containing any unwanted witness. The root-subarc choices are already included in that count. Add the path exceptional set from Lemma 8.3, with parameter $\lambda = \epsilon$.

Every remaining path-and-trial configuration satisfies the deterministic implication (33). That weak-bridge statement is a property of γ alone. If a path violates it and is not in the thin exceptional set, every trial configuration over that path must contain an unwanted witness. Therefore, writing $\mathcal{B}$ for the bad paths and $\mathcal{W}$ for the unwanted configurations,

$$\mu_H(\mathcal{B}) = \int_{\mathcal{B}} K_\gamma(\Omega_\gamma) d\mu_H(\gamma) \leq (\mu_H K)(\mathcal{W}) + \mu_H(\text{thin-excursion exception}).$$

There is no loss from projecting away the trial variables.

To pass to arbitrary subwalks, split at a lowest vertex and read both halves away from that vertex. Successively cut each half at the last alternating extremum of the remaining tail. This gives

weak bridges of nonincreasing spans. Outside another superpolynomial exceptional set, there are fewer than $\lceil \log H \rceil$ consecutive pieces in any one span dyad above $H^\beta/2$: otherwise apply the final part of Lemma 8.4 to $k = L = \lceil \log H \rceil$ of those pieces. Root locations and exterior pieces again have only fixed polynomial cost. Summing the weak-bridge length bound over decreasing dyads gives at most $CH^{2\epsilon} \log H(1+s)^{4/3}$ per half. Once the remaining span is below H^β , the whole remaining tail is thin and has length at most H^ϵ . Choose ϵ so that these bounds fit $H^\tau(1+s)^{4/3}$ for large H . Enlarging the exceptional-set constant handles bounded H . A finite union over lattice symmetries proves the simultaneous directional statement. $\square$

8.5 Many crossings

We use the all-count cylinder estimate stated among the analytic inputs. For clarity, its precise physical form is the following: on the balanced two-marker cylinder with $N = 2m$ bands, the first m band parameters are 2λ and the next m are λ . If $w_N(a)$ is the critical mass of families of exactly a mutually disjoint polygons separating the markers, then

$$w_N(a) \leq \exp\left(C \log N - \frac{ca^2}{\log N}\right) \quad (a \geq 0) \quad (37)$$

for every sufficiently large even N , with one pair of constants independent of N and a . The analytic input proves, before sector truncation, $Z_N(2 \cosh v) \leq \exp(C \log N(1+v^2))$ for every $v \geq 0$; positivity and optimization in v give (37). The quantifier over all a is needed below, since the number of crossings grows as a power of the ambient scale.

Lemma 8.5 (Many bridges in one slab). *Fix $D < \infty$. Consider ordered tuples of a mutually avoiding strict bridges across the same $j \geq 1$ bands, all contained in one abscissa slab of width Dj . Prescribe the distinct starting ports, the distinct ending ports, and their matching. The critical mass of these tuples is at most*

$$\exp\left(C_D \log(2+j) - \frac{c_D a^2}{\log(2+j)}\right). \quad (38)$$

Proof. Put $d_0 = \sqrt{3}/2$ and use triangular coordinates with generators $(1, 0)$ and $(1/2, d_0)$. Choose a fixed integer $b = b(D)$ sufficiently large and put $m = bj$. In the physical cylinder the successive half-period translations are

$$m(-1/2, d_0), \quad m(1/2, d_0).$$

Thus the total period is $(0, 2md_0)$ and the marker abscissas differ by $m/2$. For b sufficiently large, an integer horizontal translation places the entire width- Dj slab strictly between those abscissas, with a fixed positive margin. Align its bottom with a marker band level.

Label the fixed lower ports $s_1, \dots, s_a$ and the fixed upper ports $t_1, \dots, t_a$, and write π for the fixed matching: strand i joins s_i to $t_{\pi(i)}$. Choose independently $2b$ tuples from the ensemble. Put the first in bands $0, \dots, j$ and reflect the second across the line at height jd_0 , traversing it upward. Reflection across any horizontal band boundary is a lattice symmetry. The second strip takes $t_{\pi(i)}$ to the translate of s_i at height $2jd_0$. Repeat this construction in successive double strips. Each double strip therefore induces the identity on the lower port labels. After all $2b$ strips, the vertical quotient closes each strand at its own initial port.

The strip interiors are disjoint. Strict bridges meet their boundary cuts only at the fixed ports, and those ports are distinct within each tuple. Hence the resulting family contains exactly a disjoint simple polygons. Each lift travels through one full vertical period, so each polygon is essential. Since the whole family stays between the marker abscissas, each polygon separates the two markers.

Cutting this family at the fixed j -spaced levels recovers every input tuple. The strip index specifies its reflection, and the prescribed ports recover the strand labels and matching. Recovery is therefore injective even though the polygon family itself is unlabeled and unoriented. Ports carry no vertex weight, so the critical weights multiply exactly at every concatenation. If M is the input tuple mass, positivity and (37) give

$$M^{2b} \leq w_{2bj}(a) \leq \exp\left(C \log(2bj) - \frac{ca^2}{\log(2bj)}\right).$$

Take the $2b$ th root and use $\log(2bj) \asymp_b \log(2+j)$. For bounded j , the fixed-width slab contains only boundedly many distinct ports, so a is bounded; the finite bridge masses cover these cases by changing the constants. $\square$

8.6 Local vertex mass and the temporal modulus

Proposition 8.6 (Uniform local mass bound). *For every $\tau > 0$ and $A > 0$, there is $C_{\tau,A} < \infty$ such that for all $H \geq 2$, outside a set of μ_H -mass at most $C_{\tau,A}H^{-A}$, simultaneously for every visited vertex v and every real radius $1 \leq s \leq H$,*

$$\#(V(\gamma) \cap \overline{B}(v, s)) \leq H^\tau s^{4/3}. \quad (39)$$

Proof. Use Theorem 8.1 with a smaller slack $\tau' > 0$. It suffices to prove the statement at dyadic radii with further slack; rounding a radius upwards then absorbs the constant for all radii. There are polynomially many possible centers and $O(\log H)$ scales. Small fixed radii are harmless by the lattice area bound.

For fixed v, s , surround v by a lattice hexagonal box whose faces are at perpendicular distances between $10s$ and $20s$. For each face fix a parallel inner lattice line about $4s$ inward. Starting at the next visit to $\overline{B}(v, s)$, follow the walk until its first subsequent hit of a box face, or its endpoint; repeat at the next visit to the ball. Each such visit batch is contained in the box. Its span is $O(s)$, so the local length bound, with bounded trimming at face ports, allows at most $CH^{\tau'}s^{4/3}$ visited vertices in that batch. If (39) fails, then for a fixed small $0 < \xi < \tau - \tau'$ and large H , at least $a = \lceil H^\xi \rceil$ completed passages reach the same face. There is at most one uncompleted terminal batch.

From each selected passage retain its segment from the last hit of that face's inner line to the first hit of the face. These are mutually disjoint strict bridges, all of height $j \asymp s$, in the common box and hence a slab of width Dj , after a lattice symmetry. If there are fewer than a available ports, the event is empty. Otherwise the endpoint tuples, matchings, and chronological placements have at most $\exp(O(a \log H + \log H))$ choices. The complementary path segments cost the same order: there are $a + 1$ of them, each with polynomial full-box susceptibility, and we may drop their mutual avoidance. Apply Lemma 8.5 to each fixed choice. Since $s \leq H$, the total mass is bounded by

$$\exp\left(Ca \log H + C \log H - \frac{ca^2}{\log H}\right),$$

which is smaller than every power of H^{-1} for $a = \lceil H^\xi \rceil$. The polynomial union over centers and scales, and the exceptional set for the local length estimate, preserve the claim. $\square$

Proposition 8.7 (Uniform temporal modulus). *For every $\tau > 0$ and $A > 0$ there is $C_{\tau,A} < \infty$ such that, for every $H \geq 2$, outside a set of μ_H -mass at most $C_{\tau,A}H^{-A}$, every subwalk σ of l steps satisfies*

$$\text{diam}(\sigma) \leq H^\tau (1 + l)^{3/4}.$$

Proof. A violation has diameter $d > H^\tau(1+l)^{3/4}$. Some lattice normal projection has span at least cd . Between its extreme levels take an intervening weak bridge and extend it to a strict bridge. Its height h is at least cd and at most CH , and its length is at most $l + O(1)$. Thus $h \geq cH^\tau$ and, for a smaller fixed $\tau' > 0$ and large H ,

$$l + O(1) \leq CH^{-4\tau/3}h^{4/3} \leq h^{4/3-\tau'}.$$

Apply the fast-bridge estimate (28) dyadically in h . The height lower bound is a positive power of H , and all placements and exterior portions cost only a fixed polynomial by the full-box bound. Its arbitrary failure power therefore gives the assertion. $\square$

9 Completion of the fixed-length theorem

We now collect the consequences of the full-box estimates. This also identifies exactly where their unnormalized statements become probability estimates at a specified length.

9.1 The probability transfer

Let $Z_n = c_n \rho^n$ be the critical partition function of the n -edge paths from the fixed root.

Lemma 9.1 (Normalization at every length). *For every integer $n \geq 0$,*

$$Z_n \geq 1. \tag{40}$$

Proof. The honeycomb lattice is vertex-transitive. Cutting a walk after m edges and forgetting the mutual avoidance of its two pieces gives $c_{m+n} \leq c_m c_n$. If $Z_{n_0} = a < 1$ for some positive n_0 , writing $n = qn_0 + r$, $0 \leq r < n_0$, gives

$$Z_n \leq a^q \max_{0 \leq r < n_0} Z_r \leq C e^{-cn}.$$

Trimming the two endpoint half-edges of a port bridge gives a vertex walk from one of finitely many possible neighbors of the source port; the weight ratio and inverse multiplicity are bounded. A height- h bridge has at least a constant multiple of h edges. The preceding bound would therefore imply $B_h \leq C \sum_{n \geq c'h} e^{-cn} \leq C' e^{-c'h}$, contrary to $B_h \asymp h^{-1/4}$. The zero-length case is immediate. $\square$

Every n -step path from o lies in a Euclidean ball of radius a constant times n . Choose the full-box scale $H = Kn$, with a fixed K large enough to contain that ball and the bounded port extensions used in cutting. The distinction between the critical edge and vertex weights at those extensions costs a bounded factor. If a bad collection in this full box has unnormalized critical mass at most $C_A H^{-A}$, then its subset of length n has $\mathbb{P}_n$ -probability at most

$$\frac{C_A H^{-A}}{Z_n} \leq C_A H^{-A}.$$

Thus arbitrary polynomial decay in H is preserved at *every* length. There is no averaging over lengths in this step.

For convenience we record the three simultaneous consequences used below. For every $\eta > 0$ and $k > 0$, outside an event of probability $O_{\eta,k}(n^{-k})$, all subwalks $\gamma[i, j]$ and all visited centers satisfy

$$j - i \leq C n^\eta (1 + \text{diam } \gamma[i, j])^{4/3}, \tag{41}$$

$$\text{diam } \gamma[i, j] \leq C n^\eta (1 + j - i)^{3/4}, \tag{42}$$

$$M_\gamma(z, s) \leq C n^\eta s^{4/3} \quad (1 \leq s \leq n). \tag{43}$$

Here (41) follows from the slow-path estimate (31), since a directional span is bounded by the Euclidean diameter. Equation (42) is Proposition 8.7, deduced from (28). Equation (43) is (39). The constants C and the fixed factor K^η can be absorbed by a slightly larger exponent. Those statements already count all needed positions, directions, and scales in a full box; no independence of subwalks under $\mathbb{P}_n$ is asserted.

9.2 Diameter and local mass

Proof of Theorem 1.1. Fix $\delta, k > 0$. Use (41)–(43) with a positive η much smaller than δ and with the prescribed failure exponent k . All arguments below take place on their single simultaneous good event.

Applying (41) to the entire path gives

$$1 + D \geq cn^{(3/4)(1-\eta)}.$$

Equation (42), again for the entire path, gives $D \leq Cn^{3/4+\eta}$. Taking η small enough, then n large enough, absorbs the constants and the additive one and proves (1).

For the upper local mass bound, use both the deterministic bound $M_\gamma(z, s) \leq n + 1$ and (43). For sufficiently small η and large n this gives

$$M_\gamma(z, s) \leq n^\delta \min\{n, s^{4/3}\}.$$

To prove the lower bound, fix $z = \gamma_i$ and $s \in [1, n]$, and put $m_0 = \min\{n, s^{4/3}\}$. If $n^{-\delta}m_0 \leq 1$, the vertex z itself suffices. Otherwise choose

$$q = \lfloor n^{-\delta/2}m_0 \rfloor.$$

For large n , one side of i has at least $n/2 \geq q$ edges available. Choose the contiguous block of q edges on that side. Since $m_0 > n^\delta$, we have $q \geq \frac{1}{2}n^{-\delta/2}m_0$ and $q + 1 \leq 2n^{-\delta/2}m_0$ for large n . Its diameter is at most

$$Cn^\eta(1 + q)^{3/4} \leq C'n^{\eta-3\delta/8}m_0^{3/4} \leq C'n^{\eta-3\delta/8}s \leq s,$$

provided $\eta < 3\delta/8$ and n is large. The entire block lies in $\overline{B}(z, s)$. Self-avoidance makes its $q + 1$ vertices distinct, and $q + 1 \geq n^{-\delta}m_0$ for large n . This proves the lower local mass bound simultaneously in i and s , using only the already simultaneous modulus.

Covering numbers

For a lower bound, consider any cover by b balls of radius s . Discard balls missing $V(\gamma)$. In each remaining ball choose one visited vertex z . The visited vertices in that ball lie in $\overline{B}(z, 2s)$. When $2s \leq n$, the upper local mass estimate at $2s$, with smaller exponent slack, bounds their number by

$$Cn^\eta \min\{n, s^{4/3}\}.$$

When $2s > n$, the deterministic mass bound $n + 1$ gives the same conclusion after changing C . Summing over the covering balls gives

$$b \geq cn^{-\eta} \max\{1, ns^{-4/3}\}.$$

Since $\max\{1, u\} \geq (1 + u)/2$, constants and the smaller exponent slack imply the lower bound in (3).

For the upper bound, fix a positive $\epsilon < \delta$ and put $m_0 = \min\{n, s^{4/3}\}$ as above. If $n^{-\epsilon}m_0 < 2$, use one ball for each vertex. The inequality $m_0 < 2n^\epsilon$ implies

$$n + 1 \leq Cn^\epsilon(1 + ns^{-4/3}).$$

Otherwise let $q = \lfloor n^{-\epsilon}m_0 \rfloor$. Split time into consecutive blocks of at most q edges, allowing consecutive blocks to share one endpoint. Each block has diameter at most

$$Cn^\eta(1 + q)^{3/4} \leq C'n^{\eta-3\epsilon/4}s \leq s$$

if $\eta < 3\epsilon/4$ and n is large. A ball of radius s centered at a vertex in the block covers it. The number of blocks is at most

$$1 + \frac{n}{q} \leq Cn^\epsilon(1 + ns^{-4/3}).$$

Taking $\epsilon < \delta$ absorbs the constant. The whole argument is deterministic on the good event, so the bounds hold simultaneously for all real radii in the stated range. This completes the proof. $\square$

Figure 10 illustrates the two deterministic steps that turn temporal and local-mass control into covering bounds.

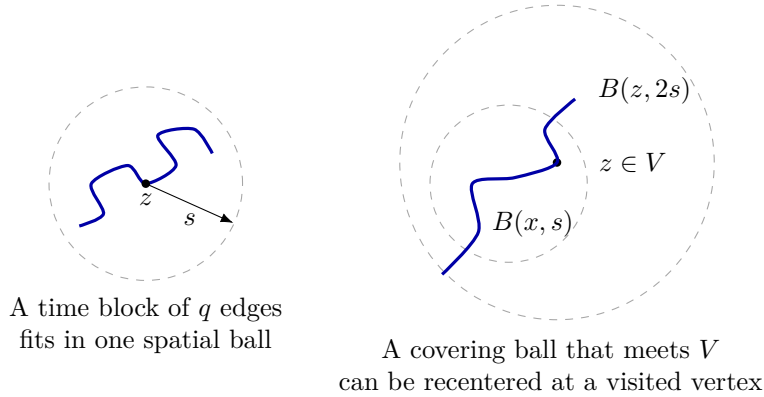


Figure 10: The two deterministic uses of local control. The modulus bounds the diameter of a contiguous block and yields the upper covering bound. The local mass upper bound, after recentering, yields the lower covering bound. The curves are schematic.

Proof of Corollary 1.2. The local-length theorem holds simultaneously in the finitely many lattice normal directions. Apply it to the whole walk after the same exact-length probability transfer. If s is any such band span, $n \leq Cn^\eta(1 + s)^{4/3}$, so $s \geq cn^{(3/4)(1-\eta)} - 1$. Convert band height to Euclidean projection using d_0 , then choose η sufficiently small relative to δ . Constants and the additive one are absorbed for all sufficiently large n . $\square$

9.3 The dimension ratio and moments

Proof of Corollary 1.3. Fix $\epsilon \in (0, 3/4)$. On the event of Theorem 1.1 with error $\delta < \epsilon/2$, uniformly for $s \leq n^{3/4-\epsilon}$,

$$\log(D/s) \geq (\epsilon - \delta) \log n, \quad \log \mathcal{N}(s) = \log n - \frac{4}{3} \log s + O(\delta \log n) + O(1).$$

The second assertion follows because $ns^{-4/3} \geq n^{4\epsilon/3}$; replacing $1 + ns^{-4/3}$ by $ns^{-4/3}$ changes its logarithm by $O(1)$. Also

$$\frac{4}{3} \log(D/s) = \log n - \frac{4}{3} \log s + O(\delta \log n).$$

The difference of the two quantities, divided by $\log(D/s)$, is $O(\delta/\epsilon) + o(1)$. Given a desired error, first choose δ small enough and then let $n \rightarrow \infty$. The bad event has probability tending to zero, proving the Corollary. $\square$

Proof of Corollary 1.4. The exact identity

$$R_g^2 = \frac{1}{2(n+1)^2} \sum_{i,j=0}^n |\gamma_i - \gamma_j|^2$$

shows $R_g \leq D$. To get the lower bound, fix $0 < \epsilon < 3/4$ and put $r = n^{3/4-\epsilon}$. Use the upper local mass bound with slack $\eta < 4\epsilon/3$. Uniformly in i , the number of j within distance r of γ_i is at most

$$n^\eta r^{4/3} = n^{1+\eta-4\epsilon/3} = o(n).$$

For large n , at least $(n+1)/2$ indices j are farther than r from each γ_i . Hence $R_g^2 \geq r^2/4$. Arbitrarily small ϵ , together with the diameter upper bound and absorption of the constant, proves the claimed probability law.

Finally both D and R_g are deterministically at most a constant times n . For fixed $p > 0$, choose the failure exponent $k > p + 1$. Their p th moments on the bad event are $O(n^{p-k})$. On the good event, their p th powers lie between $n^{3p/4-p\delta}$ and $n^{3p/4+p\delta}$ after adjusting the exponent error. Taking expectations and then letting $\delta \downarrow 0$ proves both moment assertions. $\square$

A An alternative avoidance proof using asynchronous renewals

The ordered-probe theorem in the main argument yields the logarithmic adjacent-arm bound. We record a different proof of the weaker exponent estimate. It extends many pairs of asynchronously chosen renewal prefixes to a common height and uses the telescoping strip identity there. This argument uses the earlier localization and seam estimates, but is independent of the ordered-probe theorem and fast-path amplification. It makes no assumption that the two original paths share a renewal level.

The obstacle lemma below uses high moments of recoverable joining lines, in contrast to the second-moment calculation giving the exact $H^{-1/4}$ bound in Lemma 4.11; this supplies a distinct sewing argument for the avoidance proof.

Let $S_2(H)$ be the probability that two independent infinite renewal walks, launched vertically from neighboring horizontal bottom ports, are disjoint through their respective first visits to level H . This event does not require the walks to have a common renewal level. Let $D(j, j)$ denote the unnormalized mass of two disjoint height- j bridges from these ports. Joining their starts by a fixed one-band exterior arch, and translating to fix the first terminal port, produces an arch counted at height $j + 1$ but not at height j . The input is recoverable with bounded multiplicity. The strip winding identity therefore gives

$$D(j, j) \leq C(B_j - B_{j+1}). \tag{44}$$

Lemma A.1 (Sewing past an obstacle). *Fix $D < \infty$ and $\delta > 0$. A port with abscissa x on level u is to be joined to level v , where $0 \leq v - u \leq H$. Suppose the obstacle in this slab lies to the left of $x + DH$, and in its first $\min(\delta H, v - u)$ bands lies to the left of $x - \delta H$. The mass of bridges to a free endpoint at level v avoiding that obstacle is at least $H^{-1/4-o(1)}$, uniformly over these data. The reflected statement also holds.*

Proof. For $v - u \leq \delta H/2$, use a vertical tube and Lemma 4.5. Zero gap uses the empty path, and bounded positive gaps use fixed elementary bridges. For larger gaps construct a corridor that initially rises vertically, moves far right within a small fraction of the first δH bands, then rises beyond $x + (D + 1)H$ to the target. It consists of allowed normal directions: up, right-up at 30° , alternating right-down and right-up at $-30^\circ, 30^\circ$, and finally up. Consecutive directions have positive dot product. Choose each zigzag segment a small fixed multiple of δH , with equal lengths in alternating pairs, and a sufficiently large fixed number of pairs to pass the obstacle. All lengths are bounded above and below by positive multiples of H ; the constants may depend on D, δ . The corridor has strict slack from the forbidden boundaries except at its endpoint walls.

At each joint vary a cut perpendicular to the next direction over a small fixed-proportion range. Before a corner C insert a parallel joint $F = C - \ell n_1$, where n_1 is the current unit normal and ℓ is a sufficiently small fixed multiple of H . The stages are long parallel-to-parallel tubes and short corner pieces from the F -cut, normal to n_1 , to the C -cut, normal to n_2 , where $n_1 \cdot n_2 = 1/2$. Each carries mass $\geq cH^{-1/4}$. For long stages this is Lemma 4.5. For a corner use the convex lattice domain between the start line and a parallel far line with $n_1 \cdot (p - C)$ near $\ell/100$, bounded also by

$$-(.5 + .02)\ell < (n_1 - n_2) \cdot (p - C) < .02\ell,$$

with lattice rounding. This domain contains a thin normal tube past the target cut if starting errors, cut ranges and widths are small relative to ℓ . Convex cutting in Lemma 4.1 pays the short piece by the mass of that tube.

The corner piece stays within $O(\ell)$ of C ; its exit satisfies $n_1 \cdot (p - F) > .4\ell$. The next long tube, from that actual exit, has width small enough that throughout it $n_1 \cdot (p - F) > .3\ell$. Choose the successive short scales increasing so that accumulated errors are small fractions of the next scale, while the final scale is still tiny compared with the original segments. There are only finitely many segments, so these choices are consistent, and every cut range still has $\asymp H$ levels. Adjacent segments are separated at their common cut: before switching at F , the incoming tube lies strictly on its near side by the positive dot product. Nonconsecutive segments are separated by the small widths. Thus concatenated paths are simple and avoid the obstacle.

We check the multiplicity of these movable joints. Only the two adjoining stages visit the possible port region of a joint. At F ports are much closer to F than ℓ , clear of the outgoing long tube; at C they are within $O(\ell)$, clear of the incoming long tube. Thus split times for different joints occur in their fixed temporal order in every representation. If two representations split differently at one joint, the intervening arc is a bridge between the two cut lines. It also remains strictly between the neighboring cut ranges: at F it lies in the cut band and the incoming restricted tube, hence very close to F ; at C the outgoing restricted tube gives $n_1 \cdot (p - F) > .3\ell$.

Fix a base representation and select p other admissible levels at one joint. Before and after the base level these subdivide the adjoining stages into bridges, one per ordered gap. The end pieces have bounded mass, apart from polynomial placement, by convex winding in their endpoint half-planes with distant walls. For example, the piece before the earliest selected split remains in its near half-plane; any change of split time at the preceding joint also lies strictly in that half-plane by the preceding separation argument. The same holds after the last split. Lemma 4.3 therefore gives $C_p H^{C+3p/4}$ for the joint's possible-level count, with C independent of p . A fixed level has a unique

port and split time, by the local single-crossing constraints. High fixed moments discard counts exceeding $H^{3/4+\zeta}$ with arbitrary polynomially small mass, for every $\zeta > 0$.

For m stages the construction sum is at least $cH^{m-1-m/4}$. There are $m-1$ joints, so division by their multiplicities leaves $cH^{-1/4-(m-1)\zeta}$. The discarded part is negligible relative to the construction sum by choosing sufficiently high moments. Since m is fixed and ζ arbitrary, this proves the lemma. $\square$

A.1 Horizontal displacement of an irreducible

For one random irreducible bridge let J be its height and W its largest absolute horizontal displacement from its starting port. The renewal law already gives $\mathbb{P}(J > x) \leq C(1+x)^{-3/4}$.

Lemma A.2 (An irreducible displacement bound). *For $x \geq 1$,*

$$\mathbb{P}(W > x) \leq Cx^{-3/4}, \quad \mathbb{E} \min\{1, (J+W)/x\} \leq Cx^{-3/4}.$$

Proof. Choose a large fixed A , to be specified below. We prove the bound for $x \geq 2A$; increasing the final constant covers $1 \leq x < 2A$. Put $r = \lfloor x/A \rfloor \geq 1$, and write $p = \mathbb{P}(W > x, J \leq r)$. Let Z count renewal levels $u \leq r$ whose next irreducible has $W > x$ and $J \leq r$. Independence of the next increment gives

$$\mathbb{E}Z = p \sum_{u=0}^r B_u \asymp r^{3/4}p.$$

For an ordered pair of distinct marks, expose the first marked jump through its completion. Its terminal height is at most $2r$. Conditional on this history, the expected number of later marks born at height at most r is zero if the endpoint exceeds r , and otherwise is at most $p \sum_{v=0}^r B_v \leq Cr^{3/4}p$. Consequently

$$\mathbb{E}[Z(Z-1)] \leq 2Cr^{3/4}p\mathbb{E}Z, \quad \mathbb{E}Z^2 \leq Cr^{3/4}p(1+r^{3/4}p).$$

Cauchy–Schwarz therefore gives $\mathbb{P}(Z > 0) \geq c \min(1, r^{3/4}p)$.

On $Z > 0$, stop immediately after completing the first marked jump. This is a stopping index in the irreducible sequence, at height at most $2r$; the fresh continuation is independent of the exposed history. It forces an excursion farther than $x/2$ from the original root: either its start is already that far away, or its displacement forces this during the jump. After detection the expected number of further renewals within another r bands is $\asymp r^{3/4}$. Every resulting bridge, ending by $3r$, has that excursion. Summing the lateral loss in Lemma 4.1 gives total mass at most $Cr^2x^{-5/4}$. Dividing by the continuation count bounds the detection probability by $C(r/x)^{5/4}$. Take A large enough to exclude the value 1 in the preceding minimum. It follows that $p \leq C_A x^{-3/4}$; adding $\mathbb{P}(J > r)$ proves the first claim. The union bound for $J+W$ and integration of its tail give the second:

$$\mathbb{E} \min(1, (J+W)/x) = x^{-1} \int_0^x \mathbb{P}(J+W > u) du \leq Cx^{-3/4}.$$

$\square$

Theorem A.3 (Adjacent-arm avoidance). *For every $\varepsilon > 0$, $S_2(H) \leq C_\varepsilon H^{-3/4+\varepsilon}$. Equivalently,*

$$S_2(H) \leq H^{-3/4+o(1)}. \quad (45)$$

Proof. Fix small $\varepsilon > 0$, put $p_* = 3/4 - \varepsilon$, and use strong induction with bound KH^{-p_*} . Choose a large fixed L , then set $H_0 = \lceil H/L \rceil$. Stop each path at its first renewal at or above H_0 . On survival through H , the event that either prefix has sum of $J+W$ greater than $H/8$ has probability at most

$$C_\varepsilon KL^{-\varepsilon} H^{-p_*}. \quad (46)$$

Indeed, group jumps starting below H_0 into dyadic height bins $[j, 2j)$, with a separate bin for height zero. For one path expose the other path, and stop the first at its first renewal at or above j . Survival through first visits to j is measurable then. The expected subsequent sum of charges $\min(1, (J + W)/H)$ from starting heights in that bin is at most $Cj^{3/4}H^{-3/4}$, uniformly in the overshoot; if it skips the bin the contribution is zero. Multiply by $S_2(j)$ and sum:

$$CH^{-3/4} \left[1 + K \sum_{j \leq H_0}^{\text{dyadic}} j^\varepsilon \right] \leq C_\varepsilon KL^{-\varepsilon} H^{-p*}.$$

If a sum exceeds $H/8$, its truncated charge sum is at least $1/8$, so this proves Equation (46). Choose L so this costs less than one quarter of the induction budget.

Fix two stopped prefixes having these size bounds and surviving through H_0 . Their future irreducibles remain independent. Under the law of these two continuations, we shall exclude an event of arbitrarily small prescribed probability, uniformly over the fixed prefixes. Outside that event, continued survival through H supplies at least $cH^{3/4}$ suitable renewals on each side, in $[\eta H, 3H/4]$, such that every cross-pair can be extended to a disjoint equal-height pair at cost $H^{-1/4-o(1)}$.

We first retain many renewals in this interior height window while controlling their horizontal range. Take the next $\lceil c'H^{3/4} \rceil$ jumps after either stop. Lemma A.2, applied to their truncated heights, shows that all these renewals lie within an additional $H/2$ bands except with probability $O(c')$. Renewals within ηH of the stop have expected number $O((1 + \eta H)^{3/4})$, and can be discarded. The full first-hit paths to H have horizontal range at most $D_0 H$, apart from conditional probability $O(D_0^{-3/4})$: sum the truncated charges of jumps starting below H , whose expected number is $O(H^{3/4})$.

For right renewals expose also the full left first-hit path. Its future is independent of the newly sampled right future. At height u , let $R(u)$ be its rightmost abscissa within $[u, u + \delta H]$. Lemma 4.2 implies

$$\sum_{j \geq \eta H} \|b_j\|_\infty \leq C \sum_{j \geq \eta H} (B_j - B_{j+1}) \leq CB_{[\eta H]}.$$

Thus the expected number of right renewals, at least ηH above their stop, with $|x - R(u)| \leq 2eH$, is at most $C_\eta(e + H^{-1})H^{3/4}$. Discard also renewals whose path up to the next renewal more than δH above them travels horizontally by more than eH . Restarting there and using the truncated-charge bound gives expected discarded count at most $C(\delta/e)^{3/4}H^{3/4}$.

On survival, the disjoint first-hit arcs have fixed left-right order across the slab. At a given height no point of the left arc can lie to the right of the right arc's rightmost point: that region is in the right component of its complement. At a retained right renewal the right arc in the next δH bands has abscissa at most $x + eH$. Hence $R(u) \leq x + eH$; exclusion of $|x - R(u)| \leq 2eH$ forces $R(u) < x - 2eH$. The reflected argument holds for left renewals. Choose first c' , then η , then e , and then $\delta < e$, sufficiently small, and finally D_0 large. Markov's inequality and a union bound, applied to the two tests without conditioning either on success of the other, leave $cH^{3/4}$ renewals per side with any prescribed fixed small failure probability. Lemma A.1 applies to every cross-pair: the shorter bridge is extended above its renewal level, so it cannot meet its own prefix, while the other prefix is the obstacle. This proves the assertion.

Choose the just-described failure probability less than $\frac{1}{4}L^{-p*}$, with a harmless adjustment for the ceiling in H_0 . Its total contribution is then at most one quarter of KH^{-p*} , by the induction bound on $S_2(H_0)$.

Finally, under the original product renewal law, let Z_H count pairs of disjoint renewal prefixes in the stated height and horizontal ranges for which the obstacle lemma's clearance conditions

allow the prefix of lesser height to be extended to the other's height. Equal heights use the empty extension. Set $Z_H = 0$ unless the first-hit paths survive through H . These geometric conditions depend only on the two finite prefixes, and their extension masses have a common lower bound $a_H = H^{-1/4-o(1)}$. A pair of prefixes occurs with exactly the product of its bridge weights. Dropping the subsequent survival condition and appending every allowed extension therefore gives

$$a_H \mathbb{E} Z_H \leq \sum_{\Gamma} w(\Gamma) M(\Gamma),$$

where Γ is an output pair of disjoint equal-height bridges, $w(\Gamma)$ is the product weight, and $M(\Gamma)$ counts its input representations. The common output height lies in $[\eta H, H]$. Equation (44) telescopes to total output mass $O_\eta(H^{-1/4})$. For a fixed output, an input is recovered by choosing which bridge was extended and a single-crossing seam in that bridge; its unextended companion fixes the common height. The two end pieces in the seam decomposition are strip bridges. Dropping avoidance with the companion bounds its mass by $B_j \leq 1$, and summing the common height costs only $O(H)$, independently of the moment order. Lemma 4.3 bounds the multiplicity by $H^{3/4+\zeta}$ outside arbitrarily small polynomial mass. Raw seam counts are $O(H)$, so these exceptional contributions remain negligible. Dividing by the extension mass gives

$$\mathbb{E} Z_H \leq H^{3/4+o(1)}.$$

On the retained survival event, $Z_H \geq c^2 H^{3/2}$. Its probability is therefore $H^{-3/4+o(1)}$, less than the remaining half of the induction budget for sufficiently large H . Increase K to cover the finitely many smaller heights. The induction proves the theorem. $\square$

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